Acuitas System Administration Guide

Release 9.4.0.0
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Preface

This document describes how to install, configure, operate, and maintain the Acuitas Service Management System.

Note: *The Affirmed Networks Mobile Content Cloud™ is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is written for the Administrator who installs, configures, operates, and maintains the Acuitas Service Management System.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

This guide contains the following chapters:

Chapter/Appendix	Describes
Chapter 1, Preparing the System for Installation/Upgrade	How to prepare the system to install/upgrade the Acuitas server and database software
Chapter 2, Standalone Systems	How to administer Acuitas on a standalone system
Chapter 3, High Availability Systems	How to administer Acuitas on a High Availability (HA) system
Chapter 4, Other Administrative Information	Additional administrative information
Appendix A, OS Upgrade Considerations	How to upgrade the OS and how to restore the OS to the pre-upgrade level

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	**Times New Roman bold font**	Use **configure_ha** to configure... Enter **y**. To continue, press **Enter**.
Selecting a menu item	**Menu** => **Option**	Select **Security** => **Group Management**.
Variable parameters for which you supply values	***<courier bold italic blue font>***	**configure_dr convert slave --masterhost *<host IP>***
User input	**Courier bold blue font**	**cd /opt/Affirmed/NMS/bin**
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	**[ ]**	**cluster [*id*] node [*id*] reboot**
Optional keywords and arguments in curly brackets	**{ }**	***{cdr-file-name}***
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.	|	Is this correct? [y|n]
Keywords representing a single element in angle brackets	**< >**	***<host name>***
Asterisk after a CLI object indicates a required object	*****	**name***
Variables, book and chapter titles, file path names, new terms, and emphasized text	*Times New Roman italic font*	See the *Acuitas User's Guide*. *Workflow* is an application running on the network element.

CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

WARNING: Proceed carefully to avoid possible personal injury.

ESD: *Follow proper electrostatic guidelines.*

Note: *Additional information that may apply to the subject text.*

Tip: Helpful suggestions.

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Chapter 1

Preparing the System for Installation/Upgrade

Acuitas Server Hardware Specifications for Systems Shipped from Affirmed Networks

This section lists the physical specifications of the Acuitas server hardware platform that can be ordered from Affirmed Networks. Affirmed Networks supports the Acuitas server running on any of the following machines:

- HP ProLiant DL380 Gen9
- HP ProLiant DL360 Gen10 (with 64GB, 4 x 300GB drives)
- Blade of the HP BladeSystem c7000

Bare Metal Installation: If the server supplied by Affirmed Networks is for a bare metal installation, RAID 1+0 is used.

VM installation: If the server supplied by Affirmed Networks is for a VM installation, RAID 5 is used.

HP ProLiant DL380 Gen9

The following table lists the specifications for the SFF model of the HP ProLiant DL380 Gen9 server:

Table 1-1 HP ProLiant DL380 Gen9 Server Specifications

Specification	DL380 Gen9 Server
Physical	
Dimensions	■ Height: 3.44 in. (8.73 cm) ■ Width: 17.54 in (44.55 cm) ■ Depth: 26.75 in (67.94 cm)
Weight	Minimum: 32.6 lb (14.759 kg) Maximum: 51.5 lb (23.6 kg)

Table 1-1 HP ProLiant DL380 Gen9 Server Specifications

Specification	DL380 Gen9 Server
Power: Requirements per power supply	
	2 x 800 W power supplies
Voltage	Range line voltage: 100 to 120 VAC 200 to 240 VAC
Current	Power supply output: Rated Steady-State Power For 800W Power Supply: 800W (at 100 VAC), 800W (at 240 VAC)
Frequency	Maximum Peak Power For 800W Power Supply: 800W (at 100 to 127 VAC), 800W (at 200 to 240 1VAC), 800W (at 240 VAC)
Cooling	
BTUs per hour	3207 at 100V AC input 3071 at 200V AC input
Hardware	
Disk Size	4 x 300GB, SAS, 15K RPM, 2.5" HDD
CPU	2 x 12-core Intel Xeon Processor (E5-2680v3 2.5GHz 120W)
Memory	64GB (8 x 8 GB) Single Rank x4 DDR4-2133 CAS-15-15-15 Registered Memory
Ethernet Adapter	1GbE, 4P, 331FLR FIO Adapter
RAID	HP Smart Array P440ar/2GB/2GB FBWC controller
License	
iLO Advanced License	If you supply your own HP ProLiant DL380 Server (the system was not preconfigured from Affirmed Networks), you need to install the iLO Advanced license. To do so: **1** In a web browser, enter the iLO IP address. **2** Log into the iLO as user root. **3** Under **Administration**, click **Licensing**. **4** Enter the Advanced License Activation Key. **5** Click **Install**.

HP ProLiant DL360 Gen10

The following table lists the specifications for the SFF model of the HP ProLiant DL360 Gen10 server:

Table 1-2 HP ProLiant DL360 Gen10 Server Specifications

Specification	DL360 Gen10 Server
Physical	
Dimensions	Height: 1.69" (4.29cm) Width: 17.11" (43.46cm Depth: 27.83" (70.7cm)
Weight	Minimum: 13.04 kg, 28.74 lb Maximum: 16.27 kg, 35.86 lb
Power: Requirements per power supply	
	2 x 800 W power supplies
Voltage	Range line voltage: 100 to 120 VAC 200 to 240 VAC
Current	Power supply output: Rated Steady-State Power For 1600W Power Supply: 1600W (at 240 VAC) For 800W Power Supply: 800W (at 100 VAC), 800W (at 240 VAC) For 500W Power Supply: 500W (at 100 VAC), 500W (at 240 VAC)
Frequency	Maximum Peak Power For 1600W Power Supply: 1600W (at 200 to 240 1VAC)
Cooling	
BTUs per hour	For 800W Power Supply: 3207 BTU/hr (at 100 VAC), 3071 BTU/hr (at 200 VAC) For 500W Power Supply: 1979 BTU/hr (at 100 VAC), 1911 BTU/hr (at 200 VAC),
Hardware	
Disk Size	4 x 300GB, SAS, 15K RPM, SC DS HDD
CPU	2 x 12-core Intel Xeon Processor (4116 Xeon-S 2.1GHz, 85W)
Memory	64GB (8 x 8 GB) Single Rank x8 PC4-2666V-R
Ethernet Adapter	10/25Gb 2-port 640FLR-SFP28 Adapter

Table 1-2 HP ProLiant DL360 Gen10 Server Specifications

Specification	DL360 Gen10 Server
RAID	HPE Smart Array P408i-a SR Gen10 controller
License	
iLO Advanced License	If you supply your own HP ProLiant DL360 Server (the system was not preconfigured from Affirmed Networks), you need to install the iLO Advanced license. To do so: **1** In a web browser, enter the iLO IP address. **2** Log into the iLO as user root. **3** Under **Administration**, click **Licensing**. **4** Enter the Advanced License Activation Key. **5** Click **Install**.

To prepare the system to install/upgrade the Acuitas server and database software, complete the instructions in the section that applies to your system:

- Bare Metal System
- Virtual Machine

Bare Metal System

This section describes how to prepare a Bare Metal system. Complete the instructions in the section that applies to your system:

- HP Proliant DL380 Gen8 System
- HP Proliant DL380 Gen9 System

HP Proliant DL380 Gen8 System

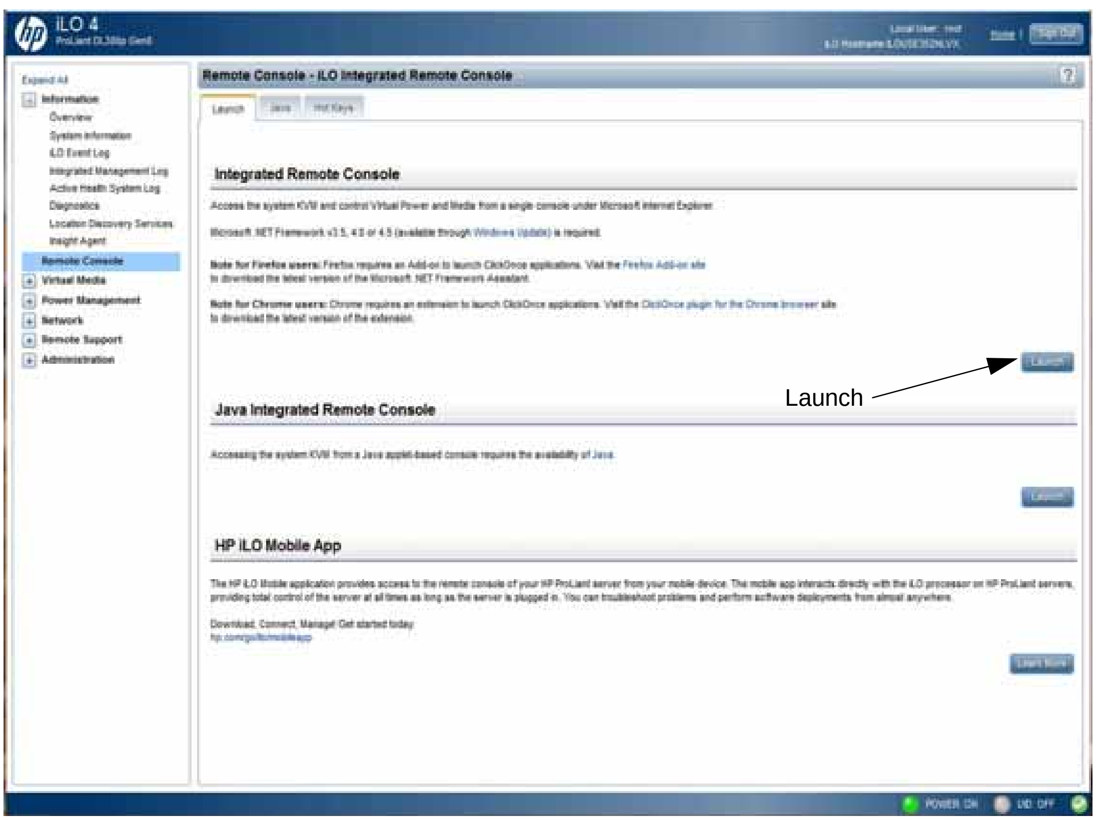
Initial Procedure to Set Up RAID

For a brand new system, perform this initial procedure to set up RAID (**Recommended:** RAID 1+0).

To set up RAID:

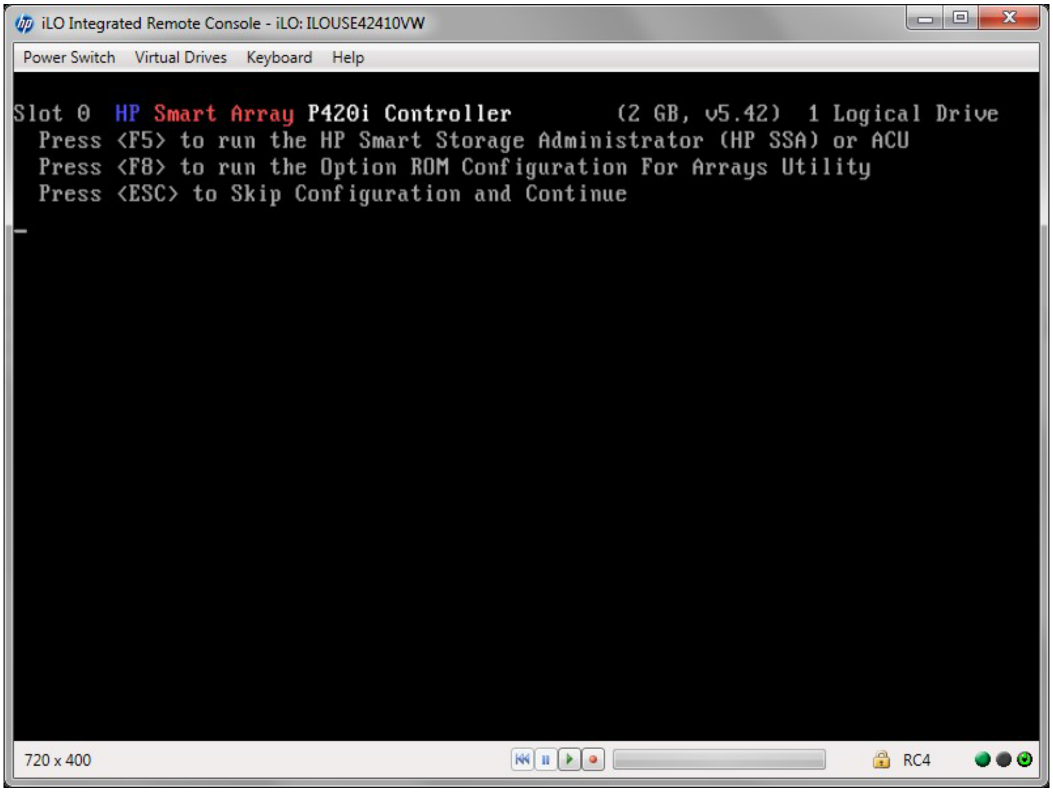
1 From a browser, log onto the iLO and click **Launch** to access the Remote Console (Firefox browser requires a Firefox Add-on to launch the Remote Console).

Figure 1-1 iLO



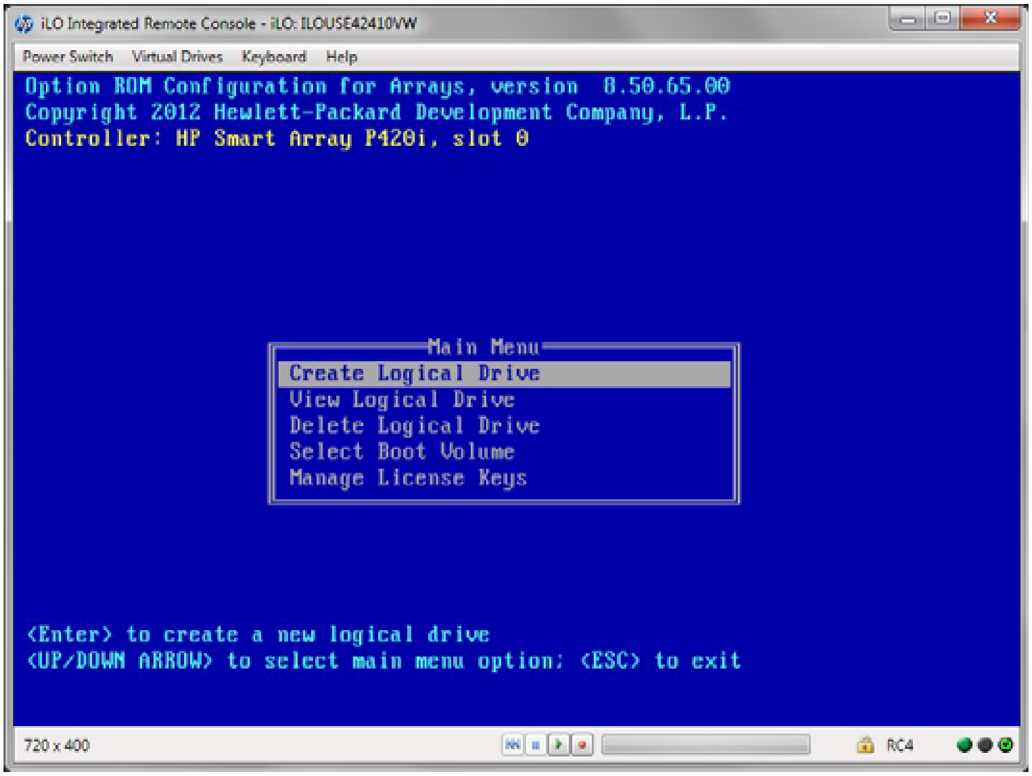
2 At the prompt HP Smart Array prompt, press **F8** to run the Option ROM Configuration Utility.

Figure 1-2 HP Smart Array Prompt



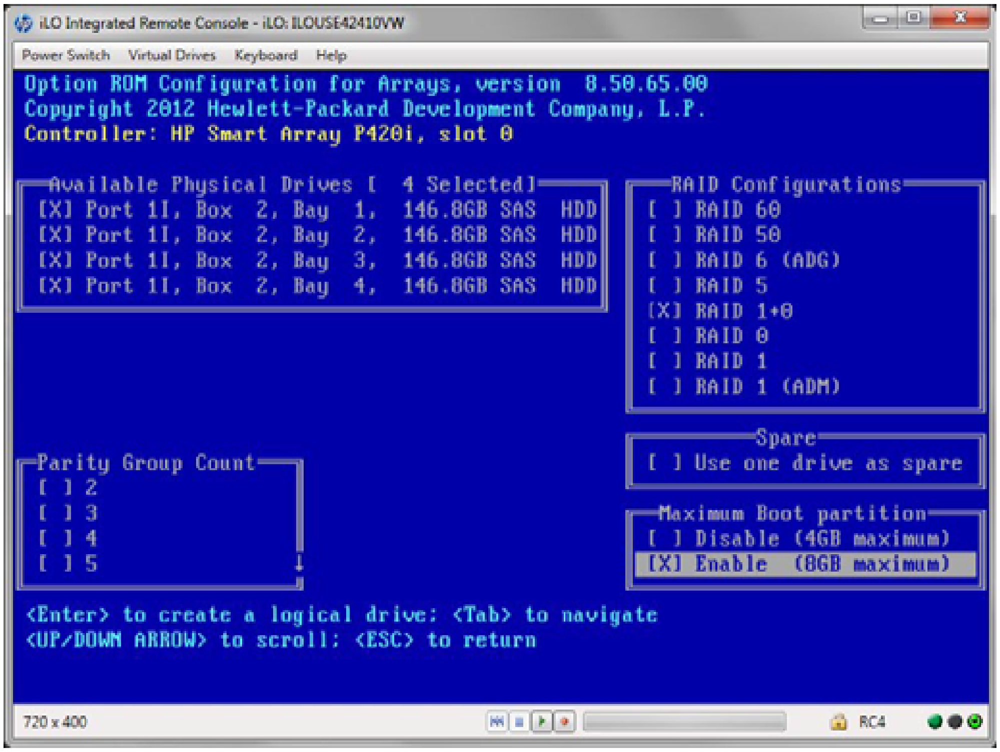
3 From the Main Menu, use the arrow key to highlight **Create Logical Drive** and press **Enter**.

Figure 1-3 Main Menu



4 Select **RAID 1+0** and **Enable Maximum Boot Partition** and press **Enter** to create the logical drive.

Figure 1-4 Logical Drive Configuration

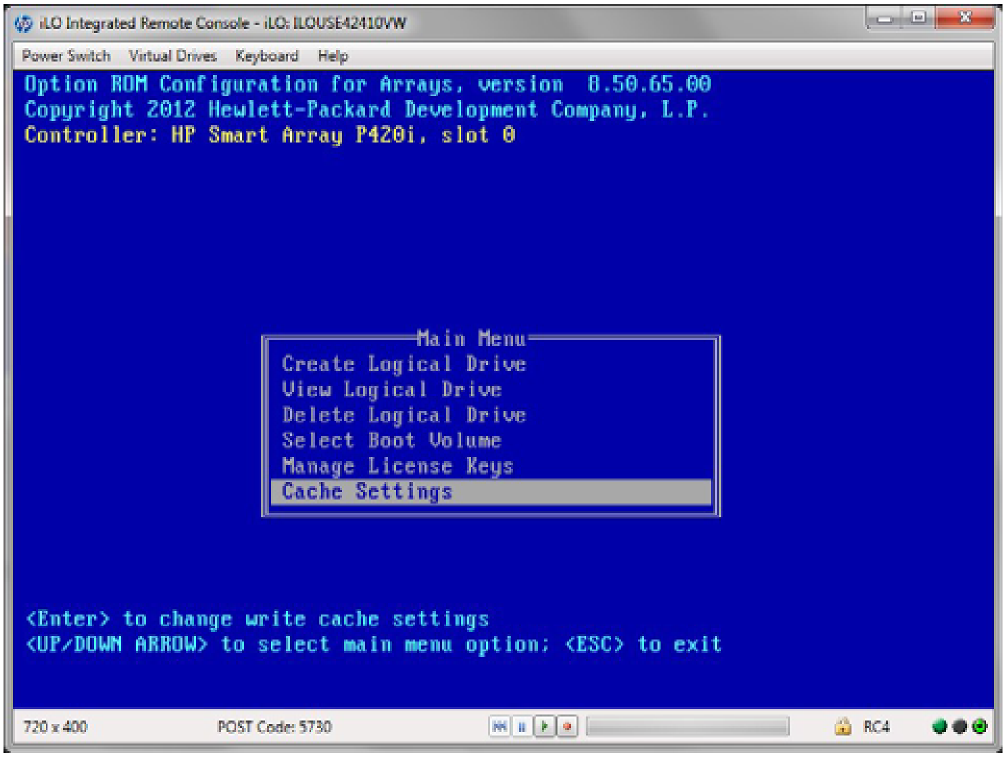


5 When prompted, press **F8** to save the configuration.

6 When prompted, press **Enter** to continue.

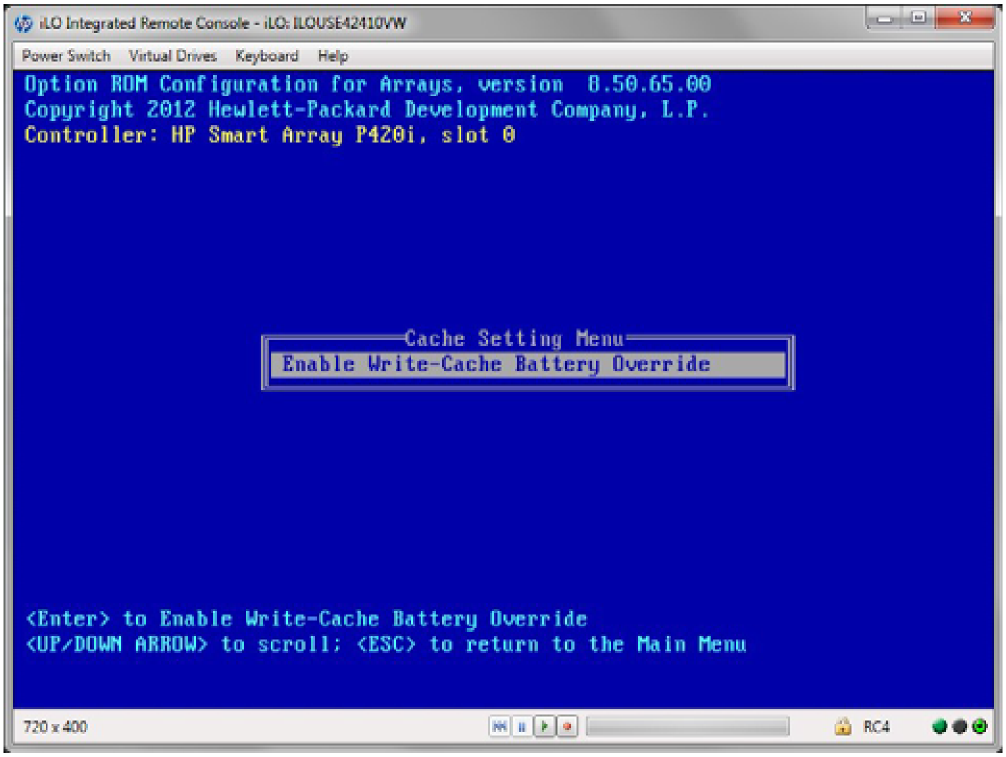
7 From the Main Menu, use the arrow key to highlight **Cache Settings** and press **Enter**.

Figure 1-5 Cache Settings



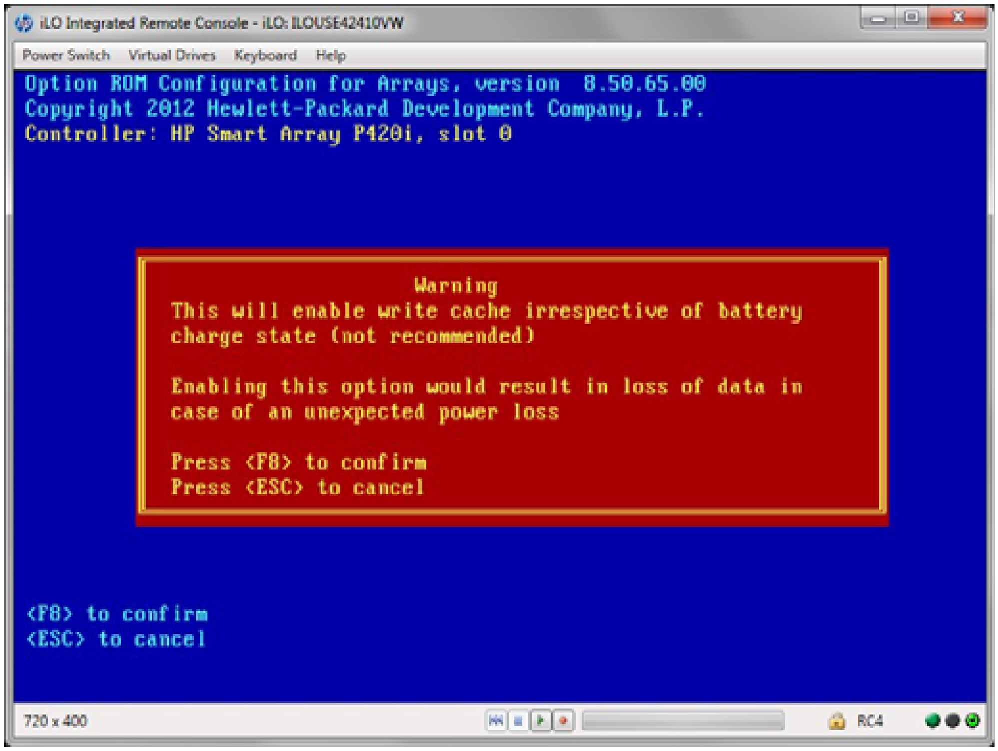
8 Use the arrow key to highlight **Enable Write-Cache Battery Override** and press **Enter**.

Figure 1-6 Enable Write-Cache Battery Override



9 At the warning prompt, press **F8** to confirm.

Figure 1-7 Warning Prompt



10 When prompted, press **Enter** to continue.

Procedure to Prepare the Bare Metal System

To prepare a Bare Metal system:

1. If the system is already running a version of Acuitas, backup the database:

   a. Stop the Acuitas server:

      `cd /opt/Affirmed/NMS/bin`

      `./emsstop`

   b. Issue the following commands to backup the database:

      `mkdir /opt/backup`

      `./emsdb backup --backupfile /opt/backup/backup.db`

   c. Move the backup of /opt/backup/backup.db to a remote server:

      `scp /opt/backup/backup.db` ***`<remote server path>`***

   d. Create a backup of /opt/Affirmed/NMS/server/ems/conf/License.dat.

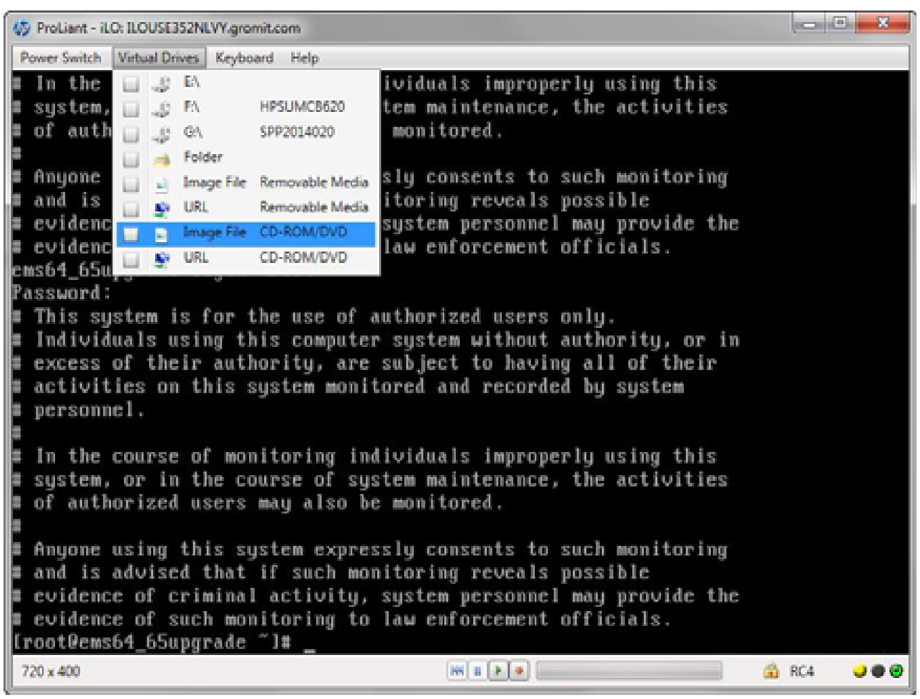
   e. Move the backup of /opt/Affirmed/NMS/server/ems/conf/License.dat to a remote server.

2. Download the ISO image from Affirmed Networks and save it to a location accessible from the server where you access iLO (such as, a local computer):

   - an-ems-3.0.*x*-rhel-7.*y*-mysql-8.0.*z*.iso

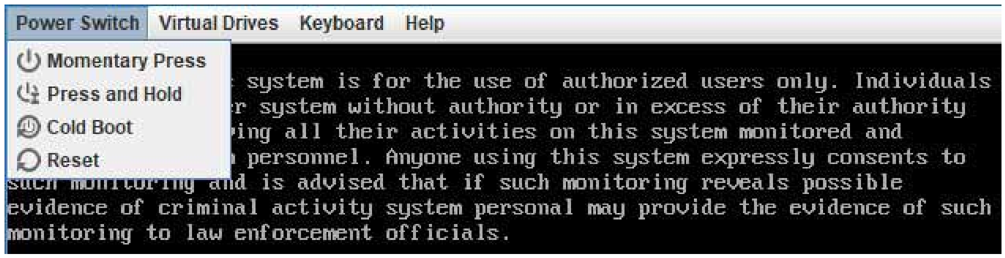
3. From the Remote Control console, select **Virtual Drives** > Image File CD/DVD drive from the Main Menu to attach to the ISO image.

Figure 1-8 Remote Control Console



4 From the Remote Control console, select **Power Switch** > Reset from the Main Menu to reboot the server:

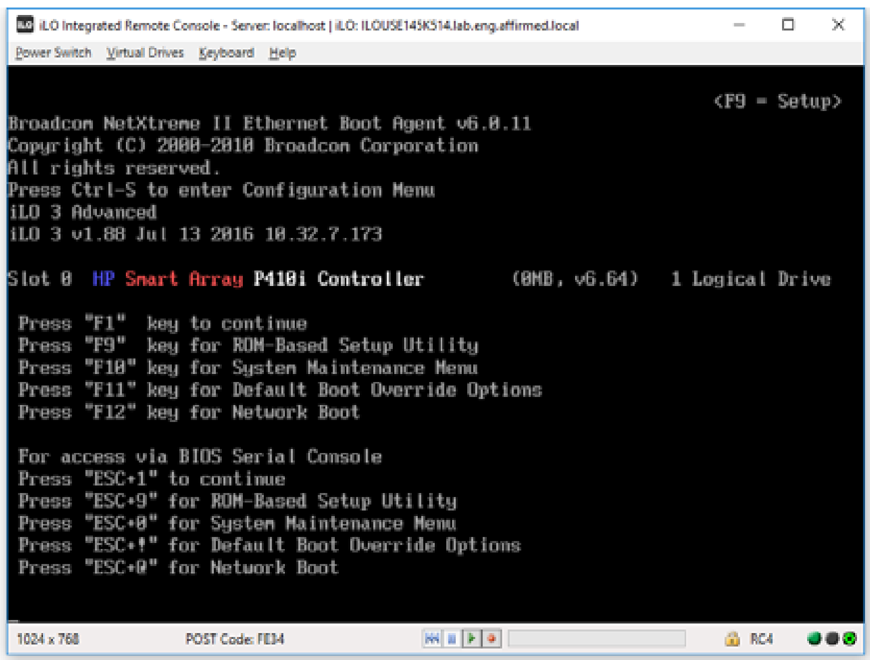
Figure 1-9 Power Switch Menu



It may take several minutes until the HP POST begins.

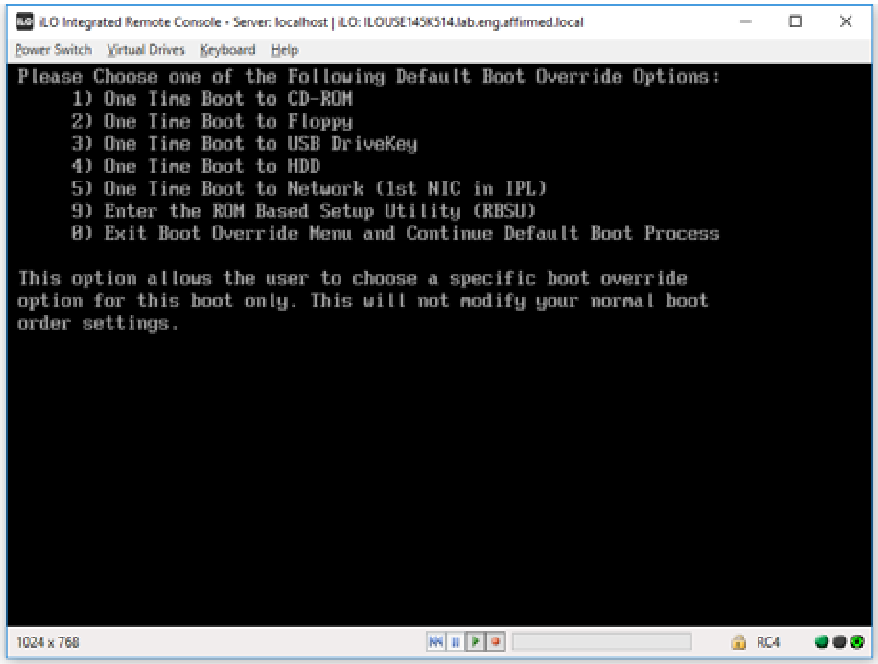
5 When the system is booting up, press **F11** at the following screen.

Figure 1-10 iLO Remote Console Screen



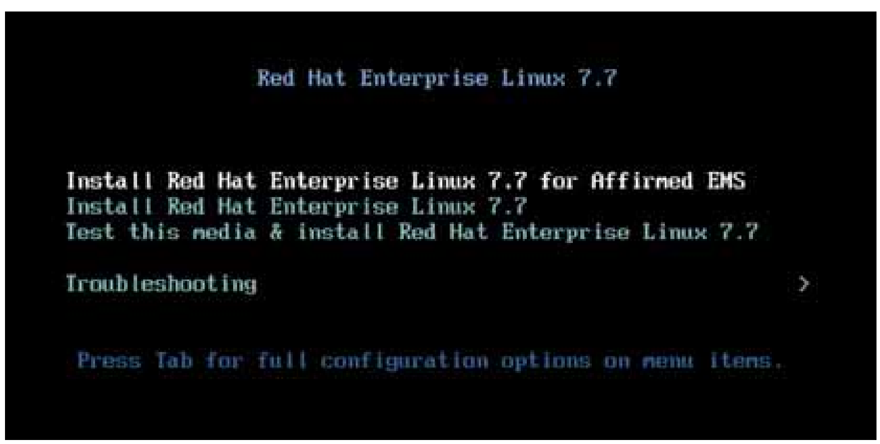
6 Press **1** to select the **One Time Boot to CD-ROM** option.

Figure 1-11 Boot Override Options



7 From the Red Hat installer screen, use the arrow key to highlight **Install Red Hat Enterprise Linux 7.7 for Affirmed EMS** and press **Enter**.

Figure 1-12 Red Hat Installer Screen

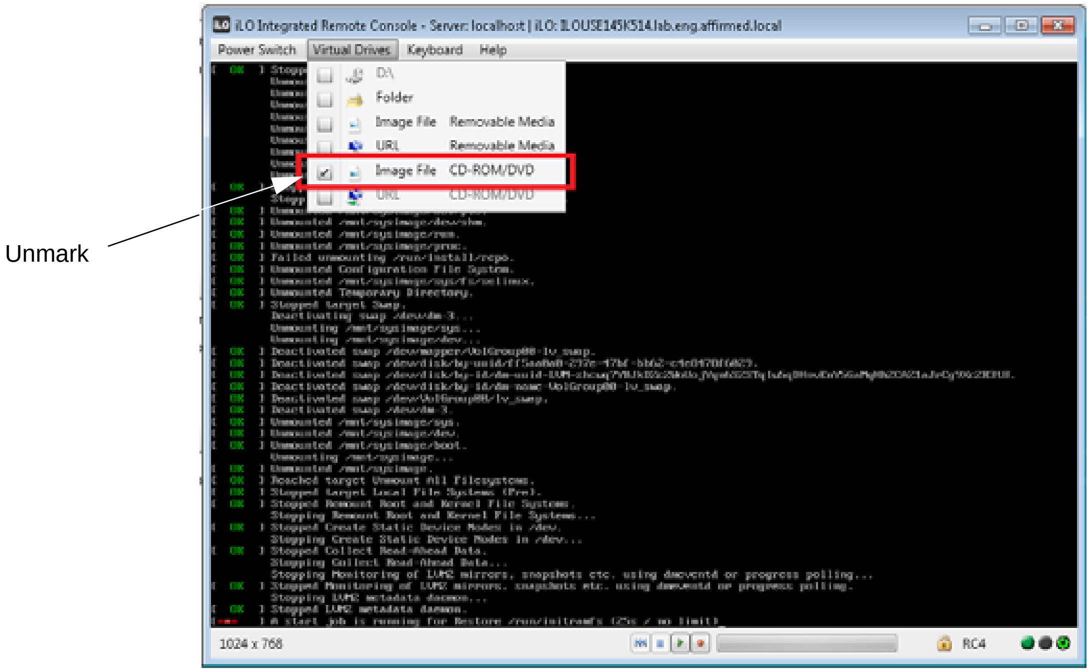


It may take several minutes for the installation to complete. When the installation is complete, the system prompts you to reboot the server.

8 Press **Enter** to reboot the server.

9 Select **Virtual Drives** from the Main Menu of the Remote Control console and unmark the Image File CD/DVD drive checkbox.

Figure 1-13   Main Menu



What's Next

The system is now ready for Acuitas installation/upgrade. Proceed to the chapter that applies to your system:

- Standalone Systems
- High Availability Systems

HP Proliant DL380 Gen9 System

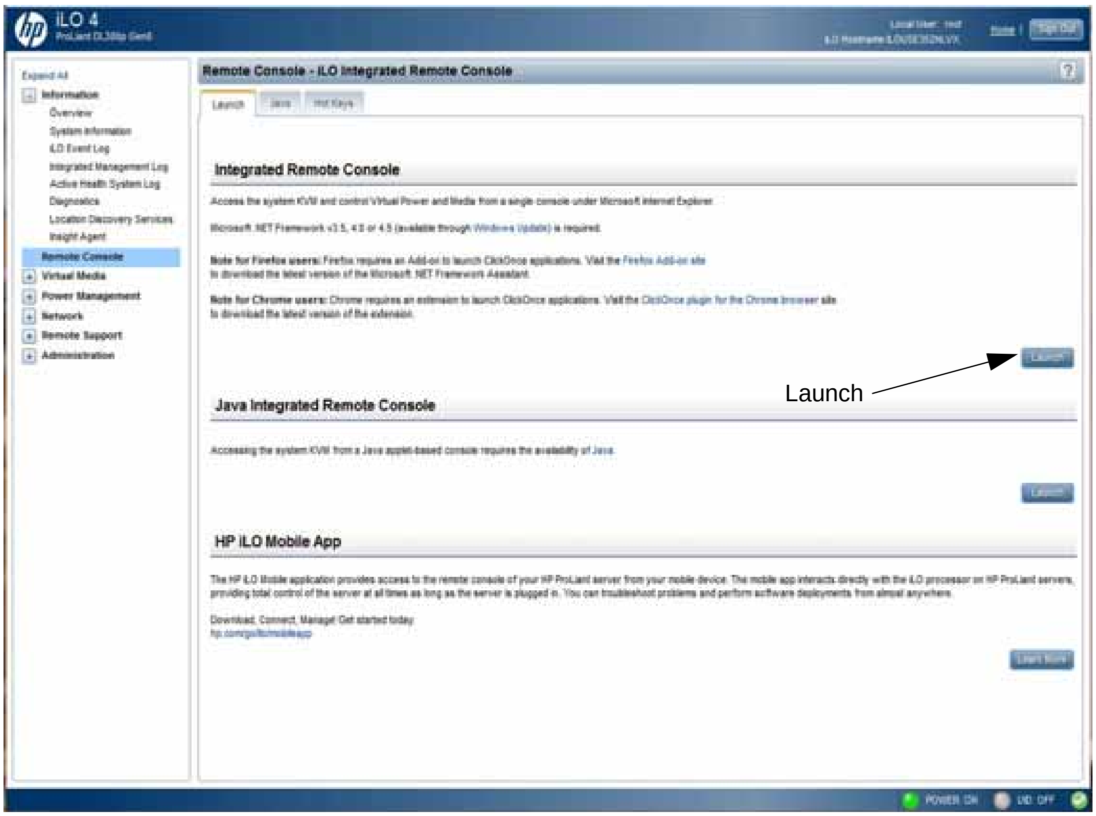
Initial Procedure to Set Up RAID

For a brand new system, perform this initial procedure to set up RAID.

To set up RAID:

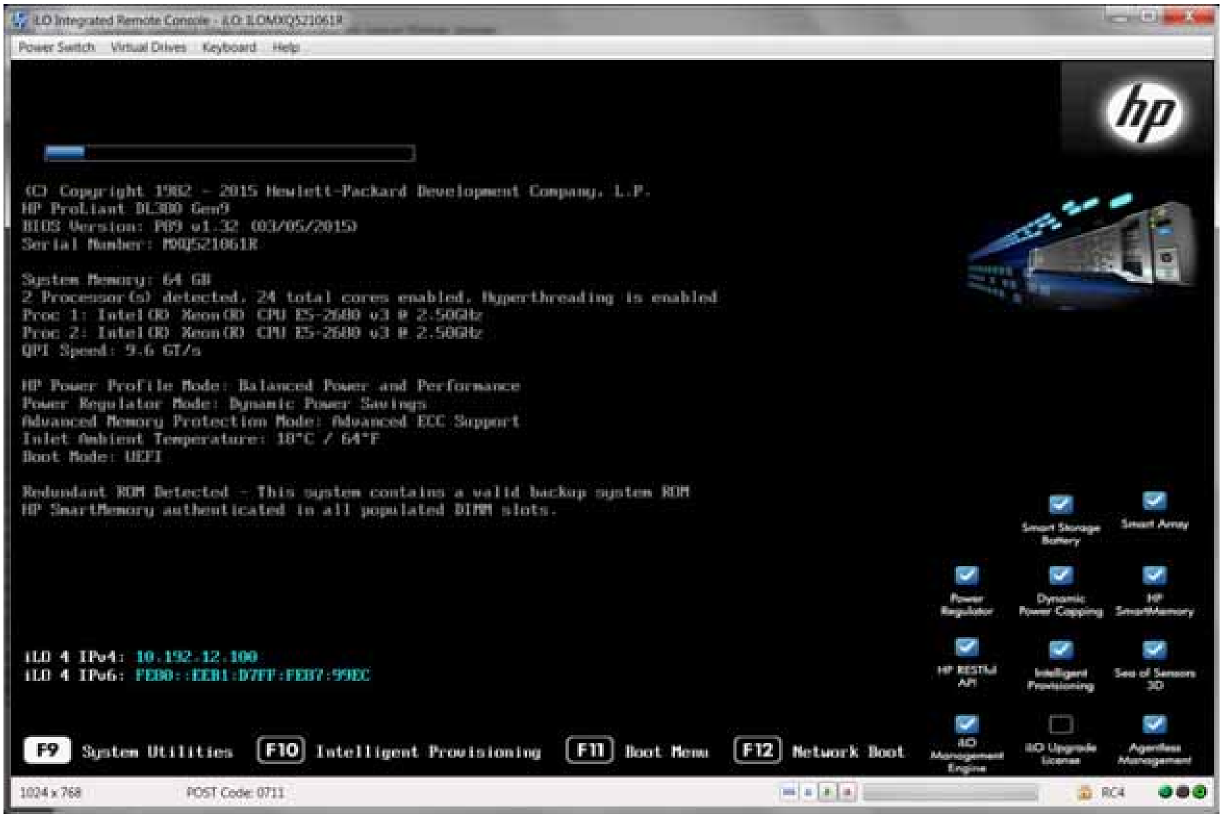
1 From a browser, log onto the iLO and click **Launch** to access the Remote Console (Firefox browser requires a Firefox Add-on to launch the Remote Console).

Figure 1-14 iLO



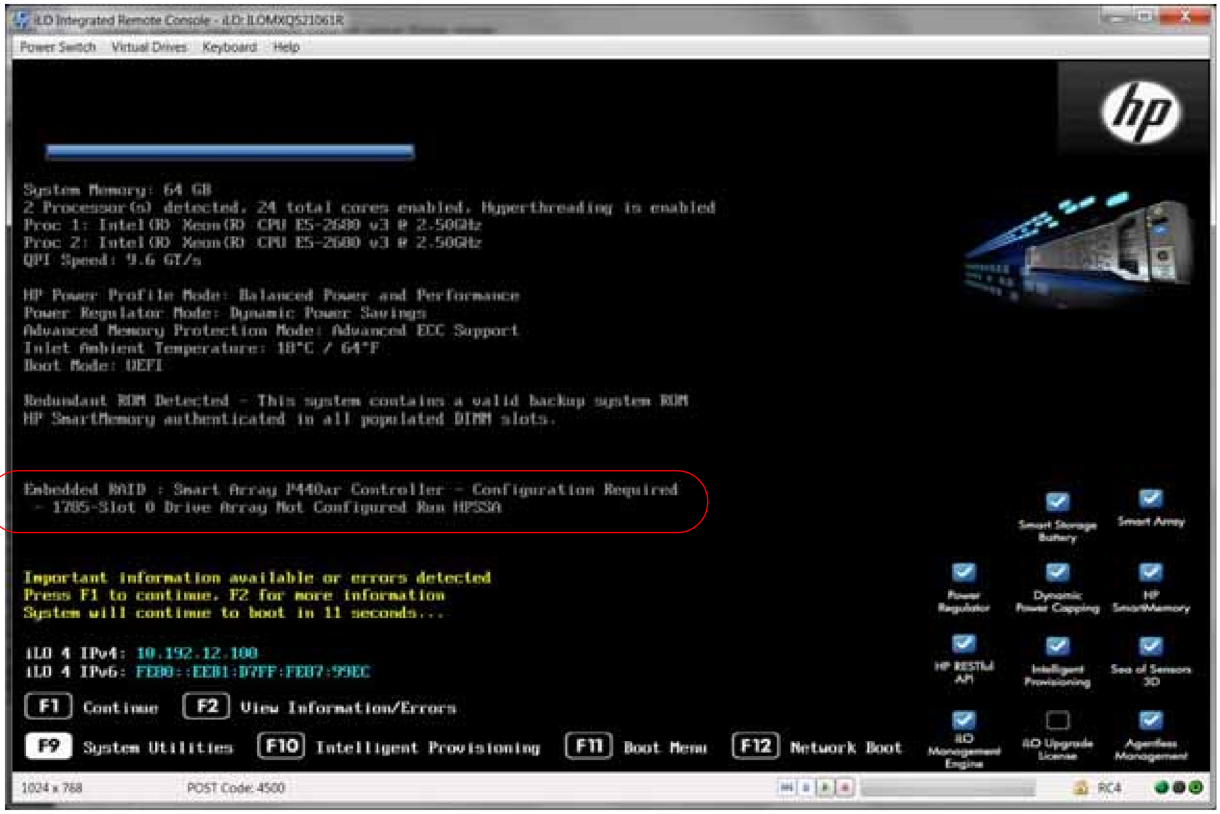
2 From the Remote Control console, press **F9** to access System Utilities.

Figure 1-15 Remote Console



3 When prompted that RAID has not been set up on the new system, press **F1** to continue.

Figure 1-16 RAID Configuration Required

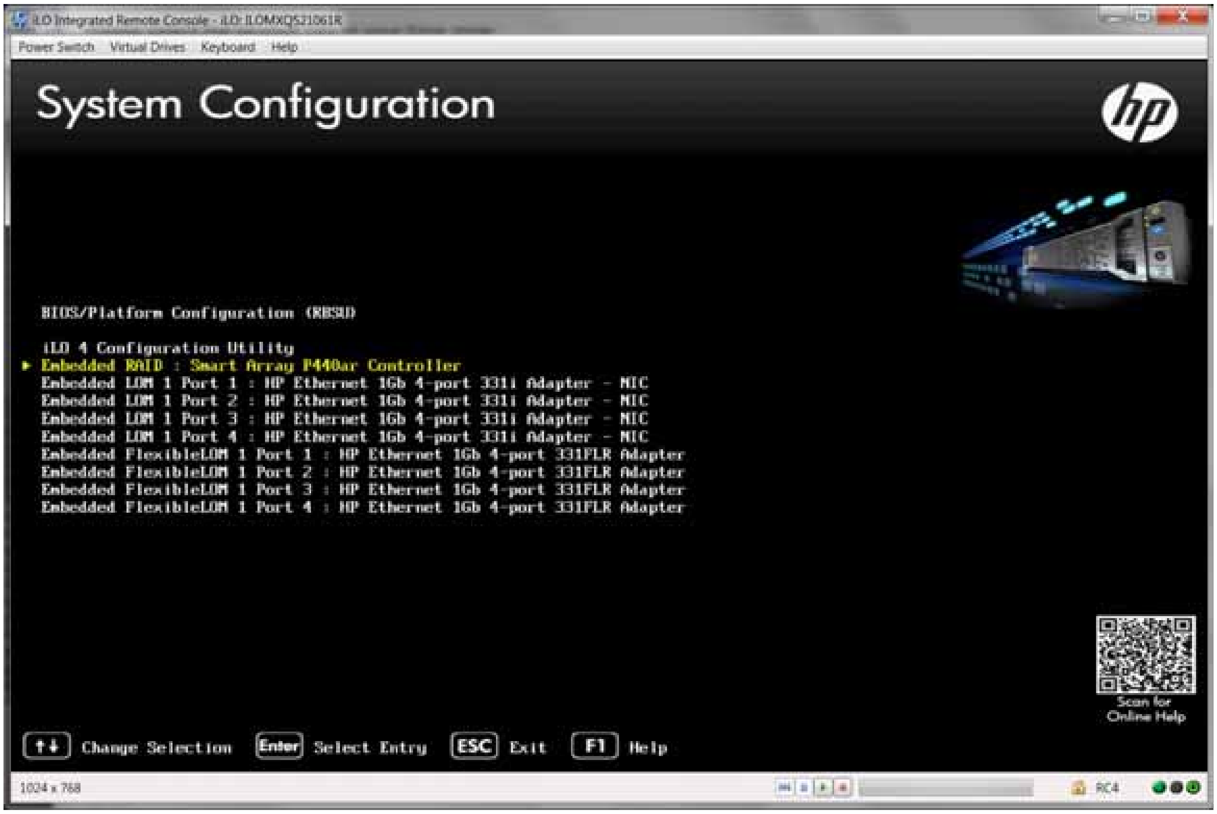


4 From the System Utilities menu, use the arrow keys to highlight **System Configuration** and press **Enter**.

Figure 1-17 System Utilities

5 Use the arrow keys to highlight **Embedded RAID: Smart Array P440ar Controller** and press **Enter**.

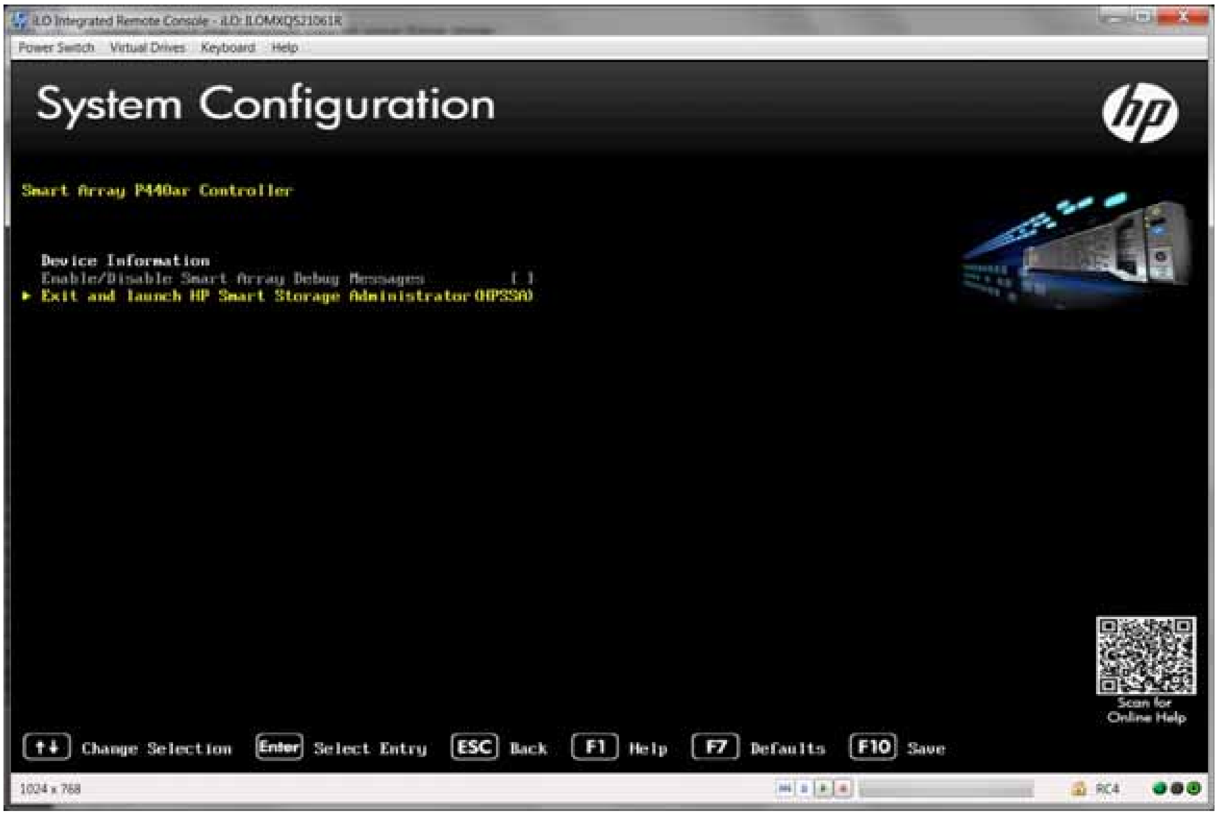
Figure 1-18 System Configuration



6 Use the arrow keys to highlight **Exit and launch HP Smart Storage Administrator (HPSSA)** and press **Enter** *only once*.

Important: Pressing **Enter** more than once launches HP Intelligent Provisioning, causing you to restart the process.

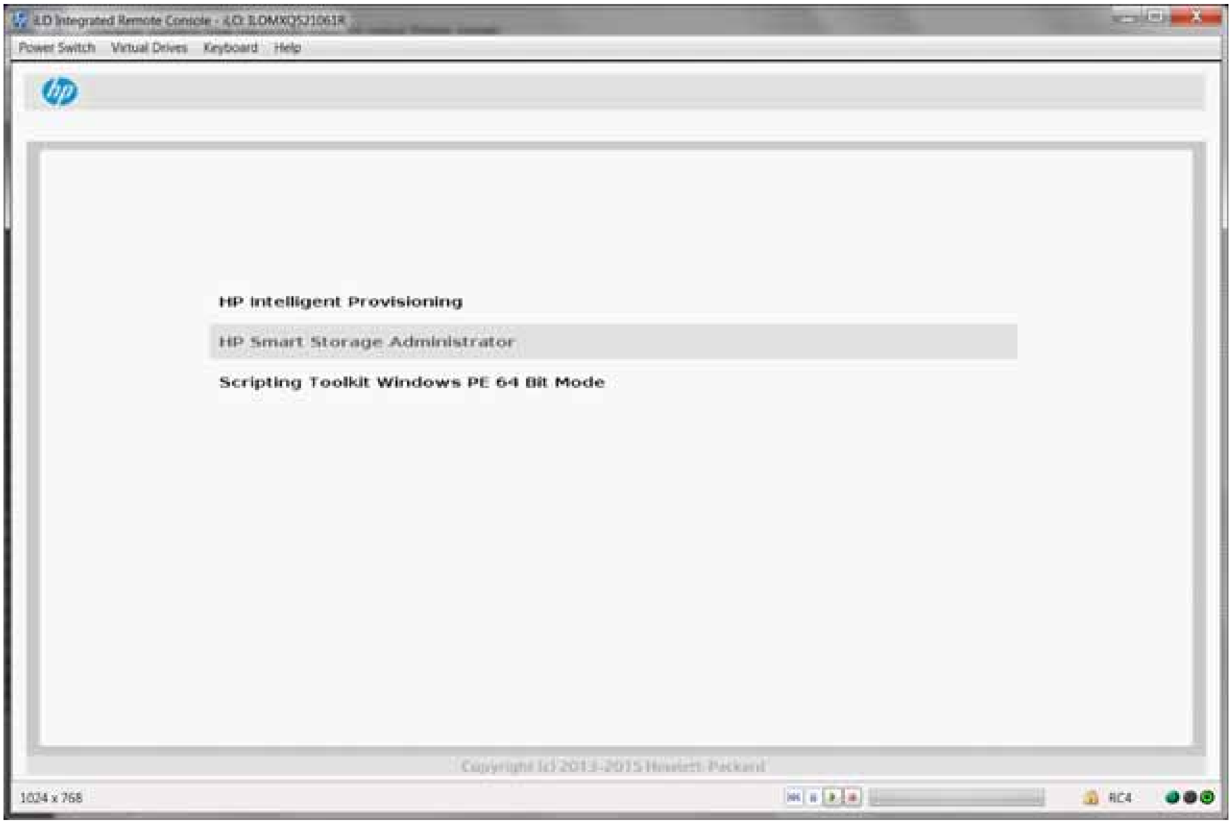
Figure 1-19 Smart Array P440ar Controller



7 Use the arrow keys to highlight **HP Smart Storage Administrator** and press **Enter**.

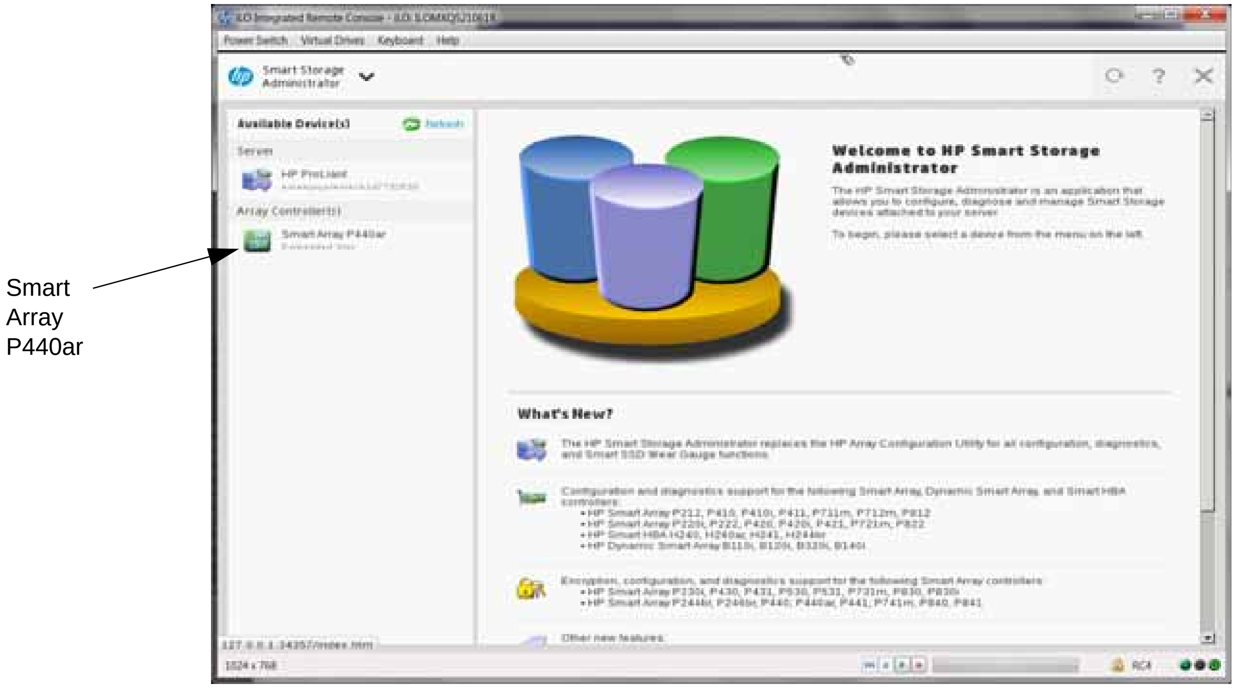
Important: You have only a few seconds to perform this step.

Figure 1-20 HP Smart Storage Administrator



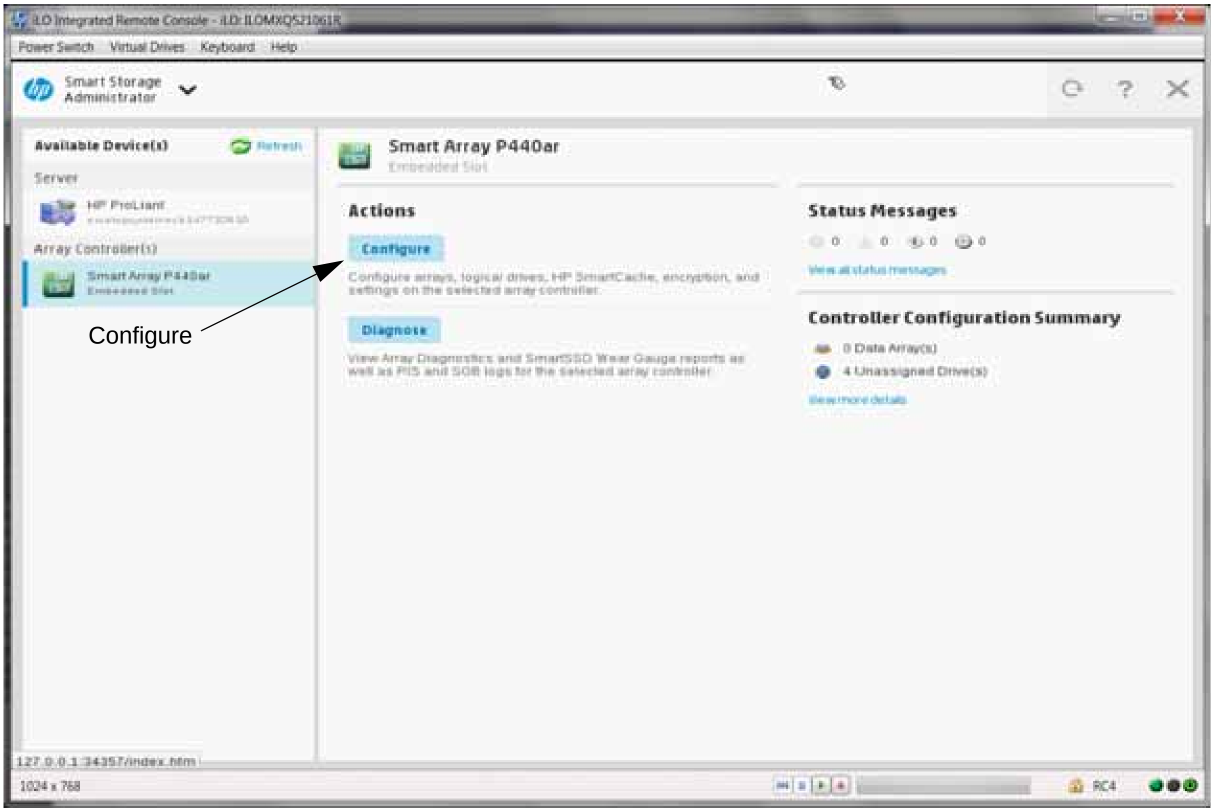
8 From the HP Smart Storage Administrator, click **Smart Array P440ar**.

Figure 1-21 Smart Array P440ar



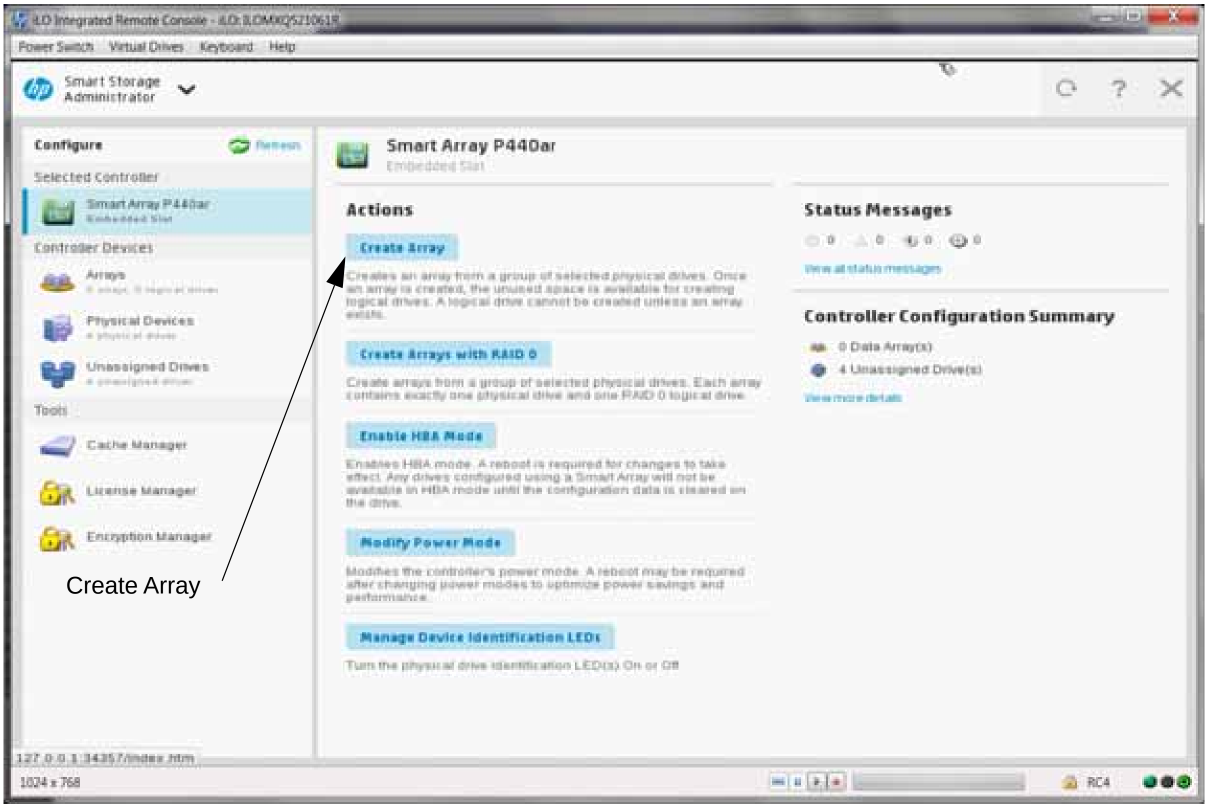
9 Click **Configure**.

Figure 1-22 Configure Smart Array P440ar



10 Click **Create Array**.

Figure 1-23 Create Array



11 Mark the **Select All (4)** checkbox and click **Create Array**.

Figure 1-24 Select All (4)

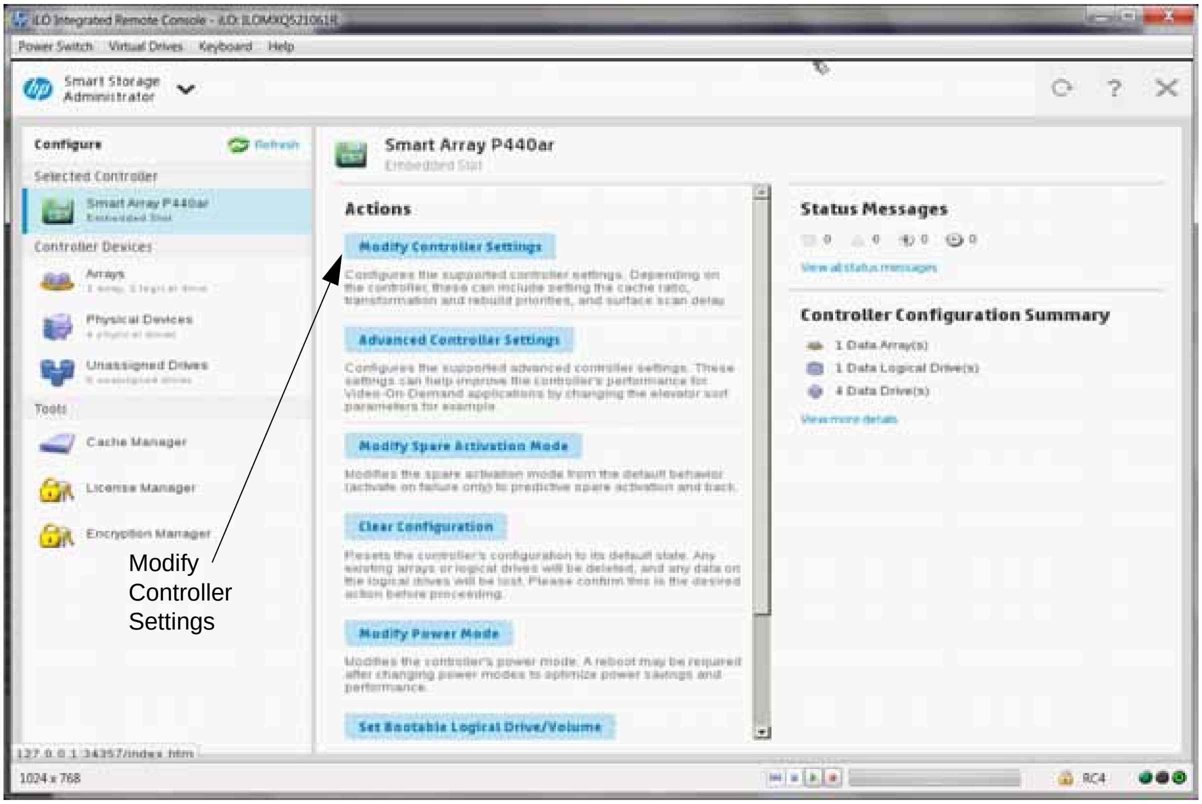
12 Mark the following selections and click **Create Logical Drive**.

Figure 1-25 Create Logical Drive Configuration

13 Click **Finish**.

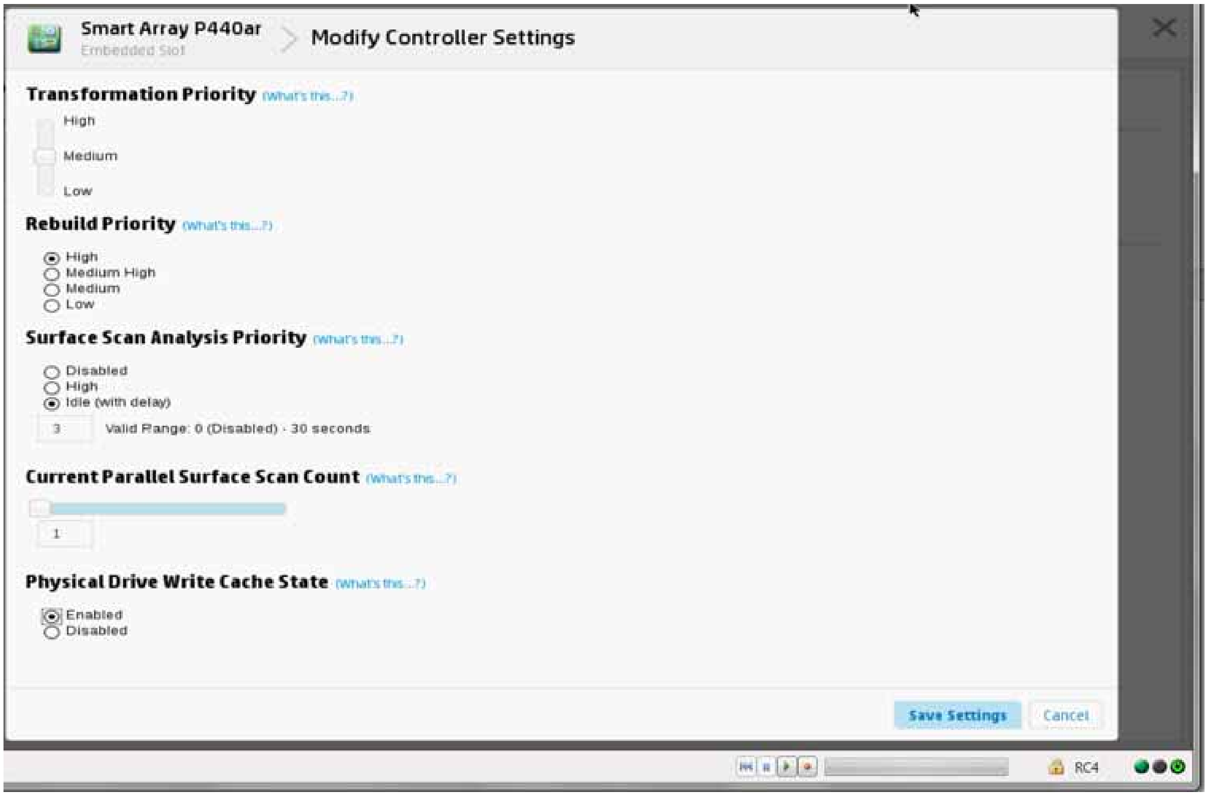
14 Click **Modify Controller Settings**.

Figure 1-26 Modify Controller Settings



15 Mark the following selections and click **Save Settings**.

Figure 1-27 Controller Settings

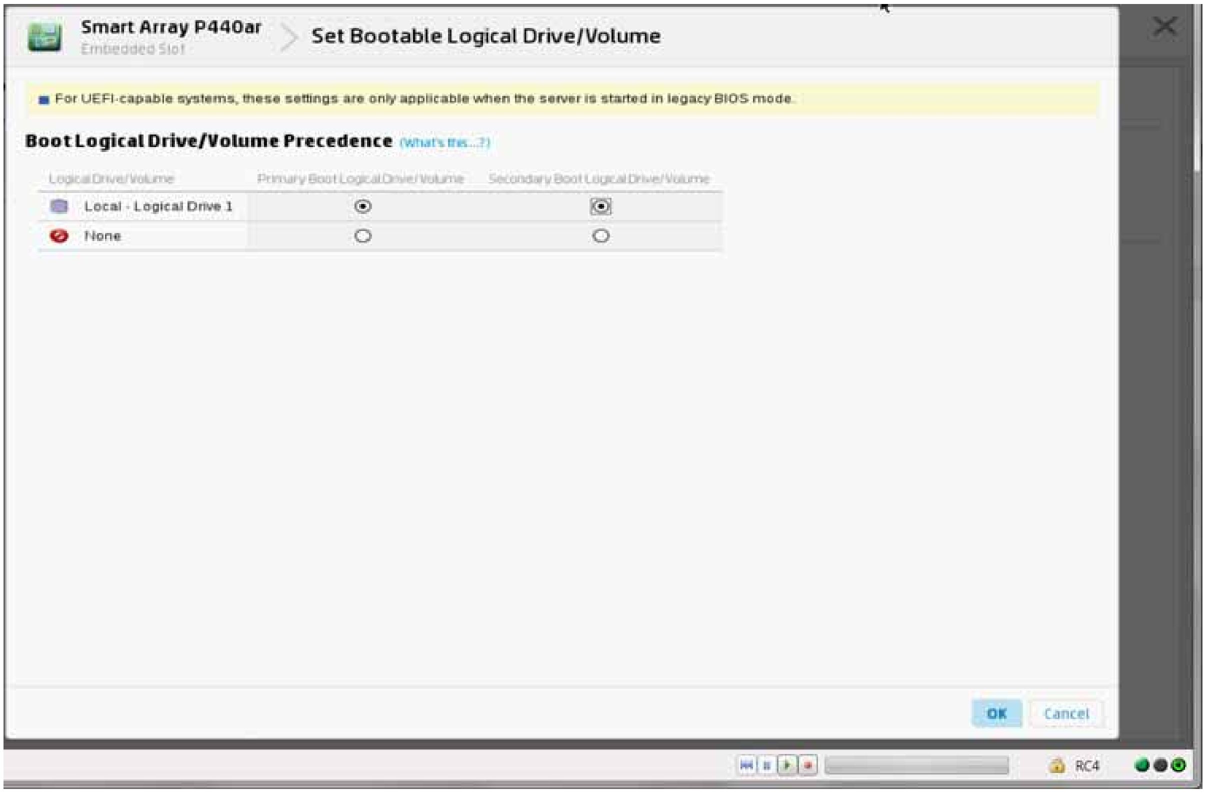


16 Click **Finish**.

17 Click **Set Bootable Logical Drive/Volume**.

18 Mark the following selections and click **OK**.

Figure 1-28 Bootable Logical Drive/Volume Setting

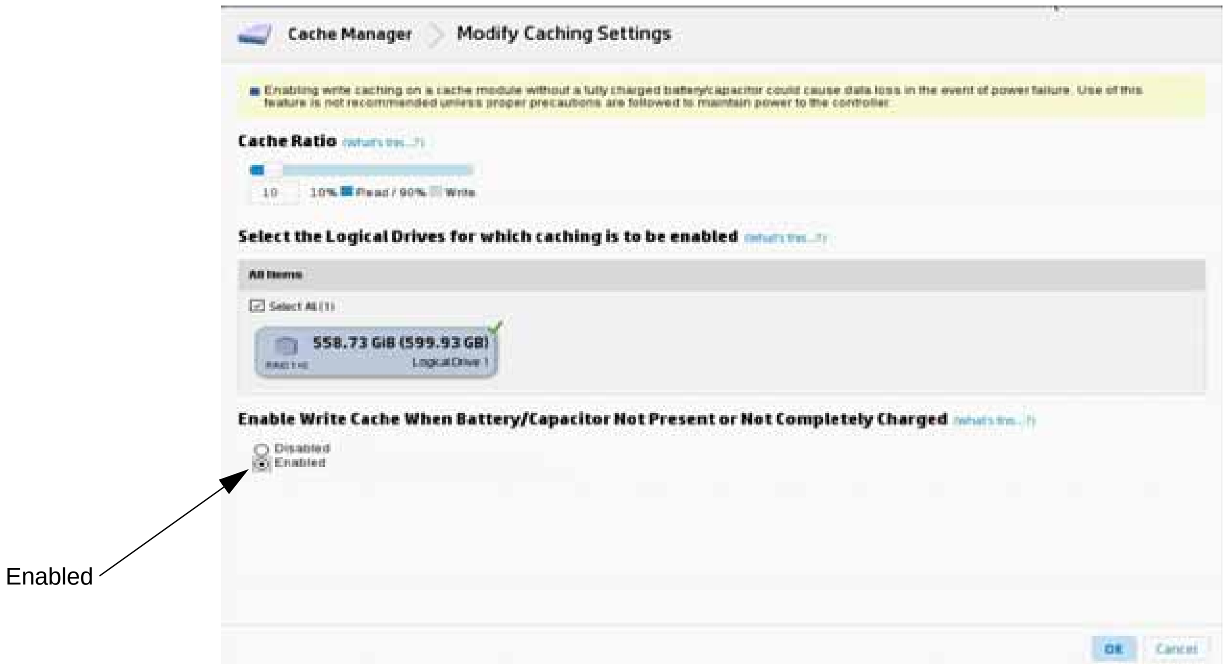


19 Click **Finish**.

20 Under **Tools**, click **Cache Manager**.

21 Under **Enable Write Cache When Battery/Capacitor Not Present or Not Completely Charged**, mark the Enabled checkbox and click **OK**.

Figure 1-29 Modify Caching Settings



22 Click **Finish**.

Procedure to Set Boot Order

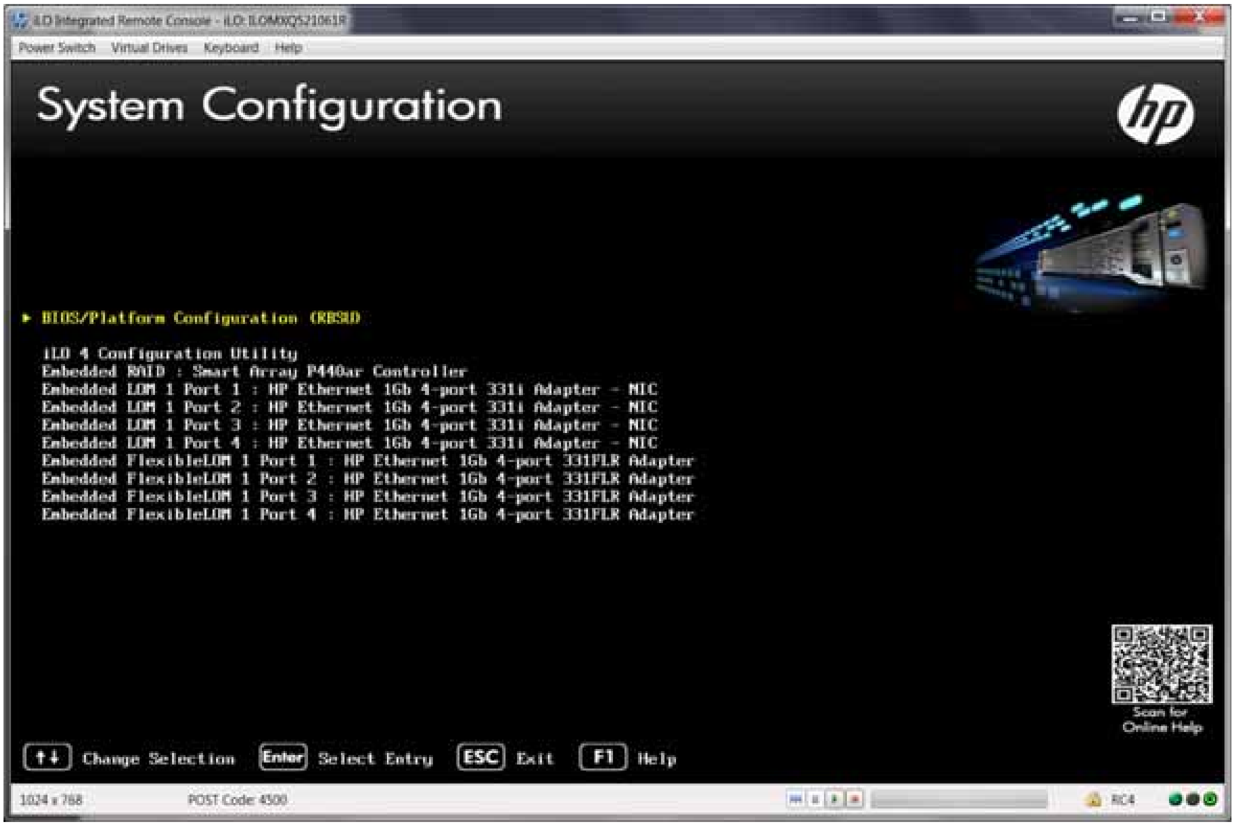
To set boot order:

1. From the Remote Control console, press **F9** to access System Utilities.
2. From the System Utilities menu, use the arrow keys to highlight **System Configuration** and press **Enter**.

Figure 1-30   System Utilities

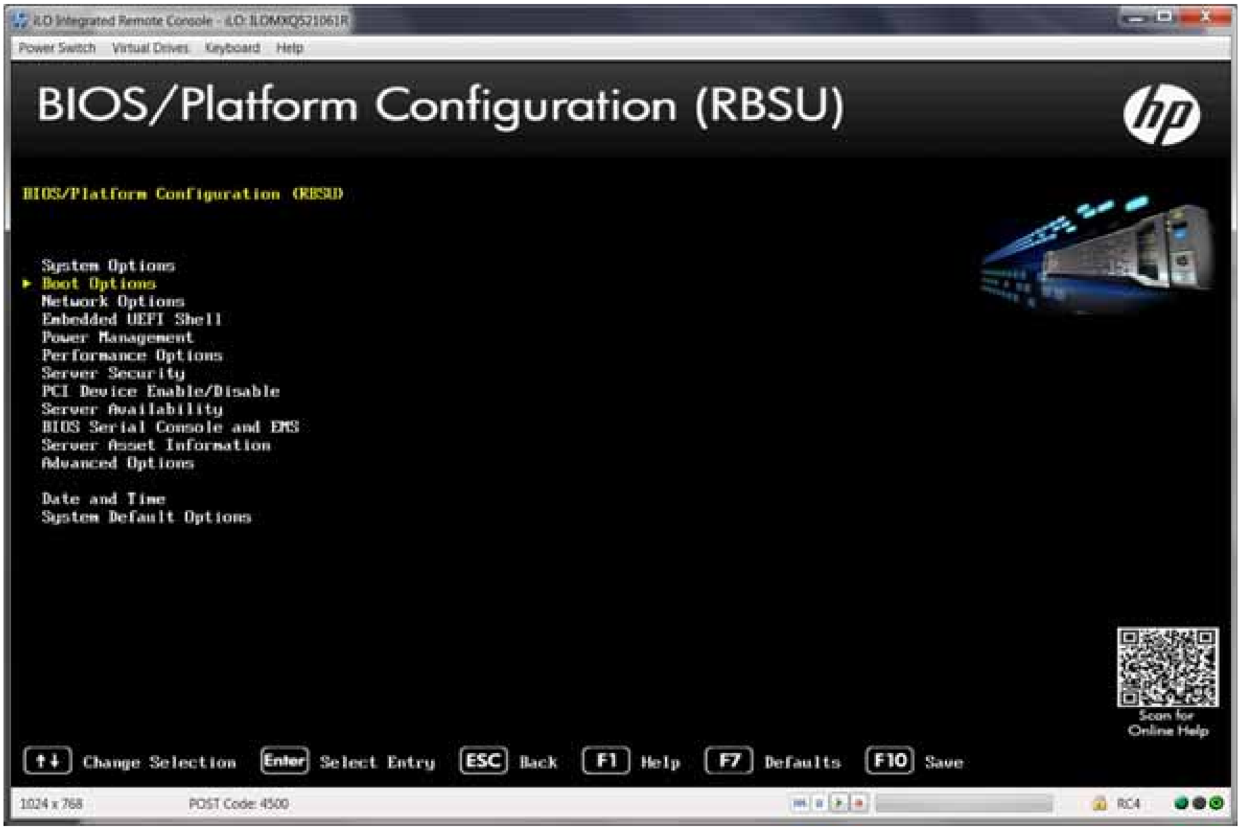
3 Use the arrow keys to highlight **Select BIOS/Platform Configuration (RBSU)** and press **Enter**.

Figure 1-31 System Configuration



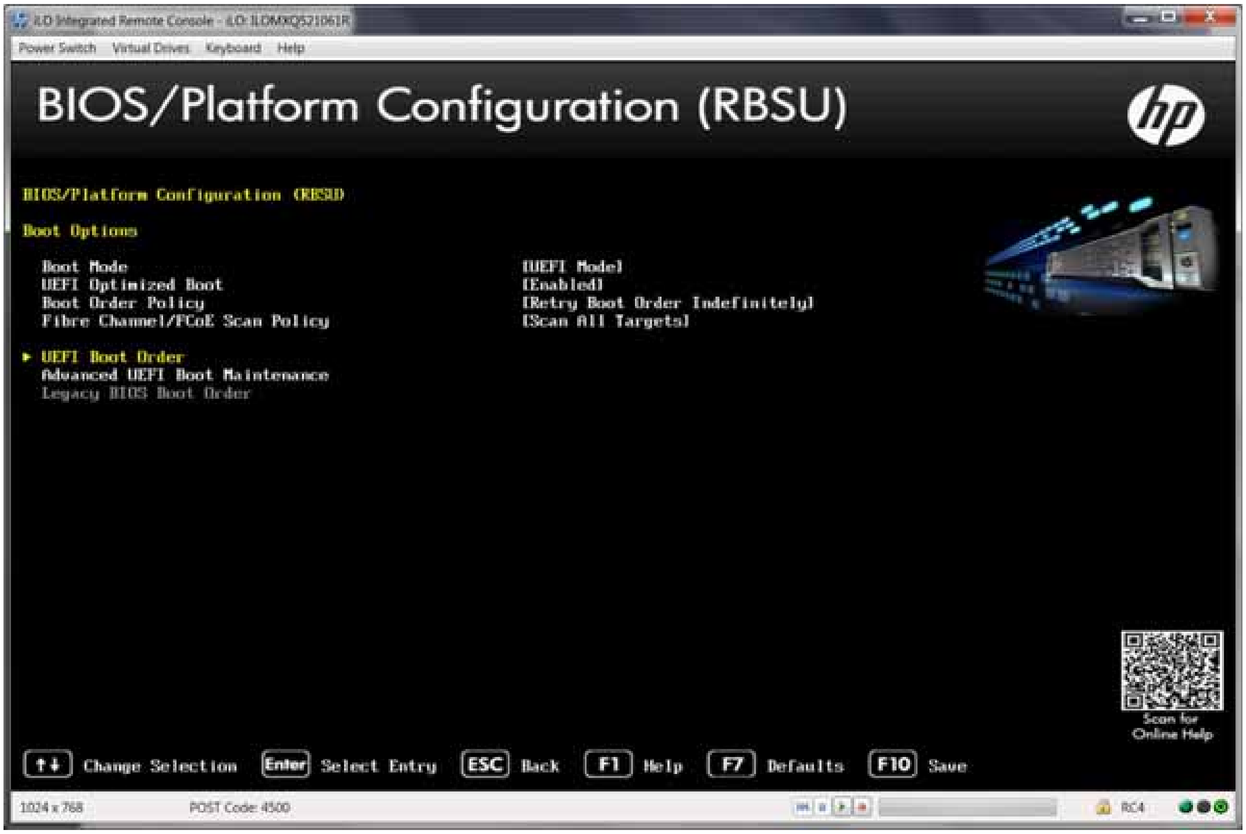
4 Use the arrow keys to highlight **Boot Options** and press **Enter**.

Figure 1-32 BIOS/Platform Configuration (RBSU)



5 Use the arrow keys to highlight **UEFI Boot Order** and press **Enter**.

Figure 1-33 UEFI Boot Order



6 Use the arrow keys to highlight **Embedded RAID**. Use <+> to move the Embedded RAID to the top of the order and press **Enter**.

7 At the confirmation prompt to save pending changes, enter **Y**.

Procedure to Prepare the Bare Metal System

To prepare a Bare Metal system:

1 If the system is already running a version of Acuitas, backup the database:

a Stop the Acuitas server:

```
cd /opt/Affirmed/NMS/bin
./emsstop
```

b Issue the following commands to backup the database:

```
mkdir /opt/backup
./emsdb backup --backupfile /opt/backup/backup.db
```

c Move the backup of /opt/backup/backup.db to a remote server:

```
scp /opt/backup/backup.db <remote server path>
```

d Create a backup of /opt/Affirmed/NMS/server/ems/conf/License.dat.

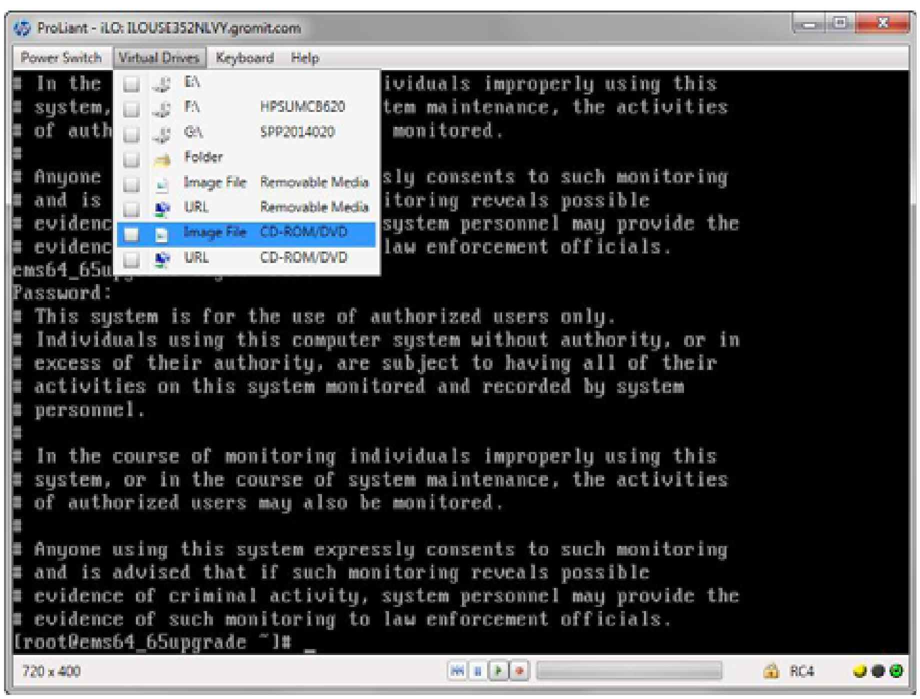
e Move the backup of /opt/Affirmed/NMS/server/ems/conf/License.dat to a remote server.

2 Download the ISO image from Affirmed Networks and save it to a location accessible from the server to be upgraded (such as, a local computer):

- an-ems-3.0.*x*-rhel-7.*y*-mysql-8.0.*z*.iso

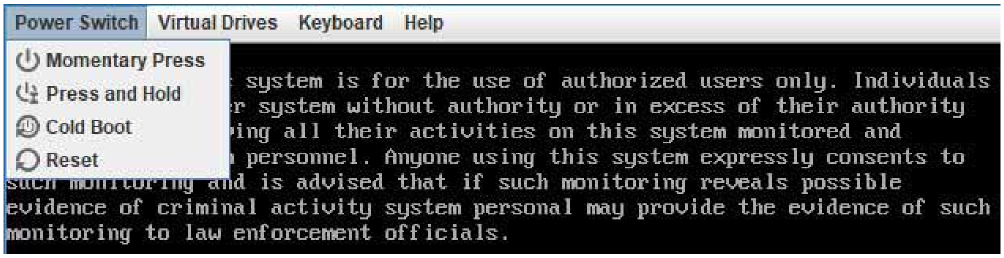
3 From the Remote Control console, select **Virtual Drives** > Image File CD/DVD drive from the Main Menu to attach to the ISO image.

Figure 1-34 Remote Control Console



4 From the Remote Control console, select **Power Switch** > Reset from the Main Menu to reboot the server:

Figure 1-35 Power Switch Menu



It may take several minutes until the HP POST begins.

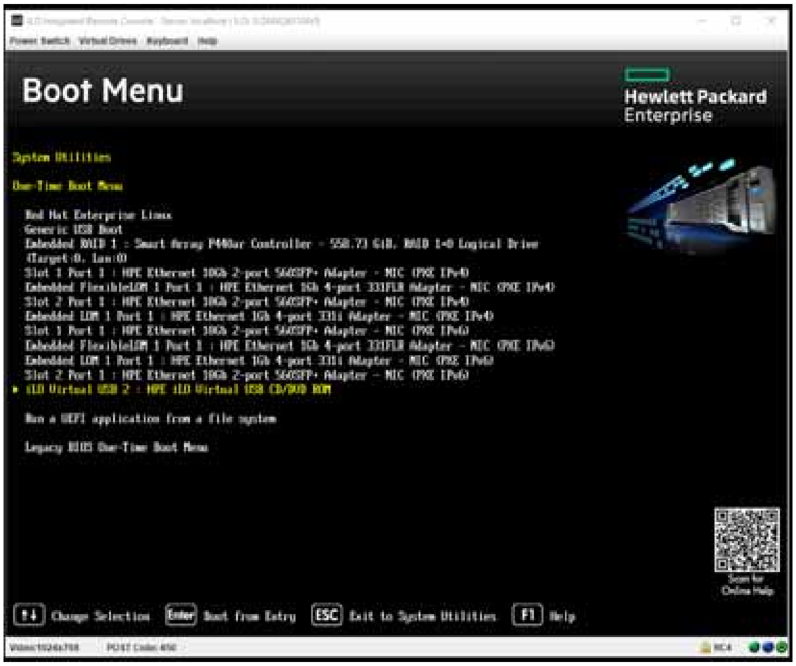
5 When the system is booting up, from the Remote Console, press **F11** to access the Boot Menu.

Figure 1-36 Remote Console

6 Press **1** to select the **One Time Boot to CD-ROM** option.

7 Use the arrow key to highlight the **iLO** and press **Enter**.

Figure 1-37 Boot Override Options



8 From the **Red Hat installer** screen, use the arrow key to highlight **Install Red Hat Enterprise Linux 7.7 for Affirmed EMS** and press **Enter**.

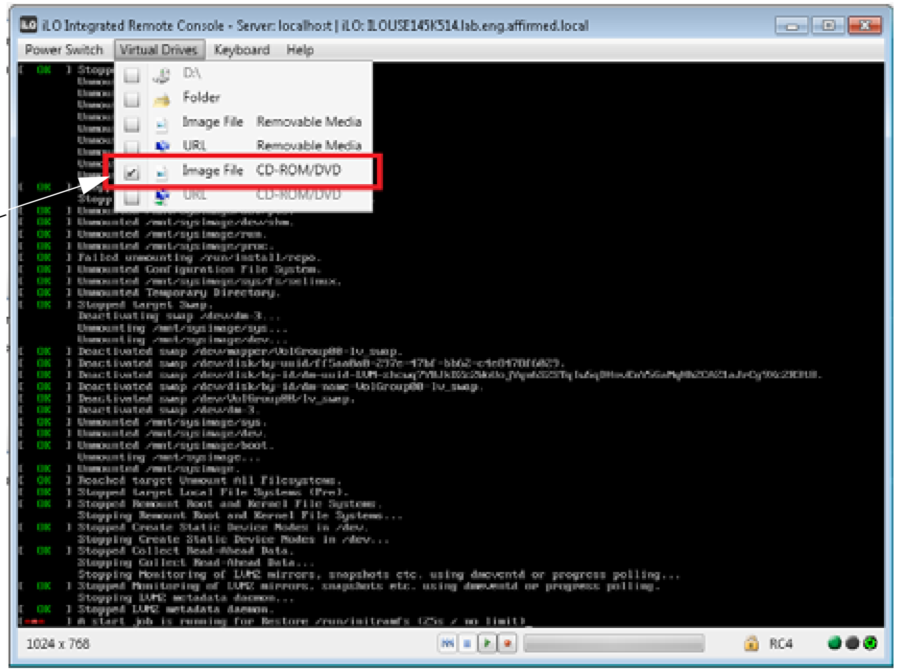
Figure 1-38 Red Hat Installer Screen

It may take several minutes for the installation to complete. When the installation is complete, the system prompts you to reboot the server.

9 Press **Enter** to reboot the server.

10 Select **Virtual Drives** from the Main Menu of the Remote Control console and unmark the Image File CD/DVD drive checkbox.

Figure 1-39   Main Menu



What’s Next

The system is now ready for Acuitas installation/upgrade. Proceed to the chapter that applies to your system:

- Standalone Systems
- High Availability Systems

Virtual Machine

This section lists the minimum specifications for Acuitas running as a virtual machine.

The following table lists the minimum virtual machine specifications.

Table 1-3 Minimum Virtual Machine Specifications

Specification	Description
Disk Provisioned Size	200GB ***Note:*** *If you have more storage available, you can increase the provisioned size once the Acuitas package has been deployed.*
CPU	16 vCPU ***Note:*** *If your system has the resources available, you can increase the number of CPUs once the Acuitas package has been deployed.*
Memory	32GB

Recommended: RAID 5.

Complete the instructions in the section that applies to your system:

- ESXi System
- KVM System
- OpenStack System

ESXi System

This section describes how to prepare an ESXi virtual machine. Be sure to perform the steps in the following order:

- Install the OVA Template Package
- Configure Security Profile
- Configure Host Size (*Recommended to configure a host size larger than the default size of 200 GB*)
- Add a Network Adapter for Interface Other than ens192 (*Optional*)
- Configure the Acuitas VM to Automatically Start Up
- Complete the Interface Setup

This procedure requires you to have VMware ESXi installed and configured, such as enabling NTP on the host. For instructions, see the *Mobile Content Cloud Deployment Guide for VMware ESXi*.

Install the OVA Template Package

To run the Acuitas server on a virtual machine, you must install the Red Hat and MySQL Open Virtualization Archive (OVA) template package on the server before you install Acuitas.

To install the OVA template package:

1. Download the OVA (an-ems-3.0.*x*-rhel-7.*y*-mysql-8.0.*z*.ova) from Affirmed Networks and save it to the Windows system where the vSphere or vCenter client is located.
2. Log into your vSphere client.
3. From the **Configuration** tab, select **File** > **Deploy OVF Template**.

Figure 1-40   Select an OVF Template Screen

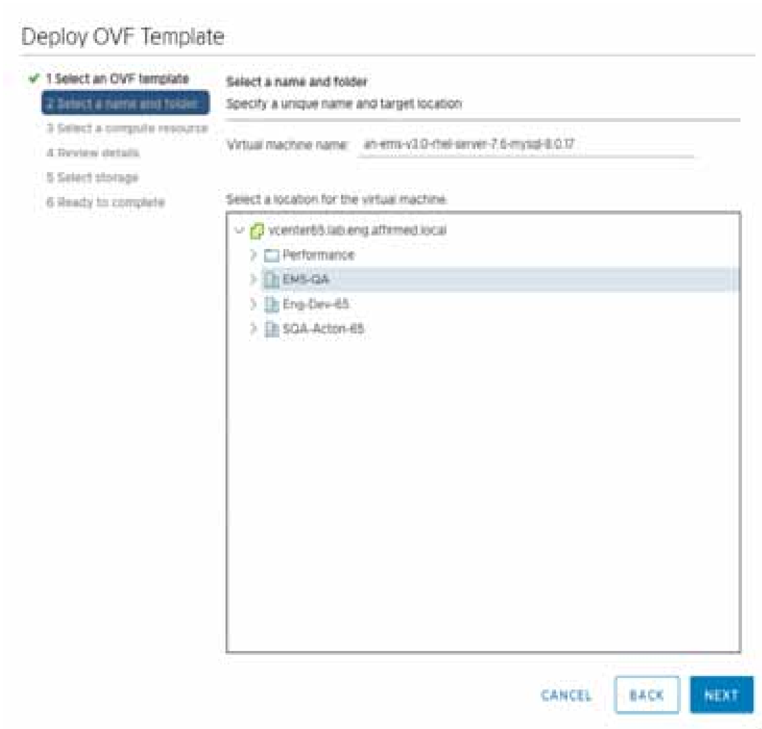
4 On the Select an OVF Template screen:

a Click **Browse** and select the OVA downloaded from Affirmed Networks.

b Click **Open**.

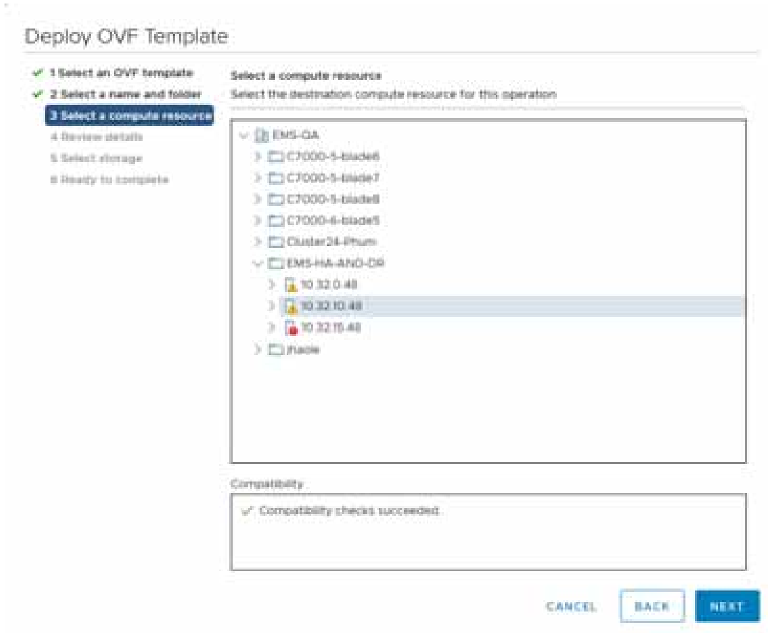
c Click **Next**.

Figure 1-41 Select a Name and Folder Screen



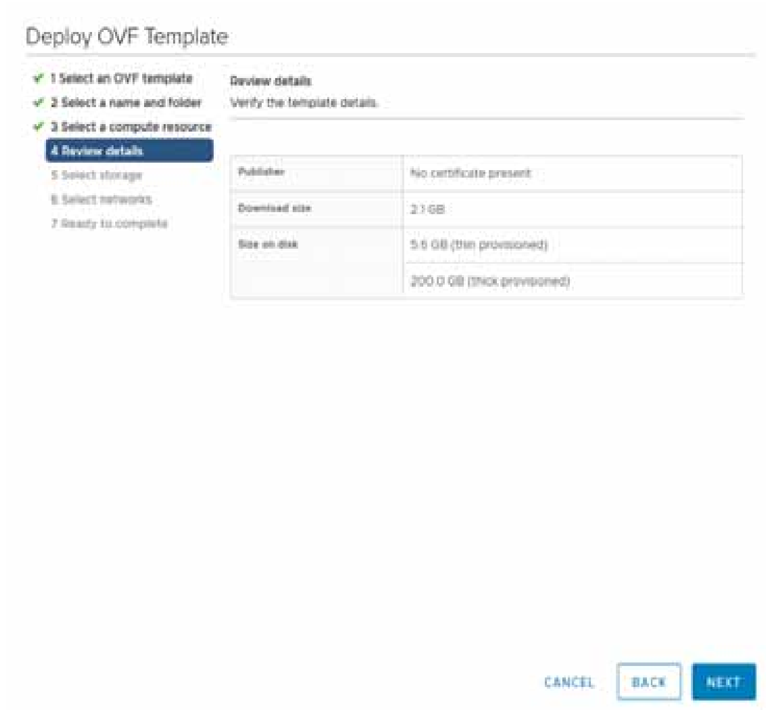
5 On the Select a Name and Folder screen, specify a unique virtual machine name and select a target location. Click **Next**.

Figure 1-42 Select a Template Resource Screen



6 On the Select a Template Resource screen, select a destination compute resource. Click **Next**.

Figure 1-43 Select a Compute Resource Screen

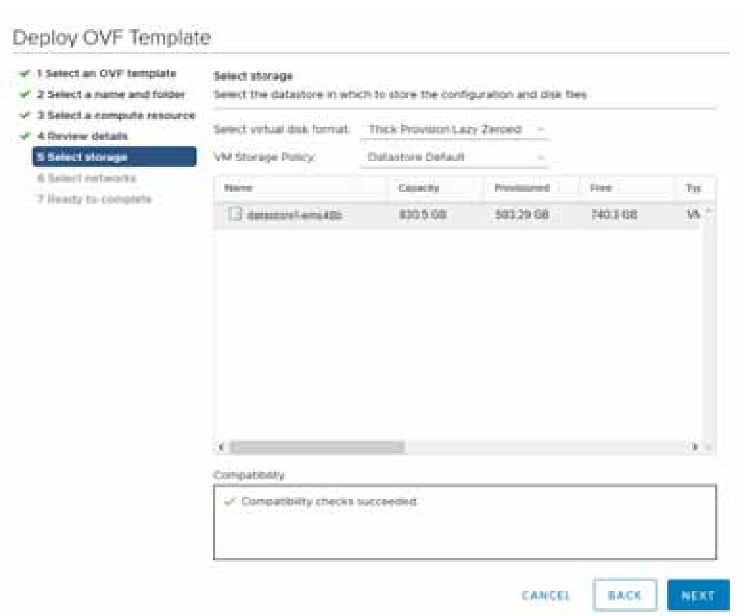


7 On the Select a Compute Resource screen, select the desired disk format. Click **Next**.

Figure 1-44 Review Details Screen

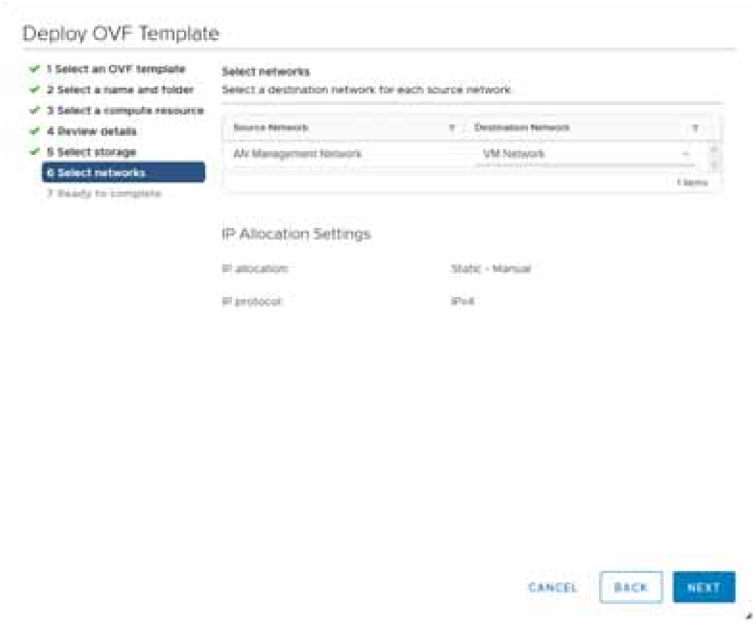
8 On the Review Details screen, verify the template details. Click **Next**.

Figure 1-45 Select Storage Screen



9 On the Select Storage screen, select the datastore in which to store the configuation and disk files. Click **Next**.

Figure 1-46 Select Networks Screen



10 On the Select Networks screen, select a destination network for each source network. Click **Next**.

Figure 1-47 Ready to Complete Screen

11 On the Ready to Complete screen, click **Finish**.

12 On the Completed Successfully screen, click **Close**.

Configure Security Profile

If you have not already done so, configure the security profile.

To configure the security profile:

1 In vSphere:

a Click the host.

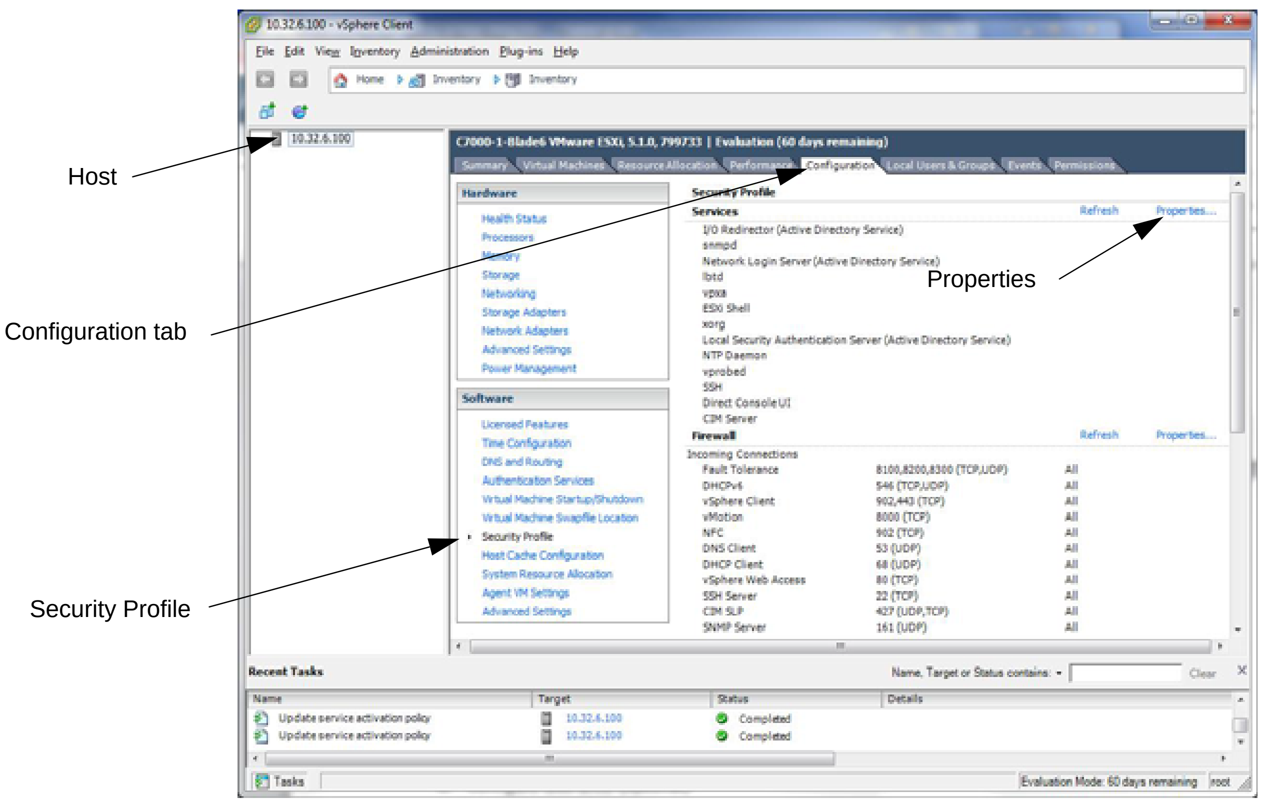
b Click the **Configuration** tab.

c In the **Software** list, click **Security Profile**.

d Under Services, click **Properties**.

A screen similar to the following appears.

Figure 1-48 Security Profile



2 To enable SSH:

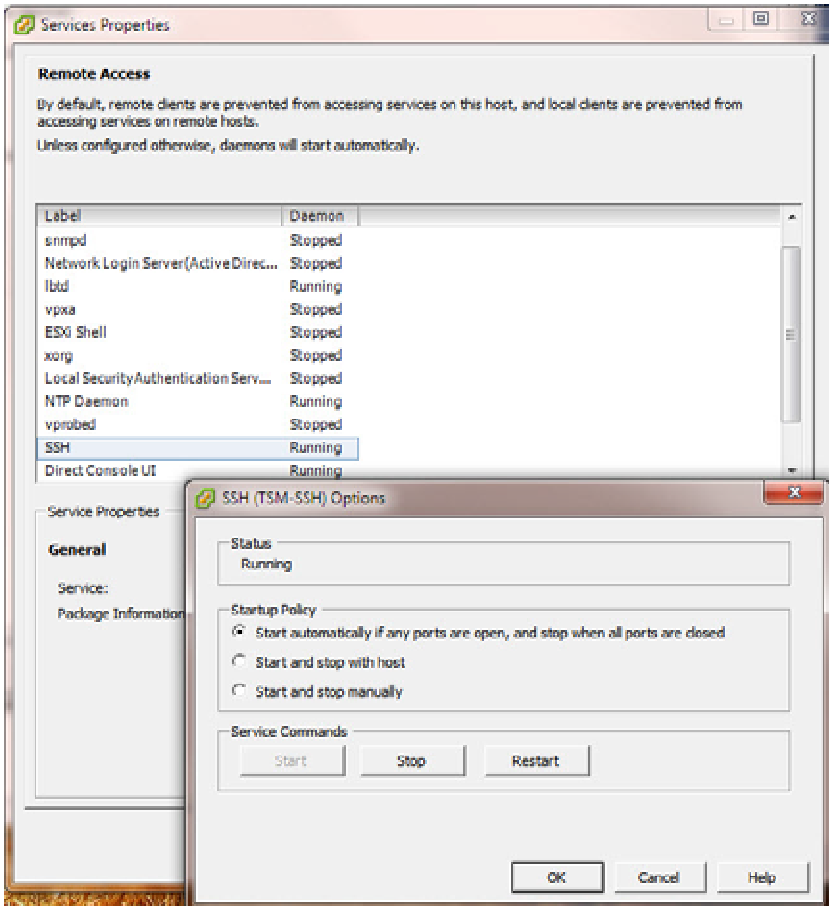
a Click **SSH**.

b Click **Options**.

c Mark **Start Automatically**.

d Click **OK**.

Figure 1-49 SSH Settings



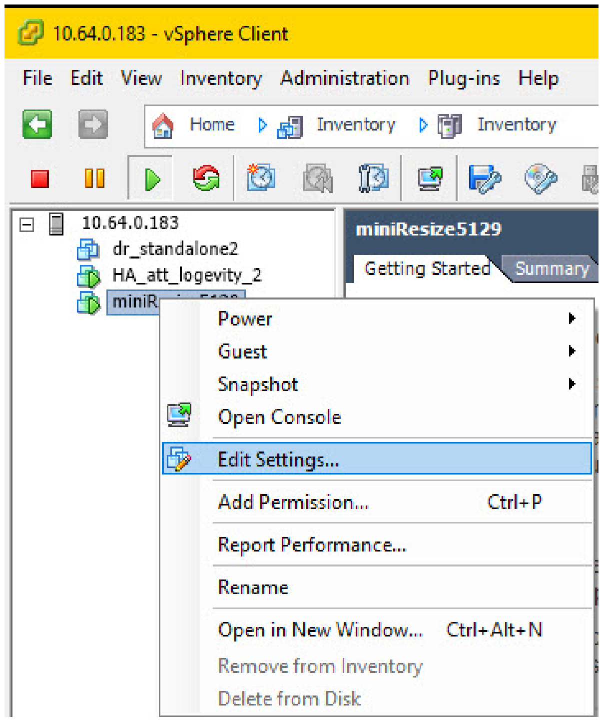
Configure Host Size

The Acuitas image is created with 200 GB size. You can configure a larger host size, if desired.

To configure host size:

1 In vSphere, right-click the Acuitas VM and click **Edit Settings**.

Figure 1-50 Edit Settings

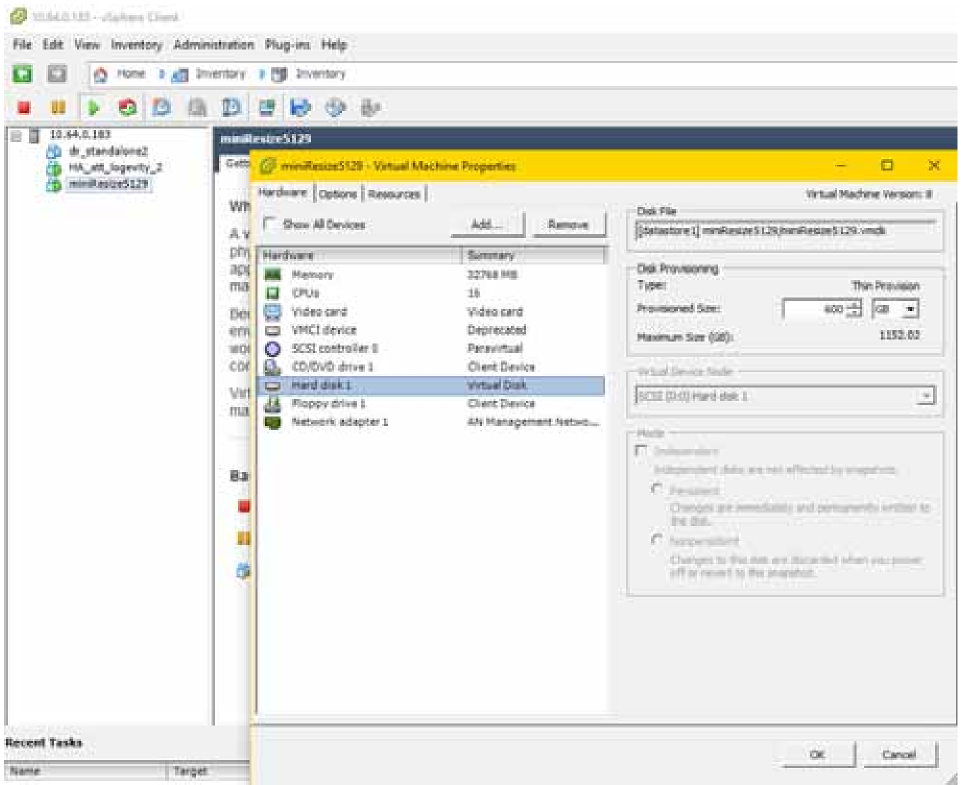


2 Click the **Hardware** tab.

3 Click Hard disk 1.

4 Under **Disk Provisioning**, in the **Provisioned Size** field, specify the VM size.

Figure 1-51 Provisioned Size Setting



5 Click **OK**.

6 Restart the VM and proceed to Complete the Interface Setup.

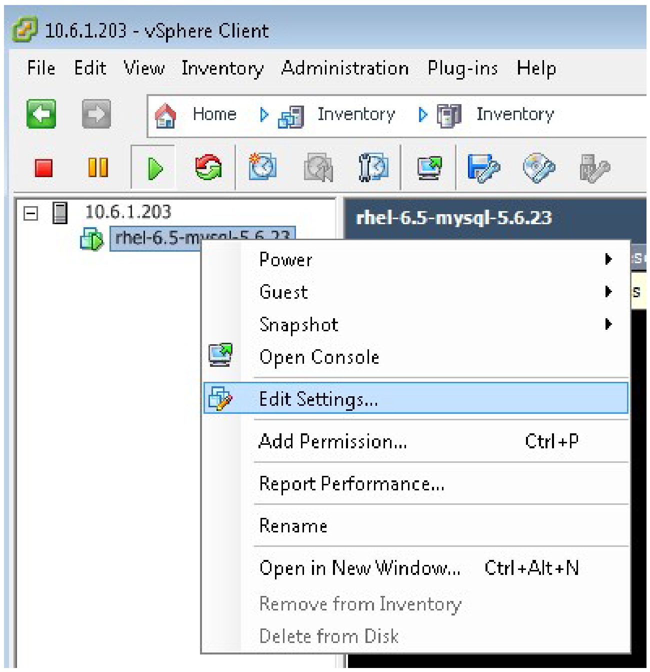
Add a Network Adapter for Interface Other than ens192

To use an interface other than ens192, add a network adapter of type VMXNET3.

To add a network adapter of type VMXNET3:

1 In vSphere, right-click the Acuitas guest VM and click **Edit Settings**.

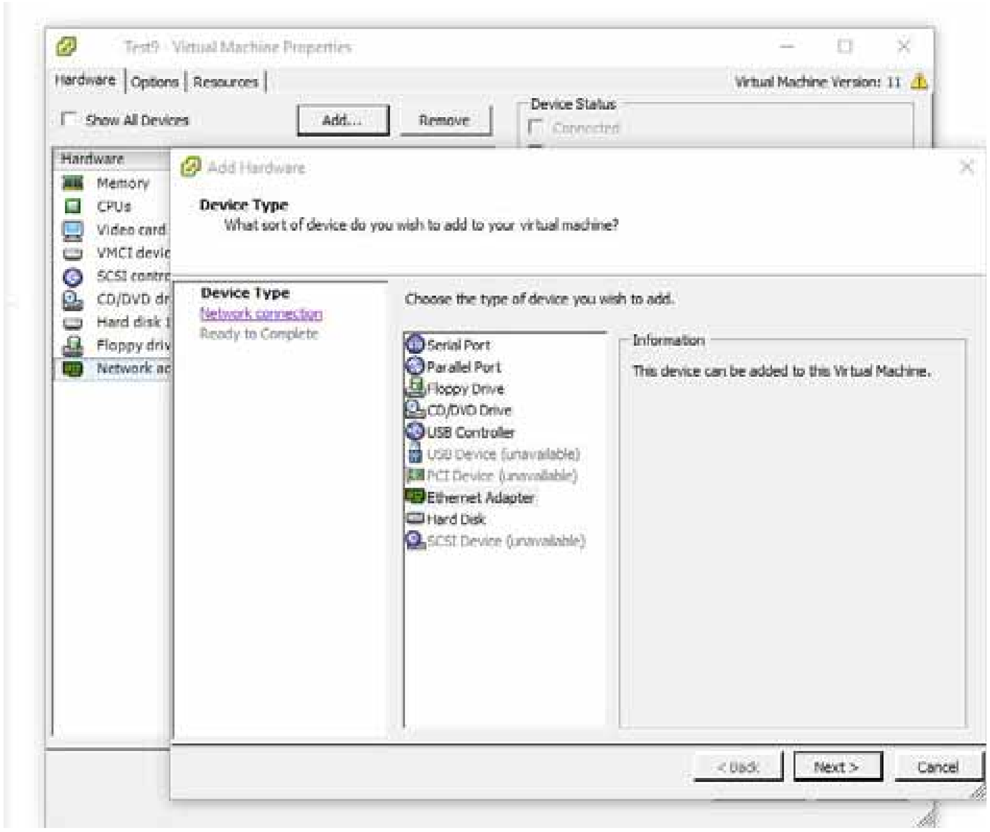
Figure 1-52 Edit Settings



2 Select **Network Adapter** and click **Add**.

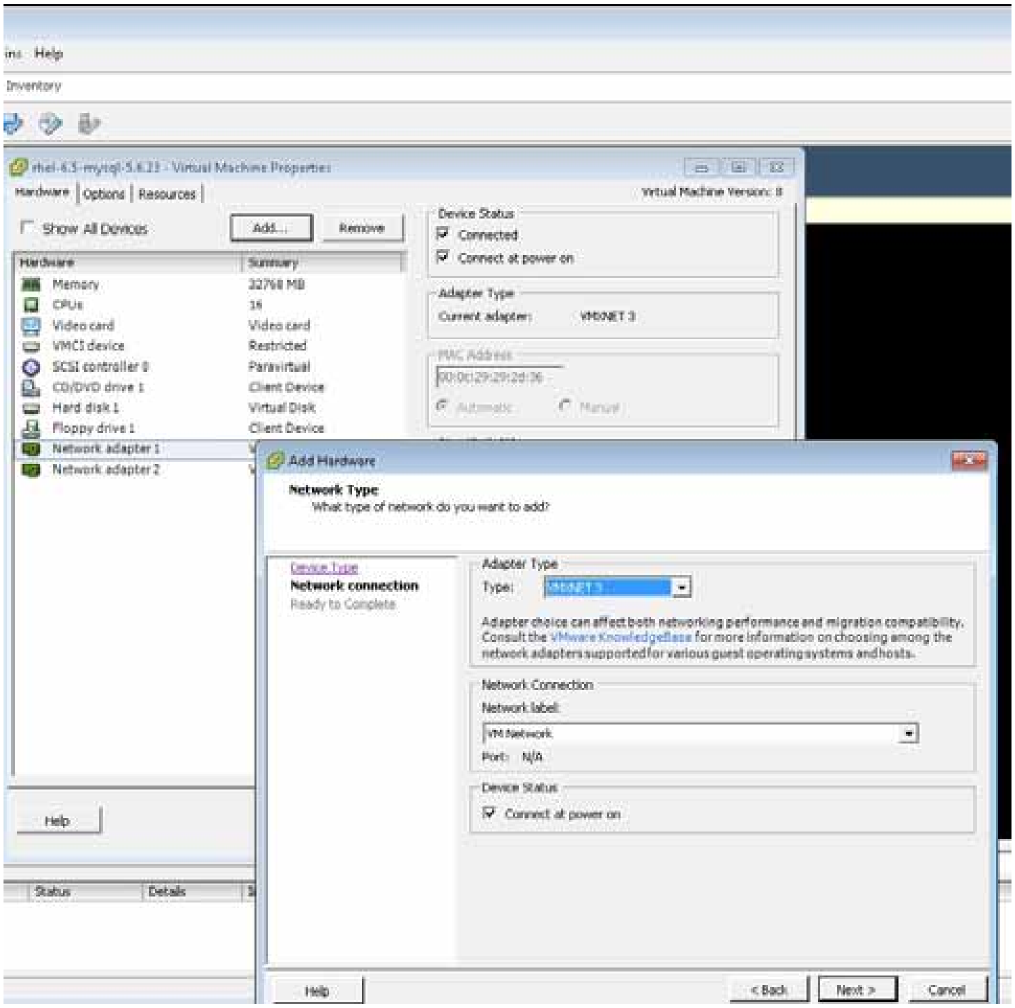
3 On the Device Type screen, select Ethernet Adapter and click **Next**.

Figure 1-53 Device Type Screen



4 In the Network Type screen, select Adapter Type VMXNET3 and click **Next**.

Figure 1-54 Add Network Adapter



5 Click **OK**.

6 Power up the Acuitas VM.

7 From the console, log in as root and issue the following command to change to the /etc/sysconfig/network-scripts directory:

```
cd /etc/sysconfig/network-scripts
```

8 Issue the following command to copy ifcfg-ens192 to a new file named ifcfg-<*interface*>:

```
cp ifcfg-ens192 ifcfg-<interface>
```

9 Use a text editor to modify ifcfg-<***interface***>:

```
vi ifcfg-<interface>
```

a Make the following changes:

   – Change the line from NAME = **`ens192`** to NAME = <***`interface`***>
   – Change the line from DEVICE = **`ens192`** to DEVICE = <***`interface`***>
   – Remove any lines containing **`uuid`**.

b Save the file and exit.

Configure the Acuitas VM to Automatically Start Up

To configure the Acuitas VM to automatically start up:

1 If the Acuitas guest VM is not setup to automatic start up when the host is started, perform these steps:

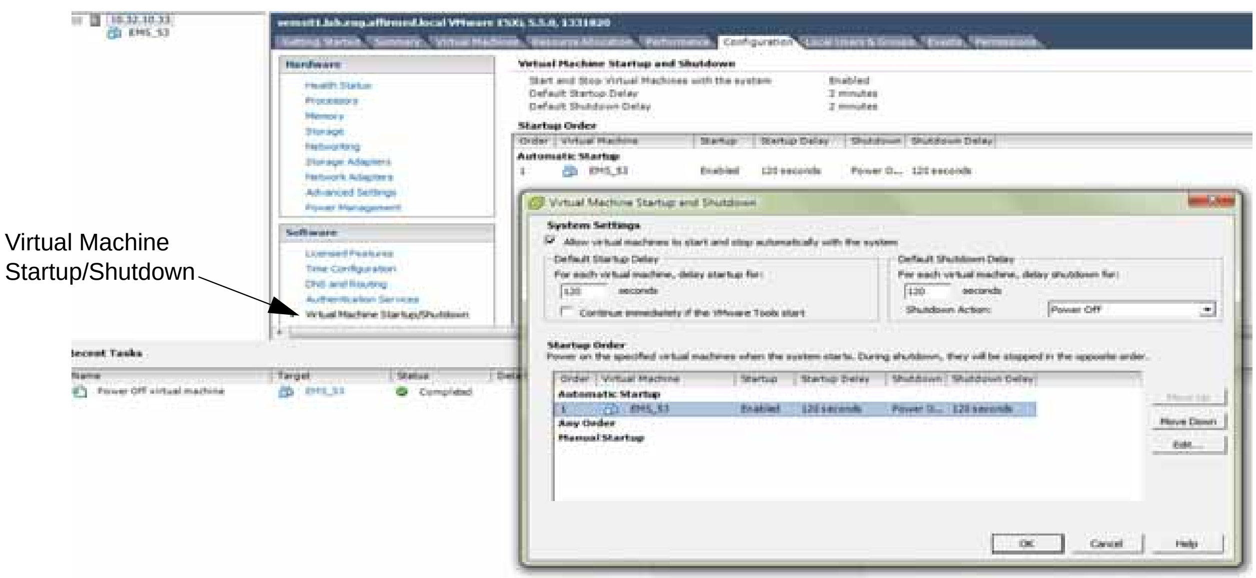
a Click the host.

b Click the **Configuration** tab.

c In the **Sofware** list, click **Virtual Machine Startup/Shutdown**.

d Click **Properties**.

Figure 1-55 Virtual Machine Startup/Shutdown



e Mark the **Allow virtual machines to start and stop automatically with the machine** checkbox.

f In **Startup Order**, click the Acuitas guest VM and click **Move Up** to move it to Automatic Startup.

g Click **OK**.

Complete the Interface Setup

To complete the interface setup:

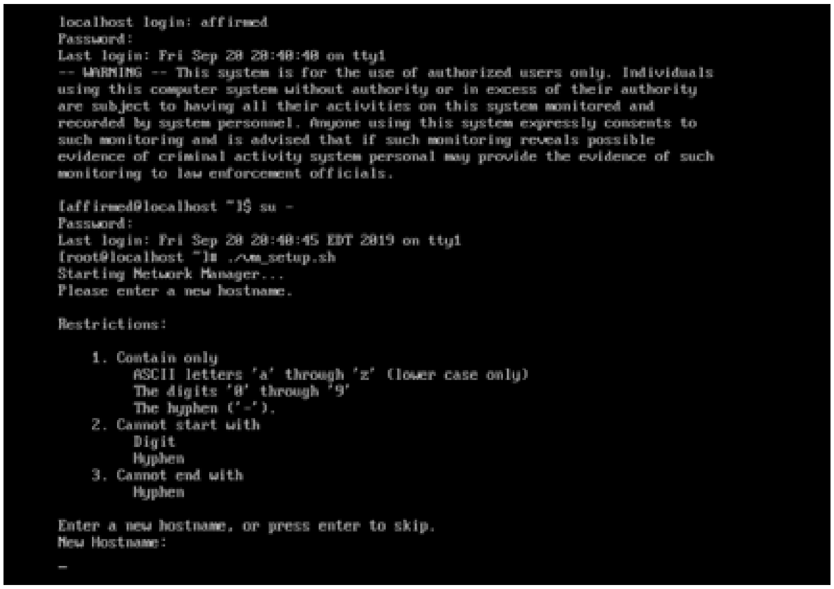
1. Power up the Acuitas VM.
2. From the console, log in as `affirmed` and issue the following command to gain super-user privileges:

   `su -`

3. When prompted, enter the super-user password.
4. Issue the following command:

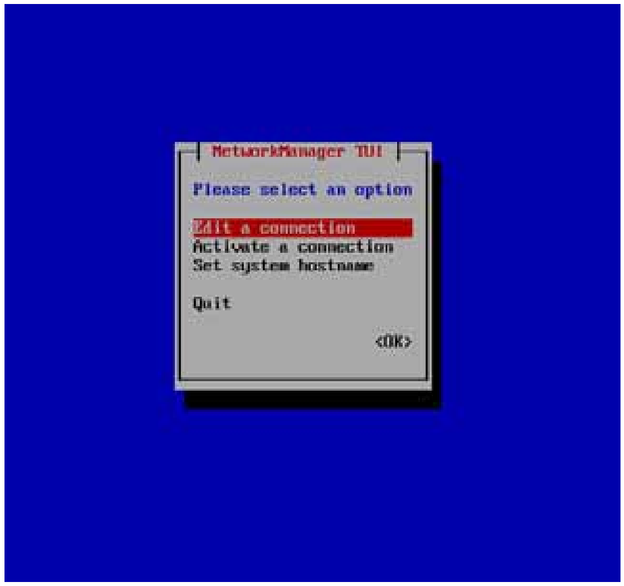
   `/root/ems_setup.sh`

Figure 1-56 Console Tab



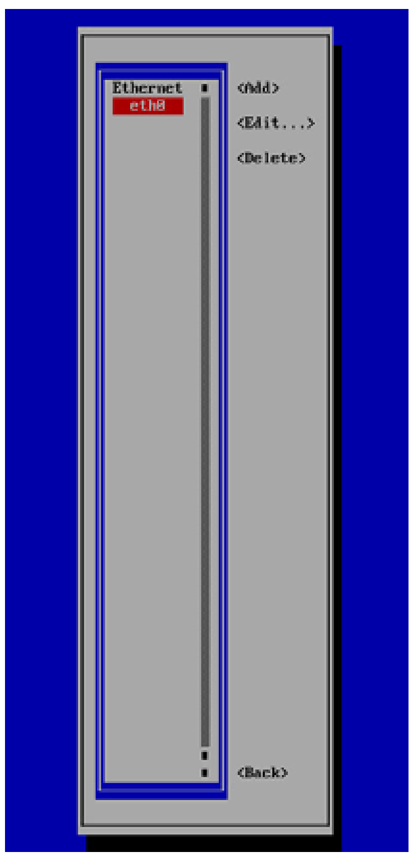
5. Enter a new hostname that adheres to the naming conventions.
6. On the NetworkManager TUI screen, select **Edit a Connection**.

Figure 1-57 NetworkManager TUI Screen



7 On the screen, select the interface to be used and press **Enter**.

Figure 1-58 Select Interface Screen



8 On the Edit Connection screen:

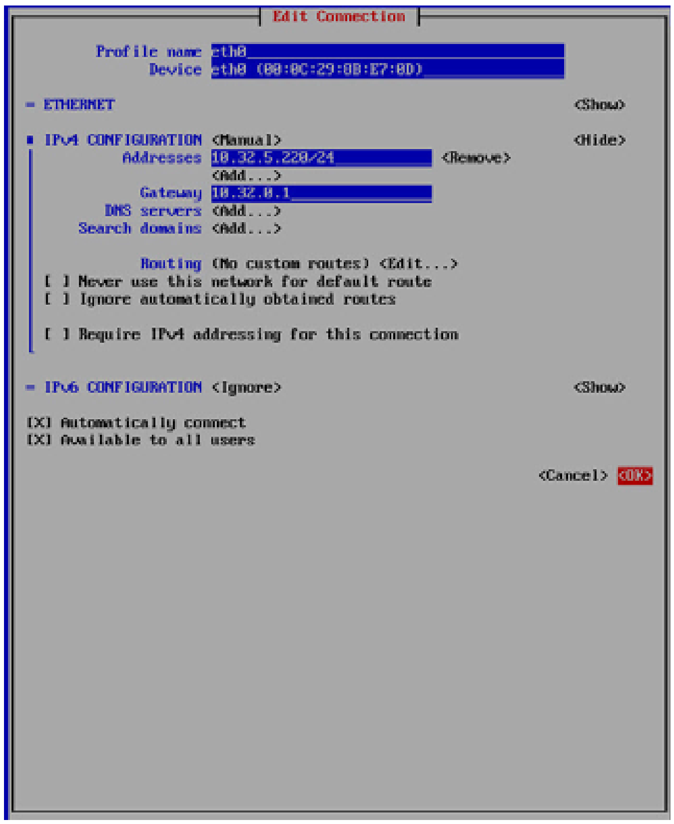
For IPv4 Configuration:

a In the IPv4 Configuration field, select **Manual** for static IP configuration.

b Enter the network configuration information, as needed.

c Click **OK**.

Figure 1-59 IPv4 Network Configuration Screen



For IPv6 Configuration:

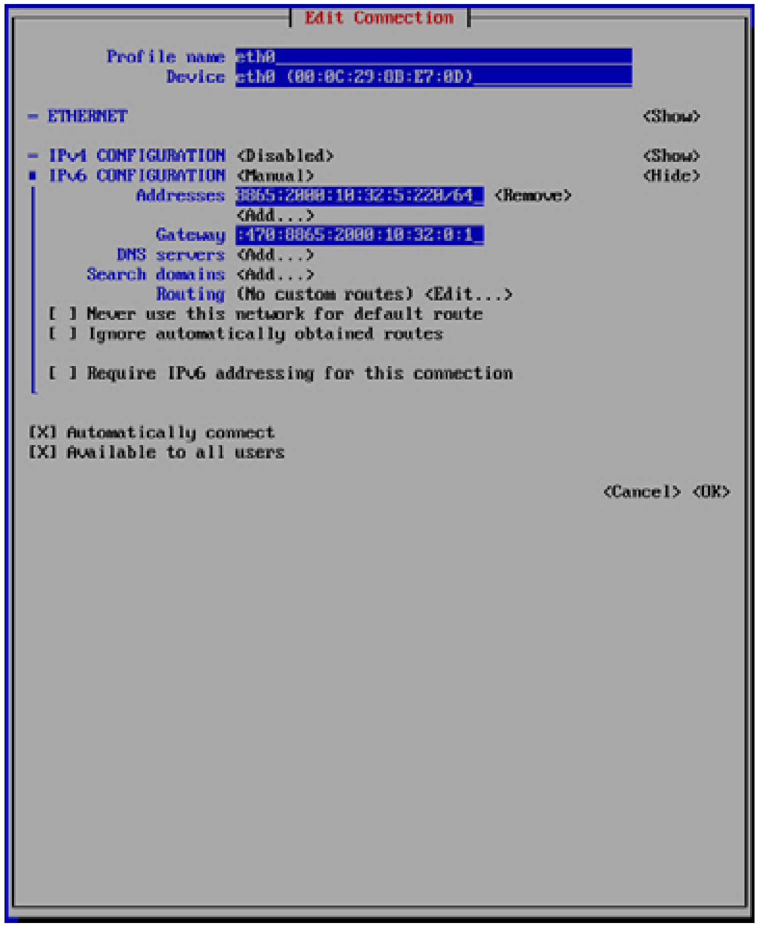
a In the IPv4 Configuration field, select **Disabled**.

b In the IPv6 Configuration field, select **Manual** for static IP configuration.

c Enter the network configuration information, as needed.

d Click **OK**.

Figure 1-60   IPv6 Network Configuration Screen



9 At the NetworkManager TUI screen, click **OK** to exit.

10 When prompted, enter the disk size you want for the VM. The disk size must match the number you configured in Configure Host Size:

```
Default disk size is 200 GB
     New disk range: 250 to 2000 (GB)
     Size value must be multiples of 50
     Or just press “Enter” to skip the resize
New disk size (GB):
Enter New Size:
```

The script adjusts the VM partitions.

11 To verify that the logical volume has been extended as expected, issue the command **`vgdisplay`** as root:

```
[root@emsha-94-a ~]# vgdisplay
  --- Volume group ---
  VG Name               VolGroup00
  System ID
  Format                lvm2
  Metadata Areas        2
  Metadata Sequence No  14
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                10
  Open LV               10
  Max PV                0
  Cur PV                2
  Act PV                2
  VG Size               549.50 GiB
  PE Size               4.00 MiB
  Total PE              140673
  Alloc PE / Size       140673 / 549.50 GiB
  Free  PE / Size       0 / 0
```

12 To verify the extra space added to the logical volume has been allocated correctly to /opt and /data partitions, issue the following command as affirmed or root:

```
[affirmed@emsha-94-a ~]$ df -h -t ext4
Filesystem                            Size  Used Avail Use% Mounted on
/dev/mapper/VolGroup00-lv_root         15G  4.7G  9.3G  34% /
/dev/sda1                             477M  125M  323M  28% /boot
/dev/mapper/VolGroup00-lv_opt         227G   11G  207G   5% /opt
/dev/mapper/VolGroup00-lv_data        229G   13G  207G   6% /data
/dev/mapper/VolGroup00-lv_var         9.8G  319M  8.9G   4% /var
/dev/mapper/VolGroup00-lv_home        976M  2.6M  907M   1% /home
/dev/mapper/VolGroup00-lv_var_tmp    4.8G   20M  4.6G   1% /var/tmp
/dev/mapper/VolGroup00-lv_tmp         9.8G   37M  9.2G   1% /tmp
/dev/mapper/VolGroup00-lv_var_log     25G   90M   24G   1% /var/log
/dev/mapper/VolGroup00-lv_var_log_audit  4.8G  1.6G  3.0G  35%
/var/log/audit
```

13 On the console tab, use a text editor to add server IP and hostname in /etc/hosts:

```
vi /etc/hosts
```

For example:

```
127.0.0.1     localhost    localhost.localdomain    localhost4
localhost4.localdomain4
```

```
::1      localhost    localhost.localdomain    localhost6
localhost6.localdomain6
```

```
10.0.0.10 qaosems05
```

14 Save and Quit out of vi /etc/hosts file.

15 *Optional:* If you want a redundant management line, perform this step:

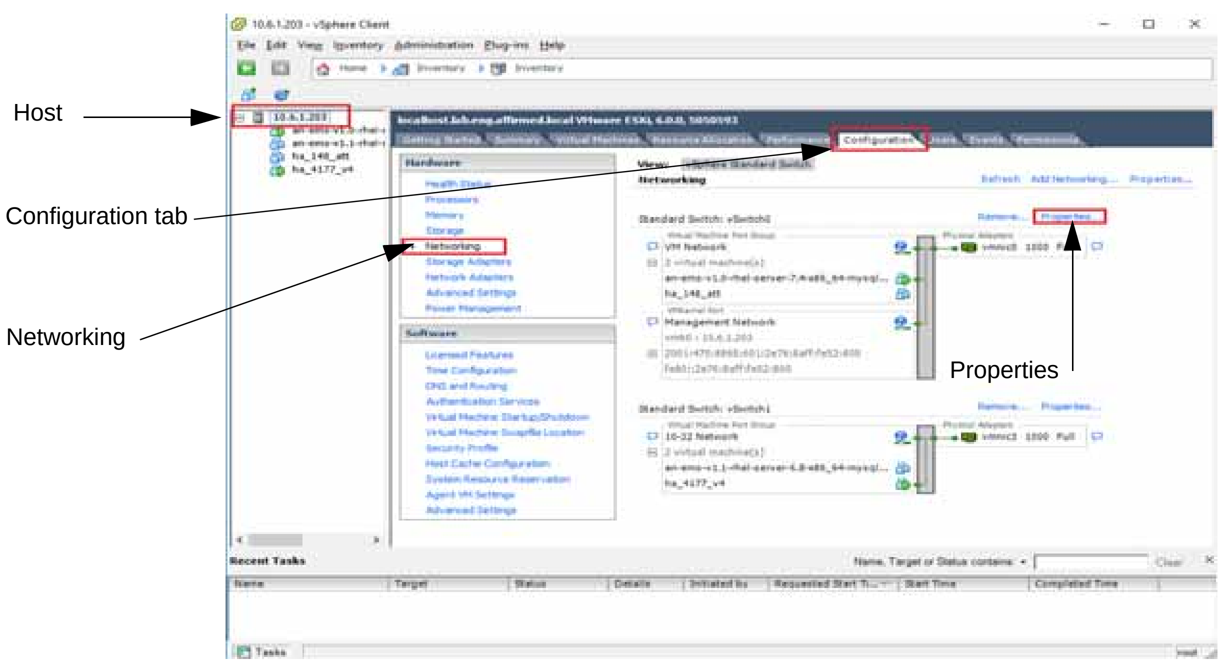
a In vSphere, click the host.

b Click the **Configuration** tab.

c In the **Hardware** list, click **Networking**.

d For the vswitch0, click **Properties**.

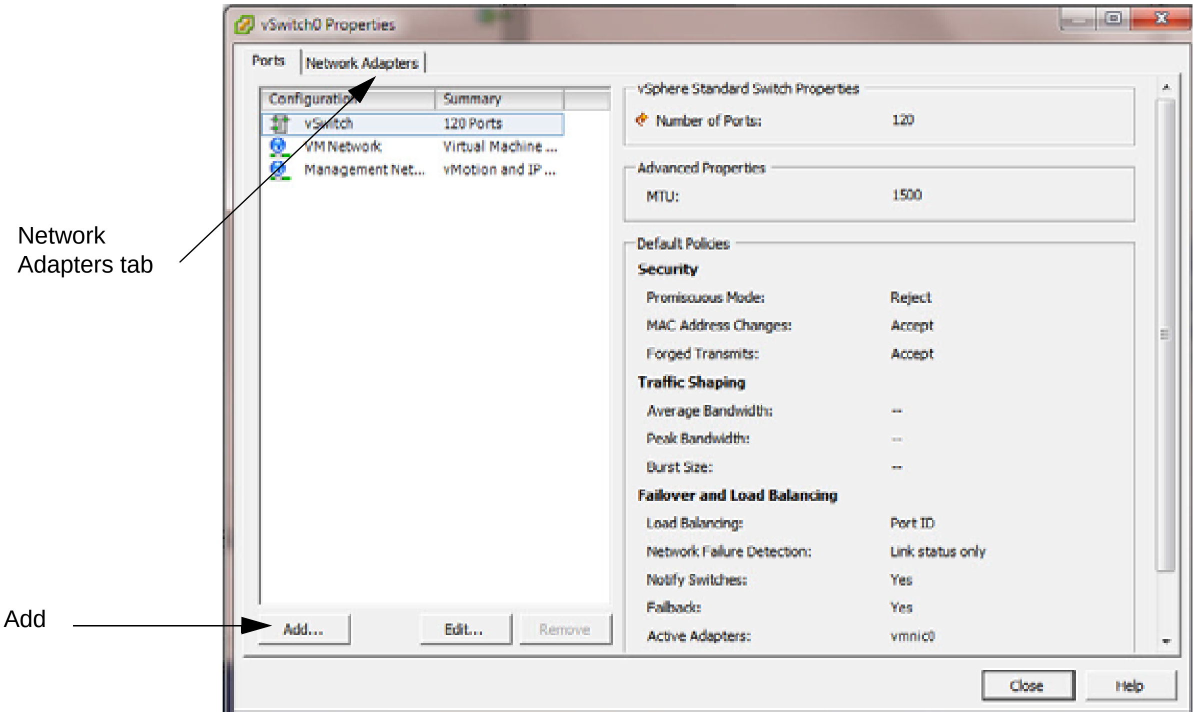
Figure 1-61 Configuration Tab



e Click the **Network Adapters** tab.

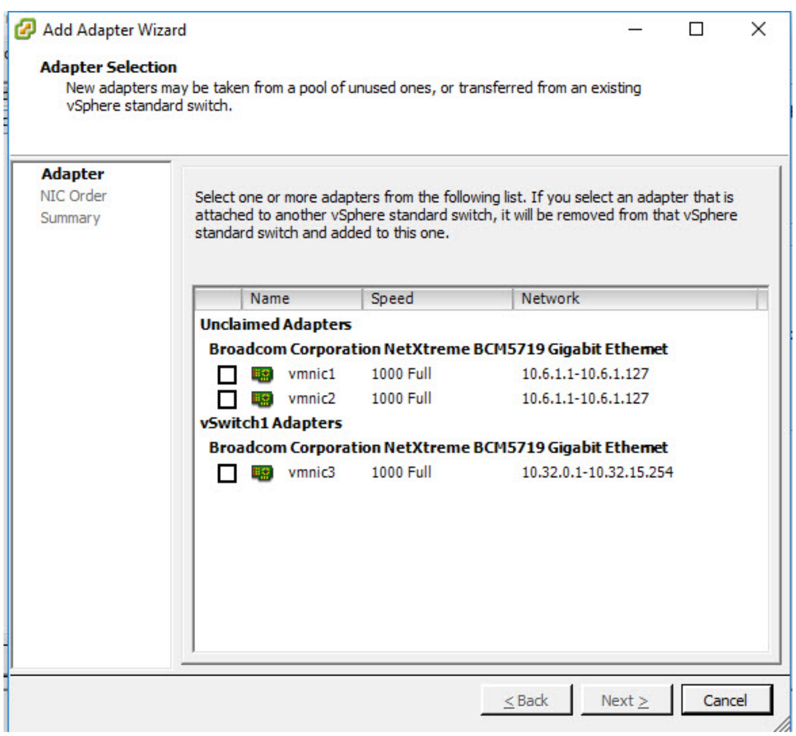
f To add vmnic1, click **Add**.

Figure 1-62 Network Adapters Tab



The system runs the Add Adapter wizard.

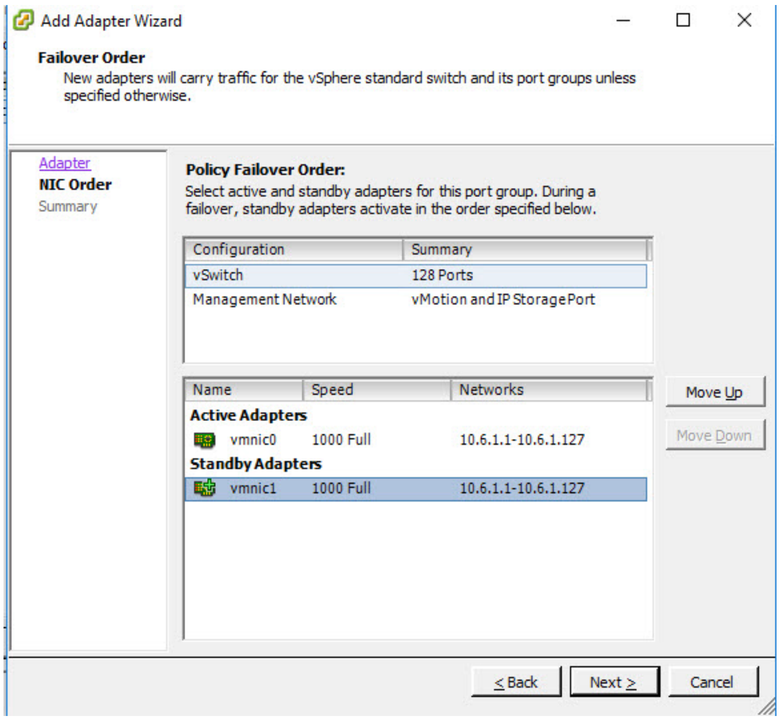
Figure 1-63 Add Adapter Wizard 1



g Mark the vmnic1 checkbox and click **Next**.

h Click vmnic1 and click **Move Down** to move it to Standby.

Figure 1-64 Add Adapter Wizard 2



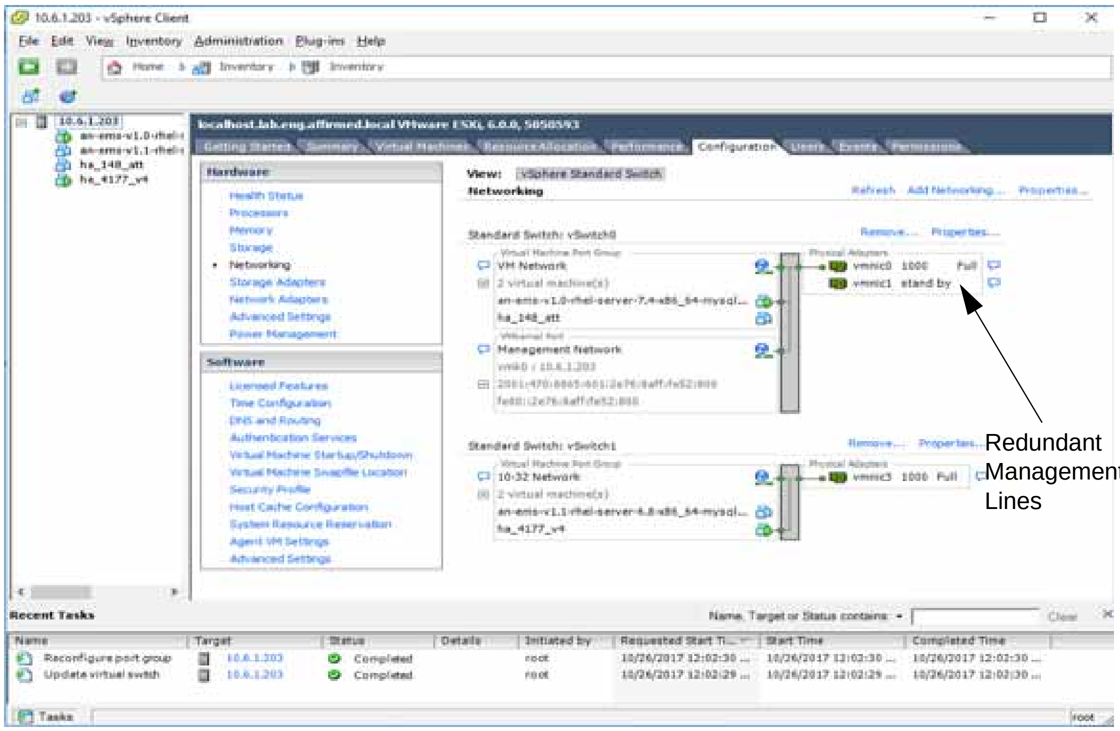
i Click **Next**.

j Click **Finish**.

k Click **Close**.

The system adds the redundant management line.

Figure 1-65   Redundant Management Lines



What's Next

The system is now ready for Acuitas installation/upgrade. Proceed to the chapter that applies to your system:

- Standalone Systems
- High Availability Systems

KVM System

This section describes how to prepare a KVM virtual machine, including:

- Install the qcow2 Disk Image
- Configure the Virtual Network Interface to Allow Multicast Forwarding (*IPv6 Only)*
- (*Required*) Configure Host Size
- Complete the Interface Setup

This procedure requires you to have KVM installed and configured. For instructions, see the *Mobile Content Cloud Deployment Guide for KVM*.

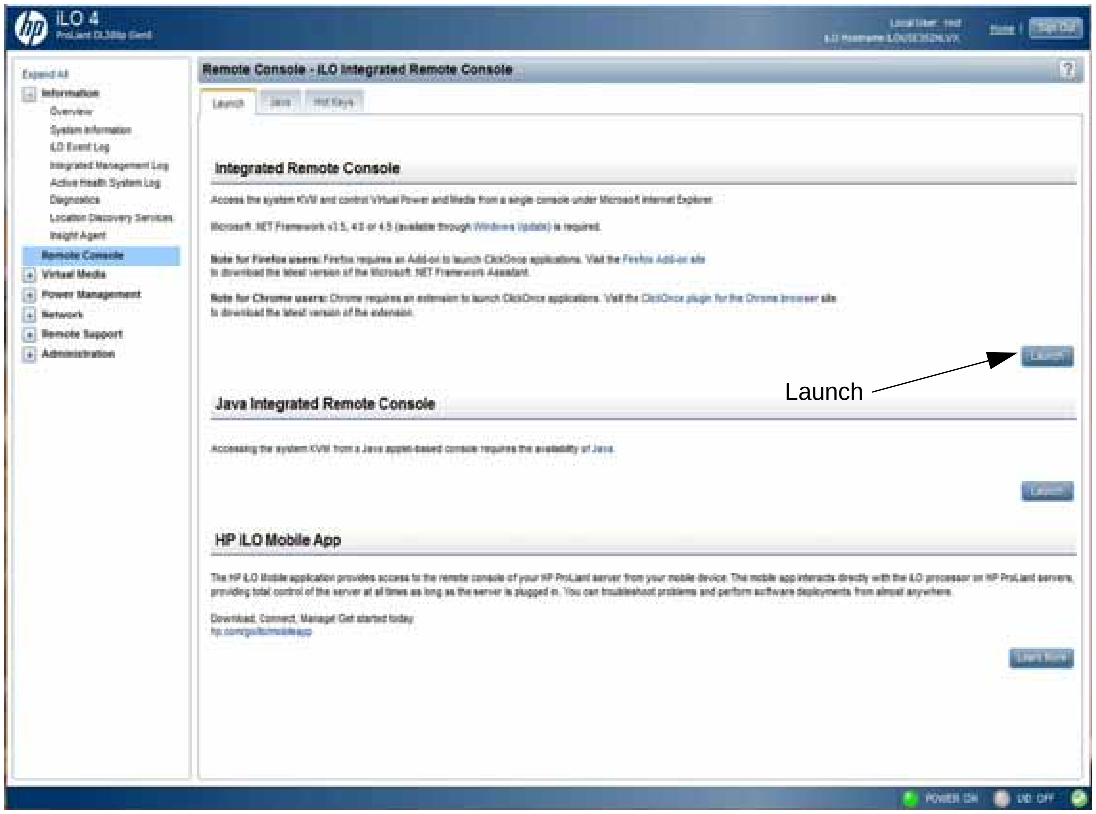
Install the qcow2 Disk Image

To install the qcow2 disk image:

1 Download the qcow2 disk image from Affirmed Networks: an-ems-3.0.*x*-rhel-7.*y*-mysql-8.0.*z*.qcow2.

2 From a browser, log onto the iLO and click **Launch** to access the Remote Console (Firefox browser requires a Firefox Add-on to launch the Remote Console).

Figure 1-66   iLO



3 As root, select **Places** > **Computer** from the iLO Remote Console menu.

4 Navigate to /home/images.

5 Right-click the directory and select **Open in Terminal**.

6 Copy the qcow2 disk image file into home/images.

7 From the iLO Remote Console menu, select **Applications** > **System Tools** > **Virtual Machine Manager**.

8 From the Virtual Machine Manager menu, select **File** > **New Virtual Machine**.

9 Click **Import Existing Disk Image**.

10 Click **Browse** and navigate to /home/images for the location of the qcow2 image.

11 In the OS Type field, select **Linux**.

12 In the Version field, select **Red Hat Enterprise Linux 7.7**.

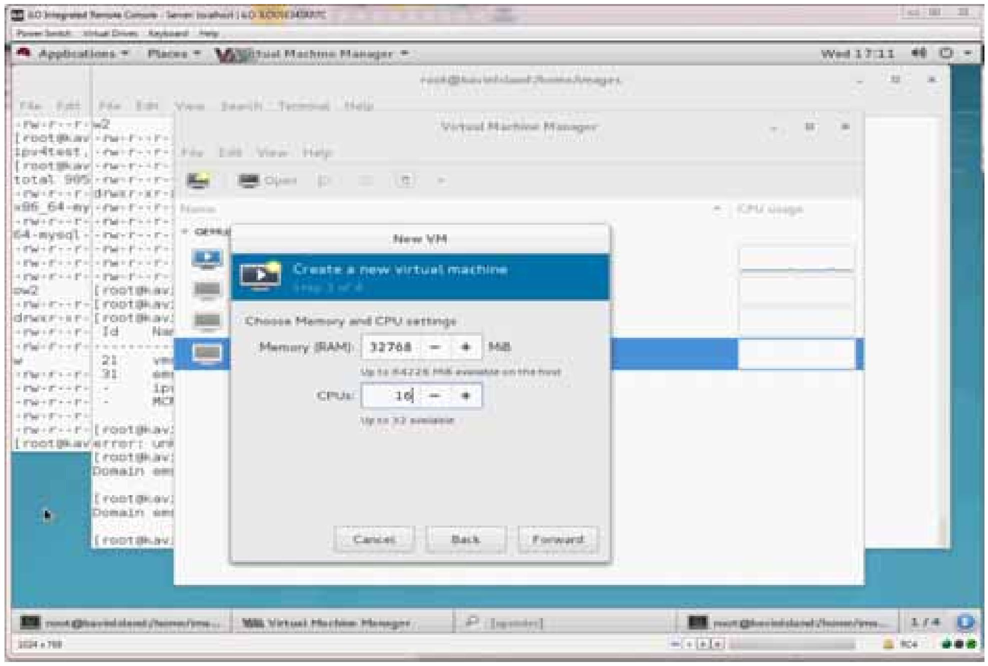
13 Click **Forward**.

14 In the Memory (RAM) field, select **32768**.

15 In the CPUs field, select **16**.

16 Click **Forward**.

Figure 1-67 Memory and CPUs

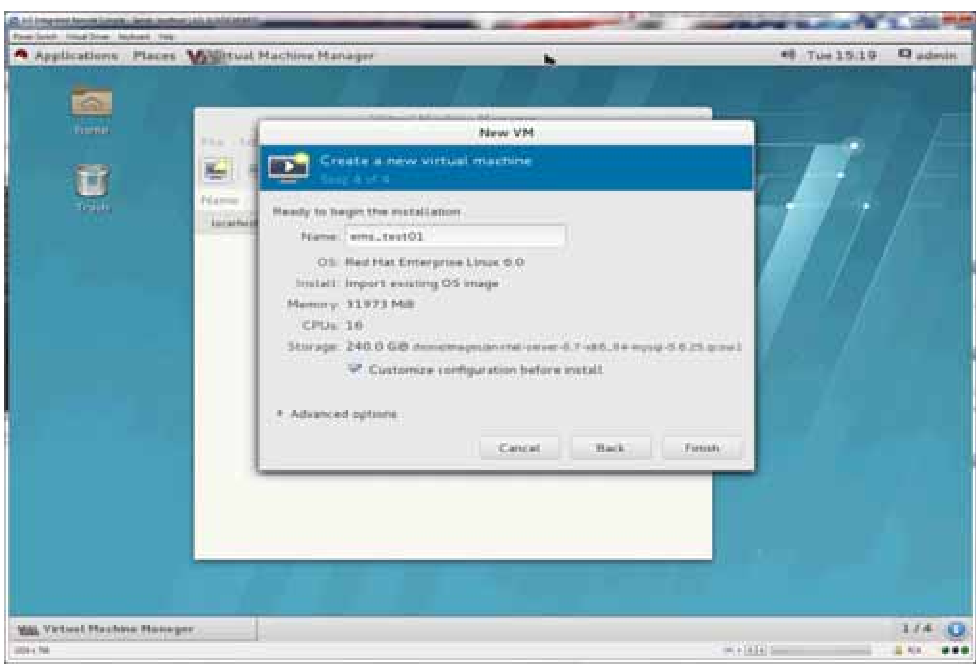


17 In the Name field, enter a name for the VM.

18 Mark the **Customize configuration before install** checkbox.

19 Click **Finish**.

Figure 1-68 Customize Configuration Before Install

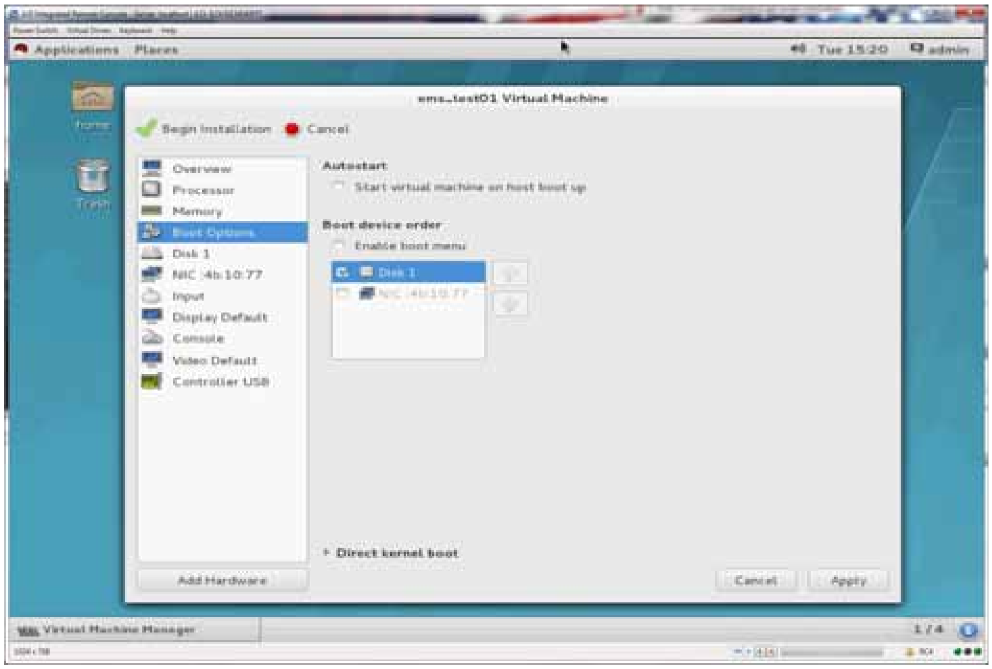


20 Highlight Boot Options.

21 In the Boot Device Order field, select Disk 1.

22 Click **Apply**.

Figure 1-69 Boot Options



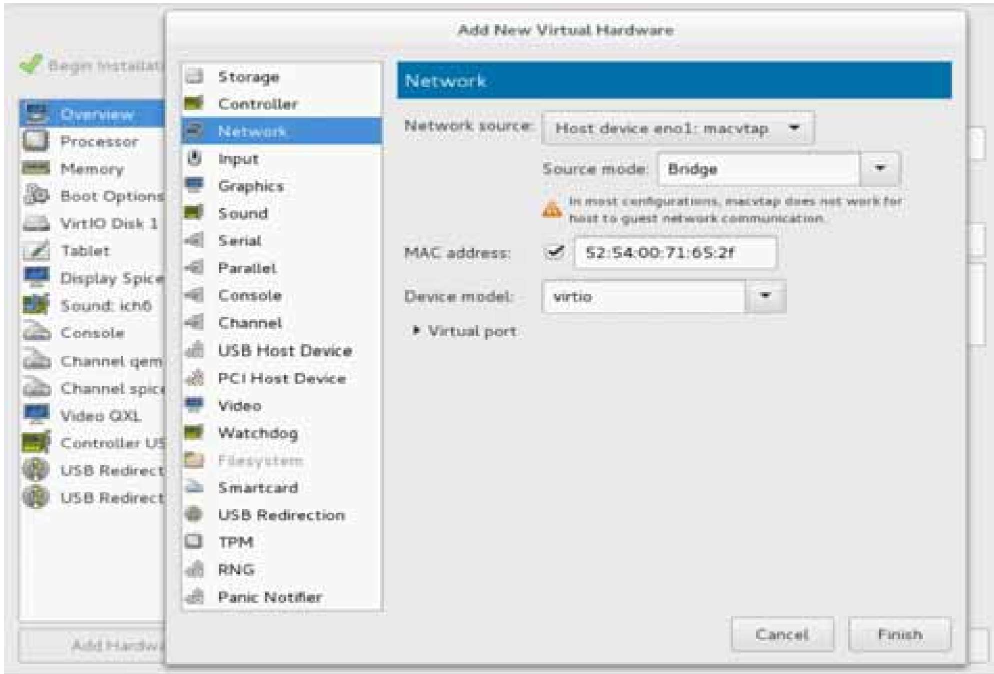
23 Click **Add Hardware** to add a network card.

24 Click **Network**.

25 In the Network Source field, select Host device eno1: macvtap.

26 In the Device model field, select virtio.

Figure 1-70 Add New Virtual Hardware



27 Click **Finish**.

28 Click **Begin Installation** to fully setup and start the VM.

Configure the Virtual Network Interface to Allow Multicast Forwarding (*IPv6 Only*)

To configure the virtual network interface to allow multicast forwarding:

1 Open a terminal window on the RedHat host. To do so, select **Applications** > **Favorites** > **Terminal** from the iLO Remote Console menu.

2 Execute the **`virsh list --all`** command to display the virtual machine instances. The Acuitas virtual machine should be in the running state, as seen in the following example:

```
[root@newIsland images]# virsh list --all

Id     Name                               State

----------------------------------------------------

21     emsserverkvm                       running
```

3 Execute the **`virsh destroy <instance>`** command to halt the Acuitas virtual machine. The command should return the `destroyed` status, as shown in the following example:

```
[root@newIsland ~]# virsh destroy emsserverkvm

Domain emsserverkvm destroyed
```

4 Execute the **`virsh edit <instance>`** command to edit the EMS virtual machine's configuration.

5 Navigate to the interface type labeled `direct` and add `trustGuestRxFilters='yes'` to the end of the line.

For example:

```
[root@newIsland ~]# virsh edit emsserverkvm


<interface type='direct' trustGuestRxFilters='yes'>
      <mac address='52:54:00:71:65:2f'/>
      <source dev='eno1' mode='bridge'/>
      <target dev='macvtap0'/>
      <model type='virtio'/>
      <alias name='net0'/>
      <address type='pci' domain='0x0000' bus='0x00' slot='0x03'
function='0x0'/>
</interface>
```

6 Save the changes and exit the edit session.

The command validates the syntax of **`virsh`** edit sessions and returns `Domain <instance> XML configuration edited` after exiting the edit session successfully.

For example:

```
Domain emsserverkvm XML configuration edited
```

7 Start the Acuitas virtual machine. For example, you can execute the **`virsh start <instance>`** command.

The command should return status `Domain <instance> started`.

For example:

```
[root@newIsland ~]# virsh start emsserverkvm
Domain emsserverkvm started
```

Configure Host Size

To configure host size:

1. If the VM automatically powered up after deployment, power it down.
2. Issue the command to resize the image. In the command, specify that additional disk size to add to the default disk size (default QCOW2 size is 200 GB) to reach the desired size. The command syntax is:

   ```
   qemu-img resize [File Path of QCOW2 Image]/[Name of QCOW2 Image] +
   [Expanded Size Amount]
   ```

   For example, to reach the recommended size of 550 GB, add 350G to the default size:

   ```
   qemu-img resize /home/images/an-ems-3.0.x-rhel-7.y-mysql-8.0.z.qcow2
   +350G
   ```

   The command returns the message `Image resized`.
3. Restart the VM and proceed to Complete the Interface Setup.

Complete the Interface Setup

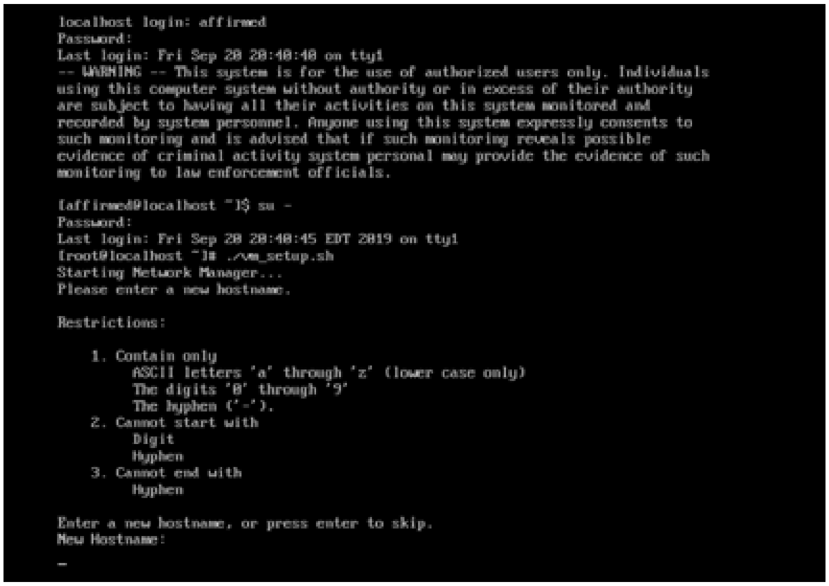
To complete the interface setup:

1. Power up the Acuitas VM.
2. From the console, log in as `affirmed` and issue the following command to gain super-user privileges:

   ```
   su -
   ```
3. When prompted, enter the super-user password.
4. Issue the following command:

   ```
   /root/ems_setup.sh
   ```

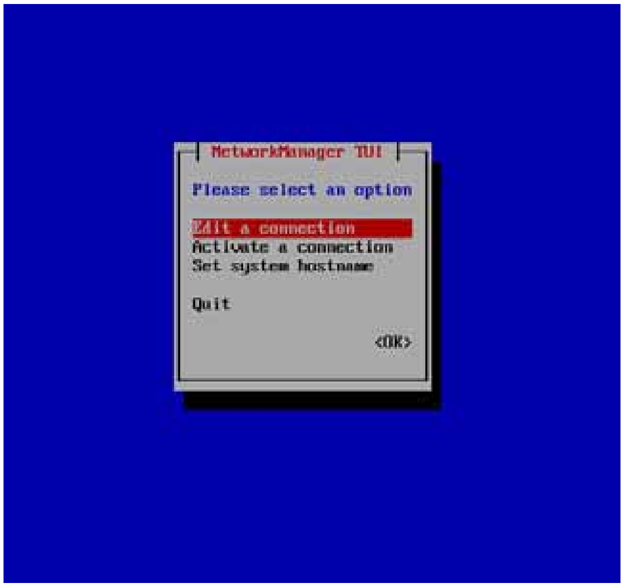
Figure 1-71   Console Tab



5 Enter a new hostname that adheres to the naming conventions.

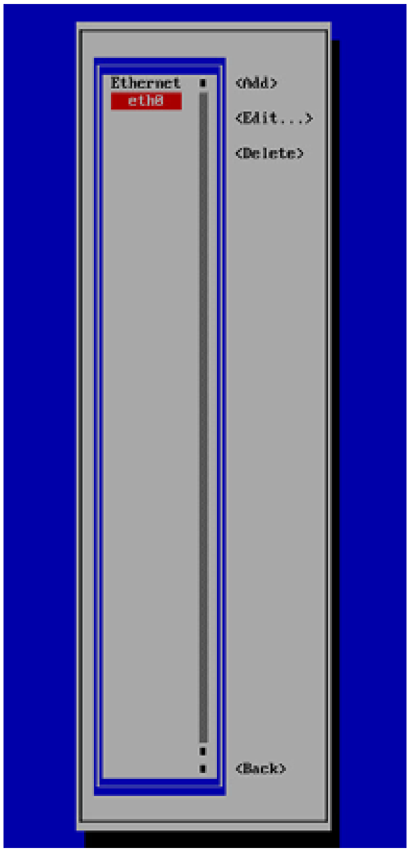
6 On the NetworkManager TUI screen, select **Edit a Connection**.

Figure 1-72   NetworkManager TUI Screen



7 On the screen, select the interface to be used and press **Enter**.

Figure 1-73 Select Interface Screen



8 On the Edit Connection screen:

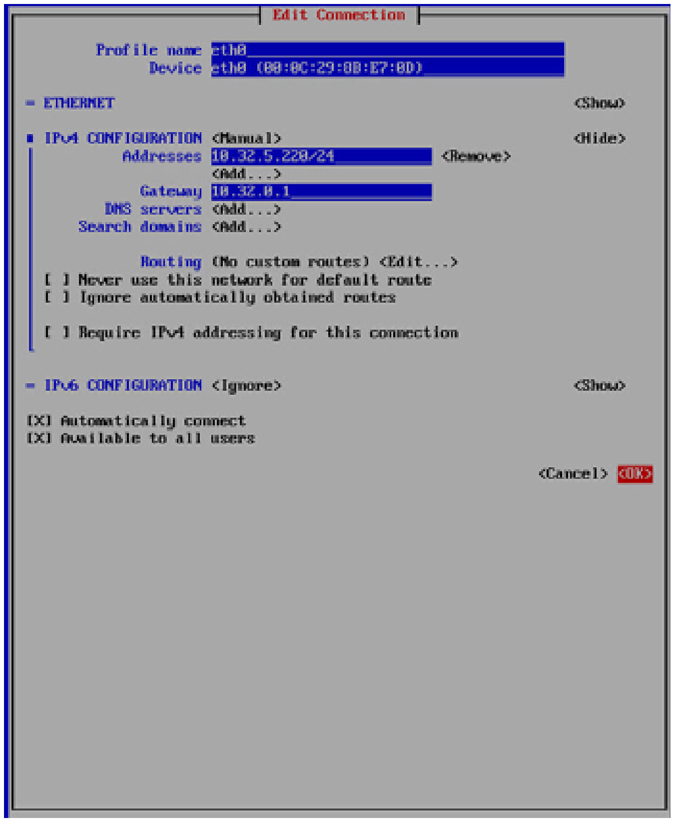
For IPv4 Configuration:

a In the IPv4 Configuration field, select **Manual** for static IP configuration.

b Enter the network configuration information, as needed.

c Click **OK**.

Figure 1-74 IPv4 Network Configuration Screen



For IPv6 Configuration:

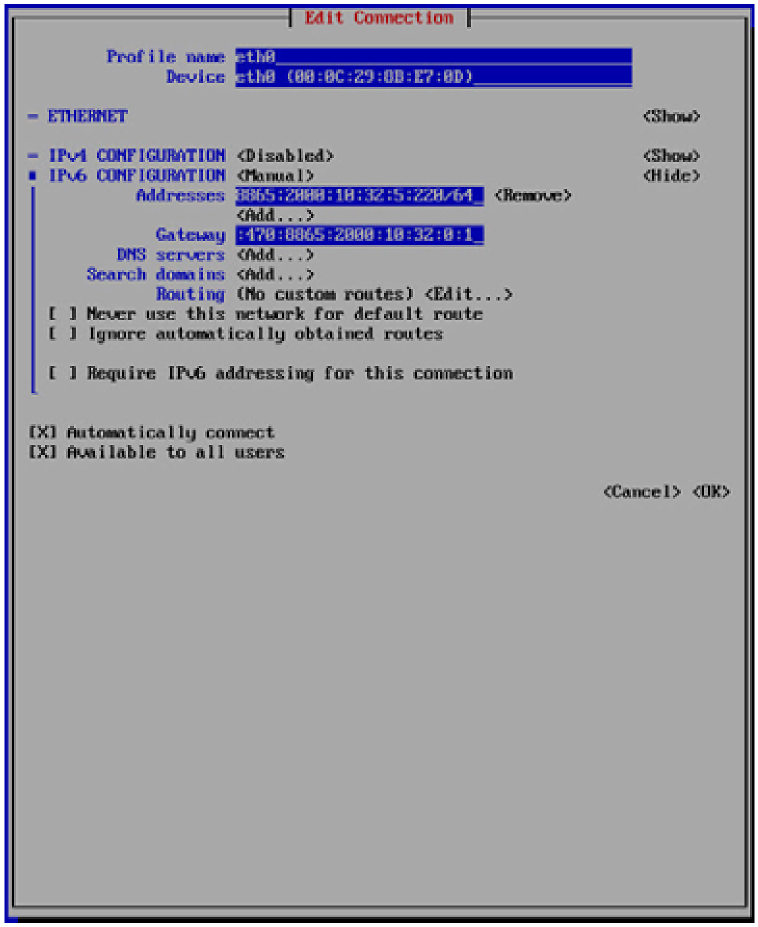
a In the IPv4 Configuration field, select **Disabled**.

b In the IPv6 Configuration field, select **Manual** for static IP configuration.

c Enter the network configuration information, as needed.

d Click **OK**.

Figure 1-75 IPv6 Network Configuration Screen



9 At the NetworkManager TUI screen, click **OK** to exit.

10 When prompted, enter the disk size you want for the VM. The disk size must match the number you configured in Configure Host Size:

```
Default disk size is 200 GB
     New disk range: 250 to 2000 (GB)
     Size value must be multiples of 50
     Or just press “Enter” to skip the resize
New disk size (GB):
Enter New Size:
```

The script adjusts the VM partitions.

11 To verify that the logical volume has been extended as expected, issue the command **`vgdisplay`** as root:

```
[root@emsha-94-a ~]# vgdisplay
  --- Volume group ---
  VG Name               VolGroup00
  System ID
  Format                lvm2
  Metadata Areas        2
  Metadata Sequence No  14
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                10
  Open LV               10
  Max PV                0
  Cur PV                2
  Act PV                2
  VG Size               549.50 GiB
  PE Size               4.00 MiB
  Total PE              140673
  Alloc PE / Size       140673 / 549.50 GiB
  Free  PE / Size       0 / 0
```

12 To verify the extra space added to the logical volume has been allocated correctly to /opt and /data partitions, issue the following command as affirmed or root:

```
[affirmed@emsha-94-a ~]$ df -h -t ext4
Filesystem                            Size  Used Avail Use% Mounted on
/dev/mapper/VolGroup00-lv_root          15G  4.7G  9.3G  34% /
/dev/sda1                              477M  125M  323M  28% /boot
/dev/mapper/VolGroup00-lv_opt          227G   11G  207G   5% /opt
/dev/mapper/VolGroup00-lv_data         229G   13G  207G   6% /data
/dev/mapper/VolGroup00-lv_var          9.8G  319M  8.9G   4% /var
/dev/mapper/VolGroup00-lv_home         976M  2.6M  907M   1% /home
/dev/mapper/VolGroup00-lv_var_tmp     4.8G   20M  4.6G   1% /var/tmp
/dev/mapper/VolGroup00-lv_tmp          9.8G   37M  9.2G   1% /tmp
/dev/mapper/VolGroup00-lv_var_log       25G   90M   24G   1% /var/log
/dev/mapper/VolGroup00-lv_var_log_audit  4.8G  1.6G  3.0G  35%
/var/log/audit
```

13 On the console tab, use a text editor to add server IP and hostname in /etc/hosts:

```
vi /etc/hosts
```

For example:

```
127.0.0.1     localhost    localhost.localdomain    localhost4
localhost4.localdomain4
```

```
::1     localhost    localhost.localdomain    localhost6
localhost6.localdomain6
```

```
10.0.0.10 qaosems05
```

14 Save and Quit out of vi /etc/hosts file.

What’s Next

The system is now ready for Acuitas installation/upgrade. Proceed to the chapter that applies to your system:

- Standalone Systems
- High Availability Systems

OpenStack System

This section describes how to prepare an OpenStack virtual machine, including:

- Prerequisites
- Access the Acuitas VM Console
- Complete the Interface Setup

Prerequisites

This procedure requires you to have OpenStack installed and configured. Using the *Mobile Content Cloud Deployment Guide for RHOSP 13 Director - SR-IOV East/West North/South Deployments*, follow the instructions in these sections:

1 Complete all the instructions in Chapter 2, *Deploying RHOSP 13 Director*.

2 Complete the following instructions in Chapter 3:

- *Uploading the EMS Image:*

a Download the qcow2 image from the Affirmed Networks FTP site and copy it to the Director VM to /tmp.

b Issue the following commands:

```
# ssh root@<Director IP address>
# source /root/keystonerc_admin
```

c Issue the following commands:

```
# glance image-create --name <EMS image name>
--disk-format qcow2 --container-format bare --progress --visibility
public --file /tmp/<EMS image name>
# rm /tmp/<EMS image name>
# glance image-list
```

- *Creating the EMS Flavor:*

  On the OpenStack Controller, issue the following command to create a flavor that specifies the VM size. Once the flavor is created, you can deploy a VM with the OS image that supports the resizing. The command syntax is:

  ```
  nova flavor-create [Name] auto [CPU Size] [Size of the VM] [Memory Size]
  ```

  For example:

  ```
  nova flavor-create AN-EMS auto 32768 450 16
  ```

- *Creating the AN Management Network*
- *Creating AN-Management Ports*
- *Updating AN-Management Ports with Virtual IP* – (Only for Acuitas HA)
- *Creating the AN Management Network*

- *Configuring Compute-Node Zones*

3 Issue the following commands to gather information needed to deploy the VMs:

nova flavor-list

```
+--------------------------------------+--------+-----------+------+-----------+------+-------+-------------+-----------+
| ID                                   | Name   | Memory_MB | Disk | Ephemeral | Swap | VCPUs | RXTX_Factor | Is_Public |
+--------------------------------------+--------+-----------+------+-----------+------+-------+-------------+-----------+
| 9bfa206f-3da6-4a58-ae4f-31290e62450a | AN-MCM | 32768     | 112  | 50        |      | 8     | 1.0         | True      |
| bdbea45a-8e18-4380-a8a4-c861284e4513 | AN-SSM | 57344     | 50   | 0         |      | 18    | 1.0         | True      |
| c037461a-de0b-41f5-b0f9-ea0503f5f928 | AN-CSM | 57344     | 50   | 0         |      | 16    | 1.0         | True      |
+--------------------------------------+--------+-----------+------+-----------+------+-------+-------------+-----------+
```

glance image-list

```
+--------------------------------------+------------------------------+
| ID                                   | Name                         |
+--------------------------------------+------------------------------+
| 46ba4008-2e23-4c94-9e41-c8e050a3c7c4 | an-controller-6.1.3.0-47.REL |
| 0d5c88ce-8a4d-4c7d-a940-763022398598 | an-payload¬-6.1.3.0-47.REL   |
+--------------------------------------+------------------------------+
```

neutron net-list

```
+--------------------------------------+----------+-------------------------------------------------------+
| id                                   | name     | subnets                                               |
+--------------------------------------+----------+-------------------------------------------------------+
| 09128e63-c7f8-4c42-89a7-aa6c90d43bdd | AN-MGMT  | d2dafbb4-e50b-4723-81fa-1af9997daa92 10.32.0.0/20     |
| 973a2553-47fe-4bcb-b5bd-fc50b2ebb72b | EW2      | 1605e80c-ead3-435f-9073-d52015a6926b 166.254.124.0/24 |
| 8cda9d92-e6dd-4a56-8c27-e616b05e18d6 | AN-BASE2 | f0c474d2-e583-45de-9c01-e0e340c9dd99 168.254.124.0/24 |
| 5ccc2ebe-d2d5-4bcf-99c1-01383a01ff62 | EW1      | 3b895e84-6554-4dfa-b2ae-1e37999723f5 166.254.1.0/24   |
| 5cb655ef-c17e-4a8f-92d1-ee961d6336c5 | AN-BASE1 | a73f3dbf-dbd3-403f-9e89-66423bf851f7 168.254.1.0/24   |
| 3f73cb86-6d6c-4407-8311-349bd8e89dd9 | NS1      | ad899679-1d11-4fa5-ba9d-1613afd2c7fa 10.80.1.0/24     |
| 175efd24-26e7-4071-87f8-7c0fbf77b51c | NS2      | 83be9821-c0c3-467a-85f6-12c82c2e1a0d 10.81.1.0/24     |
| 3f73cb86-6d6c-4407-8311-349bd8e89dd5 | NS3      | ad899679-1d11-4fa5-ba9d-1613afd2c7f5 10.82.1.0/24     |
| 175efd24-26e7-4071-87f8-7c0fbf77b515 | NS4      | 83be9821-c0c3-467a-85f6-12c82c2e1a05 10.83.1.0/24     |
+--------------------------------------+----------+-------------------------------------------------------+
```

neutron port-list

```
+--------------------------------------+------------+-------------------+------------------------------------------------------------------------------------------+
| id                                   | name       | mac_address       | fixed_ips
+--------------------------------------+------------+-------------------+------------------------------------------------------------------------------------------+
| 080a3dcf-d7c8-409b-9511-aaac692f52ff | ns4_vf_ssm1 | fa:16:3e:d1:1b:b5 |{"subnet_id":"b51a9c2b-4e19-4f16-9579-aa34a82f7932", "ip_address": "169.254.124.8"} |
| 0e718662-0708-4e39-920d-e32625180d89 | ns2_vf_ssm1 | fa:16:3e:92:c7:94 |{"subnet_id":"1c6488ba-7098-48f1-a7e6-227d6a9358c3", "ip_address": "166.254.124.8"} |
| 1721d98f-7db4-4bd3-9e98-a574399a9632 | ew2_vf_ssm2 | fa:16:3e:40:49:a3 |{"subnet_id":"b51a9c2b-4e19-4f16-9579-aa34a82f7932", "ip_address": "169.254.124.3"} |
| 1fd8b459-b17f-4297-b0e5-15cf512ff7eb | ew2_vf_csm1 | fa:16:3e:3b:8a:8d |{"subnet_id":"b51a9c2b-4e19-4f16-9579-aa34a82f7932", "ip_address": "169.254.124.1"} |
| 23388b02-d7ce-4e93-acab-91fc4faefb01 | ns2_vf_ssm2 | fa:16:3e:66:c9:e7 |{"subnet_id":"1c6488ba-7098-48f1-a7e6-227d6a9358c3", "ip_address": "166.254.124.6"} |
| 3cd06987-ffa9-4bd6-954f-bc8da47a441d | ns1_vf_ssm2 | fa:16:3e:4e:f0:1c |{"subnet_id":"1c6488ba-7098-48f1-a7e6-227d6a9358c3", "ip_address": "166.254.124.5"} |
| 534ca51e-5046-4220-b7ab-09fa857bf787 | ew2_vf_csm2 | fa:16:3e:8d:0d:dd |{"subnet_id":"b51a9c2b-4e19-4f16-9579-aa34a82f7932", "ip_address": "169.254.124.2"} |
| 67dfe1e3-4228-4e7f-a40e-41abf4b62934 | ew1_vf_csm1 | fa:16:3e:c9:dc:3e |{"subnet_id":"1c6488ba-7098-48f1-a7e6-227d6a9358c3", "ip_address": "166.254.124.1"} |
| 892a84f7-51b4-431d-9c74-88fccdc6e937 | ns3_vf_ssm2 | fa:16:3e:33:3d:98 |{"subnet_id":"b51a9c2b-4e19-4f16-9579-aa34a82f7932", "ip_address": "169.254.124.5"} |
| 9164996b-a81b-4d08-b0bf-27288f2cb823 | ew1_vf_ssm1 | fa:16:3e:e9:18:af |{"subnet_id":"1c6488ba-7098-48f1-a7e6-227d6a9358c3", "ip_address": "166.254.124.4"} |
| a3abde34-4b33-45cc-8641-fb665b84818e | ew1_vf_ssm2 | fa:16:3e:0e:fd:2e |{"subnet_id":"1c6488ba-7098-48f1-a7e6-227d6a9358c3", "ip_address": "166.254.124.3"} |
| a62f52a5-cd95-4f23-a70a-fdadaac6295b | ns3_vf_ssm1 | fa:16:3e:85:96:c7 |{"subnet_id":"b51a9c2b-4e19-4f16-9579-aa34a82f7932", "ip_address": "169.254.124.7"} |
| d40f56ff-c557-44b9-803a-6b2f2afd146b | ew1_vf_csm2 | fa:16:3e:7f:5f:4b |{"subnet_id":"1c6488ba-7098-48f1-a7e6-227d6a9358c3", "ip_address": "166.254.124.2"} |
| e4d19ae0-c6d1-4c65-ba14-cc85a1723586 | ew2_vf_ssm1 | fa:16:3e:df:62:61 |{"subnet_id": "b51a9c2b-4e19-4f16-9579-aa34a82f7932", "ip_address": "169.254.124.4"} |
| ecb1f760-3d49-48a4-8c9c-47a168c391e9 | ns1_vf_ssm1 | fa:16:3e:2a:5b:11 |{"subnet_id": "1c6488ba-7098-48f1-a7e6-227d6a9358c3", "ip_address": "166.254.124.7"} |
| fe865a72-418a-4cd3-9ff1-a11ef572f1e9 | ns4_vf_ssm2 | fa:16:3e:05:0b:dd |{"subnet_id": "b51a9c2b-4e19-4f16-9579-aa34a82f7932", "ip_address": "169.254.124.6"} |
+--------------------------------------+------------+-------------------+------------------------------------------------------------------------------------------+
```

nova server-group-list

```
+--------------------------------------+------------------------+--------------------+---------------+---------+
| Id                                   | Name                   | Policies           | Members       | Metadata|
+--------------------------------------+------------------------+--------------------+---------------+---------+
| c17429bb-8686-4b70-b4e1-ca038b5217b6 | mcm-anti-affinity-group | [u'anti-affinity'] | []            |   {}    |
| 8e23c8fe-0ac2-4e33-9b70-ebc1589fe68a | csm-anti-affinity-group | [u'anti-affinity'] | []            |   {}    |
| 7f971e49-aaf8-4310-9bd3-d93cb3524ca5 | ssm-anti-affinity-group | [u'anti-affinity'] | []            |   {}    |
+--------------------------------------+------------------------+--------------------+---------------+---------+
```

4 Issue the following command to deploy EMS (use the information obtained in step 3).

```
nova boot –-image <image ID or name> --flavor <flavor defined for EMS>
--security-groups default --nic port-id=<port ID of AN-MGMT-EMS>
--availability-zone zone-ems <name of the EMS>
```

For example:

```
nova boot –-image 92ef6989-838a-44ed-9246-e82ed491419f --flavor AN-EMS
--security-groups default --nic
port-id=114e8903-733c-4144-b6f7-513006995ded --availability-zone
zone-ems EMS1
```

Access the Acuitas VM Console

To access the Acuitas VM console:

1 Issue the following steps to log into Horizon:

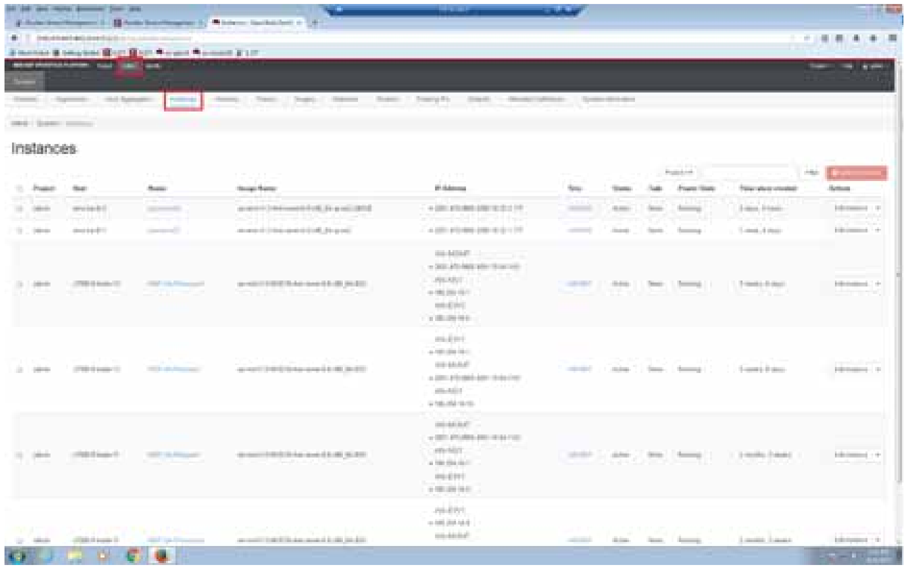
a Issue the following command to determine the admin password, ssh as root to the Controller host:

```
# cat keystonerc_admin
unset OS_SERVICE_TOKEN
export OS_USERNAME=<admin username>
export OS_PASSWORD=<admin password>
export OS_AUTH_URL=http://10.0.0.10:5000/v2.0
export PS1='[\u@\h \W(keystone_admin)]\$ '
export OS_TENANT_NAME=admin
export OS_REGION_NAME=RegionOne
```

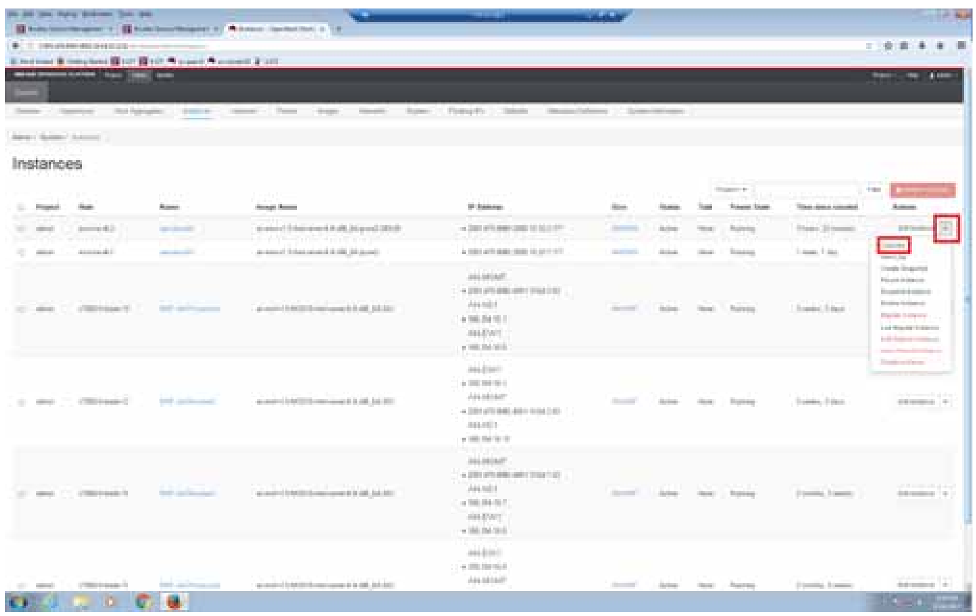
b Use a web browser to point to the Controller host IP address and login into the Horizon using the OS_USERNAME and OS_PASSWORD from above output:

```
http://10.0.0.10/
username = <admin username>
password = <admin password>
```

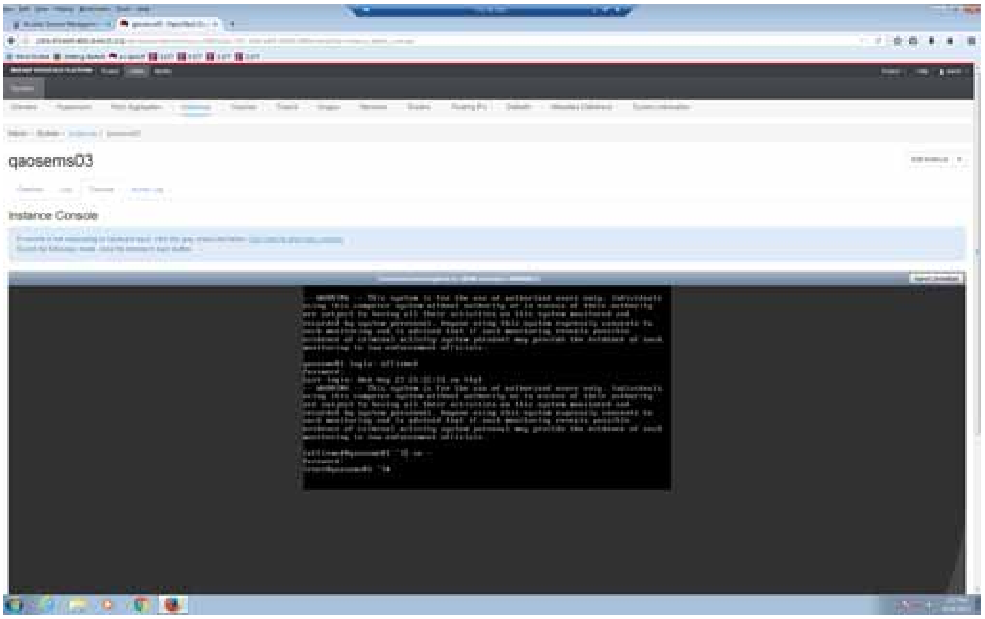
c Go to **Admin** tab > **Instances** tab as shown in the following figure.



d Access the console of the VM by selecting **Console** from the dropdown on the right side as shown in the following figure.



e From the console, login as `affirmed` and switch user to root as shown in the following figure.



2 Proceed to Complete the Interface Setup.

Complete the Interface Setup

To complete the interface setup:

1 Power up the Acuitas VM.

2 From the console, log in as `affirmed` and issue the following command to gain super-user privileges:

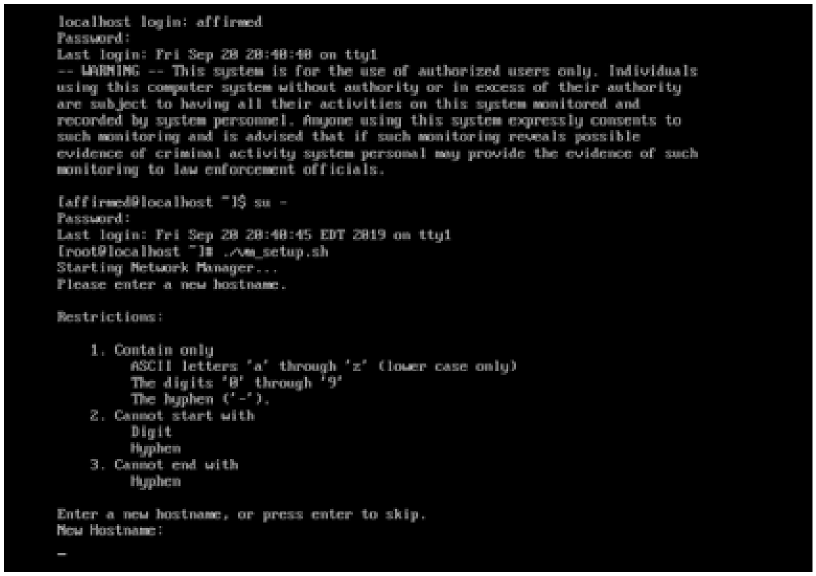
```
su -
```

3 When prompted, enter the super-user password.

4 Issue the following command:

```
/root/ems_setup.sh
```

Figure 1-76   Console Tab



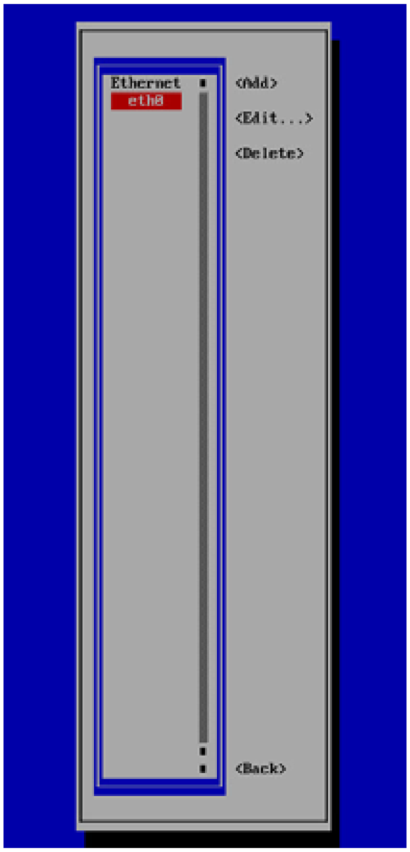
5 Enter a new hostname that adheres to the naming conventions.

6 On the NetworkManager TUI screen, select **Edit a Connection**.

Figure 1-77   NetworkManager TUI Screen

7 On the screen, select the interface to be used and press **Enter**.

Figure 1-78 Select Interface Screen



8 On the Edit Connection screen:

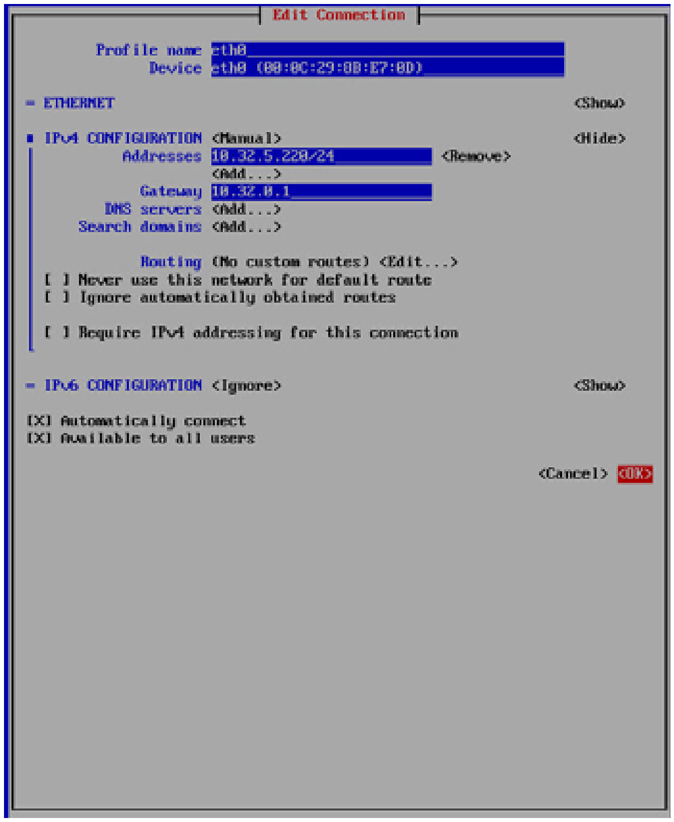
For IPv4 Configuration:

a In the IPv4 Configuration field, select **Manual** for static IP configuration.

b Enter the network configuration information, as needed.

c Click **OK**.

Figure 1-79 IPv4 Network Configuration Screen



For IPv6 Configuration:

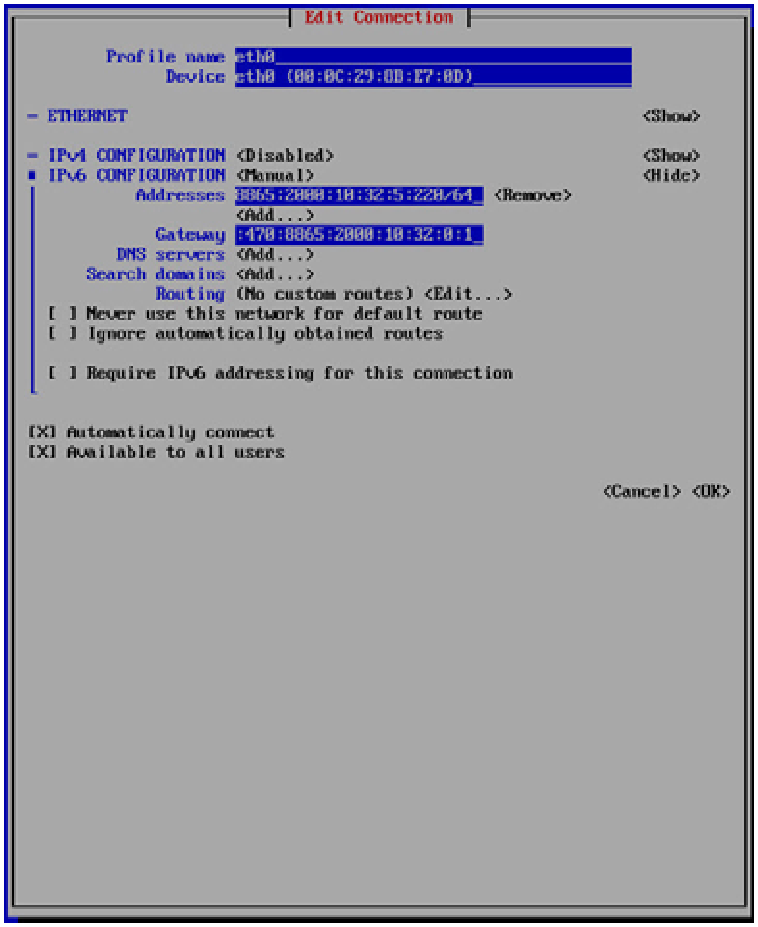
a In the IPv4 Configuration field, select **Disabled**.

b In the IPv6 Configuration field, select **Manual** for static IP configuration.

c Enter the network configuration information, as needed.

d Click **OK**.

Figure 1-80 IPv6 Network Configuration Screen



9 At the NetworkManager TUI screen, click **OK** to exit.

10 When prompted, enter the disk size you want for the VM. The disk size must match the number you configured in Prerequisites, under *Creating EMS Flavor*:

```
Default disk size is 200 GB
     New disk range: 250 to 2000 (GB)
     Size value must be multiples of 50
     Or just press “Enter” to skip the resize
New disk size (GB):
Enter New Size:
```

The script adjusts the VM partitions.

11 To verify that the logical volume has been extended as expected, issue the command **`vgdisplay`** as root:

```
[root@emsha-94-a ~]# vgdisplay
  --- Volume group ---
  VG Name               VolGroup00
  System ID
  Format                lvm2
  Metadata Areas        2
  Metadata Sequence No  14
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                10
  Open LV               10
  Max PV                0
  Cur PV                2
  Act PV                2
  VG Size               549.50 GiB
  PE Size               4.00 MiB
  Total PE              140673
  Alloc PE / Size       140673 / 549.50 GiB
  Free  PE / Size       0 / 0
```

12 To verify the extra space added to the logical volume has been allocated correctly to /opt and /data partitions, issue the following command as affirmed or root:

```
[affirmed@emsha-94-a ~]$ df -h -t ext4
Filesystem                              Size  Used Avail Use% Mounted on
/dev/mapper/VolGroup00-lv_root           15G  4.7G  9.3G  34% /
/dev/sda1                               477M  125M  323M  28% /boot
/dev/mapper/VolGroup00-lv_opt           227G   11G  207G   5% /opt
/dev/mapper/VolGroup00-lv_data          229G   13G  207G   6% /data
/dev/mapper/VolGroup00-lv_var           9.8G  319M  8.9G   4% /var
/dev/mapper/VolGroup00-lv_home          976M  2.6M  907M   1% /home
/dev/mapper/VolGroup00-lv_var_tmp      4.8G   20M  4.6G   1% /var/tmp
/dev/mapper/VolGroup00-lv_tmp           9.8G   37M  9.2G   1% /tmp
/dev/mapper/VolGroup00-lv_var_log       25G   90M   24G   1% /var/log
/dev/mapper/VolGroup00-lv_var_log_audit 4.8G  1.6G  3.0G  35%
/var/log/audit
```

13 On the console tab, use a text editor to add server IP and hostname in /etc/hosts:

```
vi /etc/hosts
```

For example:

```
127.0.0.1     localhost    localhost.localdomain    localhost4
localhost4.localdomain4

::1     localhost    localhost.localdomain    localhost6
localhost6.localdomain6

10.0.0.10 qaosems05
```

14 Save and Quit out of vi /etc/hosts file.

What's Next

The system is now ready for Acuitas installation/upgrade. Proceed to the chapter that applies to your system:

- Standalone Systems
- High Availability Systems

Chapter 2

Standalone Systems

To install/upgrade the Acuitas server and database software, complete the instructions in the section that applies to your system:

- Fresh Installation
- Upgrade
- Standalone Disaster Recovery Systems

Prerequisites

Most Acuitas operation procedures can be performed using the default user account `affirmed`. However, if a user needs root access in the console, follow these procedures:

If you want to remote log into the server, follow this procedure:

To set up remote login:

1 As user `affirmed`, add the IP address of every server you want to access to `/etc/hosts.allow`:

```
ALL : <server IP address 1>, <server IP address 2> ...
```

The syntax for IPv6 `/etc/hosts.allow` is as follows:

```
ALL : [<IP Address>]/<netmask>
```

For example:

```
ALL: [2001:470:8865:2000:10:32:1:147]/64
```

Note: *By default, the hosts.deny file denies all hosts not specified in hosts.allow. To override deny processing, comment out the ALL: ALL line in hosts:deny.*

To gain root access in physical console:

1 From the console, log in using the default user account `affirmed` and the account password.

2 Once you are logged in, issue the following command to gain super-user privileges:

```
su -
```

3 When prompted, enter the super-user password.

Fresh Installation

Installation Instructions

You must download the Acuitas software package from the SFTP server. See Affirmed Networks for instructions on how to access the SFTP server.

Once you download and install the software, you must apply the Acuitas license to activate the software. Obtain the Acuitas license file (License.dat) from Affirmed Networks.

To fresh install the Acuitas software:

1 Log in using the default user account `affirmed` and the account password.

2 Copy the ems package *ems-release*.rpm to /data/affirmed.

3 Once you are logged in, issue the following command to gain super-user privileges:

```
su -
```

4 When prompted, enter the super-user password.

5 Execute the following command:

```
rpm -ivh /data/affirmed/ems-release.rpm
```

where *release* is in the form of *x.x.x.y*.

6 Make sure that NTP configuration, timezone, and MySQL user password have been correctly set. For instructions, see:

- NTP Server Configuration
- Changing Timezone of the Acuitas Server
- Changing MySQL root/Replication User Password

7 Issue the following command to switch back to the `affirmed` user:

```
exit
```

8 SCP the Acuitas license file (License.dat) to affirmed@<*server_ip*>:/data/affirmed.

9 Issue the following command to apply the license:

```
cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/conf/
```

10 For an IPv4 server:

a Follow the instructions in Standalone Acuitas Server Startup.

11 If you have configured an IPv6 address for your Acuitas server, you must perform this step to allow the application to communicate over IPv6:

a Log in using the default user account `affirmed` and the account password.

b Enter the following commands:

```
cd /opt/Affirmed/NMS/bin

./set_ipv6_address

  1  eth0  2001:470:8865:601::5e

  2  eth0  2001:470:8865:601:20c:29ff:fef1:7f89

Select the server IP Address from the list above: [1]
```

c Enter the number of the server IP address you want configured.

d At the confirmation prompt, enter **y**.

e You can now start the server with the configured IPv6 address. For instructions, see Standalone Acuitas Server Startup.

Upgrade

The supported upgrade paths are:

- **Release Prior to 9.4.0.0 => 9.4.0.0 or Later**

  If you are upgrading from a release prior to 9.4.0.0 to release 9.4.0.0 or later, the upgrade requires a major upgrade to the OS. Follow the instructions in OS Upgrade Considerations.

- **Release 9.4.0.0 => 9.4.x.x or later**

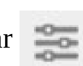
  If you are upgrading from a 9.4.0.0 to 9.4.x.x or later, follow the instructions in Upgrading from 9.4.0.0 to 9.4.x.x or Later.

Upgrading from 9.4.0.0 to 9.4.x.x or Later

This section describes how to upgrade the Acuitas server software and other required software from 9.4.0.0 to 9.4.x.x or later.

To upgrade the software and database to Release 9.4.x.x or later:

1 Trigger a database backup:

   a Click the **Settings** icon on the toolbar  and select **Tools =>**

   Database Administration.

   Acuitas displays the **Database Backup and Restore** window.

   b Click the **User Triggered DB-Backup** icon. 

   c Configure the parameters as follows:

   – **Backup Files Prefix Name** – Specify the prefix you want for the backup file, such as DB_backup.

   – **FTP Server** – Specify whether you want the backup stored on the local or remote server. It is recommended that you store the backup on the remote server, as a local backup will not be available if the system becomes out of service.

   – **Relative Path in FTP Server** – If you select to have the backup stored on remote server, specify the path where you want it stored. If you select to have the backup stored on the local server, it is stored at: */opt/Affirmed/NMS/server/databaseBackup*.

   – **Full Backup** – Mark this checkbox to trigger a full backup.

   d Click **Save**.

   e At the confirmation prompt, click **OK**.

   f Click **Close**.

2. Move the backup to a remote file server in case you later need to restore the database.
3. Upgrade to Acuitas 9.4.0.0 or later:

   Copy the ems package *ems-release*.rpm to /data/affirmed.

   ```
   ./emsupgrade /data/affirmed/ems-release.rpm
   ```

   where *release* is in the form of *x.x.x.y*.

   ```
   Do you want to upgrade the server on localhost using
   /opt/ems-x.x.x.y.rpm (Y/N)?
   ```
4. Enter **y**. The upgrade can take some time to complete.
5. Clear the browser cache before logging into Acuitas.
6. Log into Acuitas and verify that the Acuitas application is running the new version:

   a. Click the **About** icon on the toolbar. 

      Acuitas displays a window listing the version number, release date, and copyright information.

   b. Click **Close** to close the window.

   To start the server, proceed to Standalone Acuitas Server Startup.

Install Acuitas Packages for VMware

Following the upgrade of the Acuitas server, you must install Acuitas packages for VMware. Installation is required only when the MCC's are deployed by the Acuitas VNF Manager with VMware. Even if you had previously installed the packages, you need to reinstall them.

To install Acuitas packages for VMware:

1. Log into the Acuitas server as `root`.
2. Issue the following commands to untar the packages tarball:

   ```
   cd /opt/Affirmed/NMS/ctool
   ```

   ```
   tar -zxvf ./acuitas-commission-tool.tar.gz
   ```
3. Issue the following commands to install the packages:

   ```
   cd acuitas-commission-tool
   ```

   ```
   ./install.sh
   ```
4. Accept the VMware license agreement.
5. When prompted to install the RedHat pre-built packages, enter **yes**.
6. Issue the following command to remove the acuitas-commission-tool directory created by the tar command:

   ```
   cd ..; rm –rf acuitas-commission-tool
   ```

Software Downgrade Procedure

If desired, you can revert the Acuitas software back to an older version. The procedure requires you to specify an .rpm file to use for the downgrade.

Note: *If the backup file is not from the same release, incompatibility issues can result. Make sure that the backup file version and downgrade version match.*

To downgrade Acuitas software:

1 Copy the ems package *ems-release*.rpm to /data/affirmed.

2 Issue the following command:

```
rpm -Uvh /data/affirmed/ems-release.rpm --force
```

where *release* is in the form of *x.x.x.y*. *(Note: include path of the ems-release*.rpm*)*

The upgrade can take some time to complete.

To rollback the database, proceed to Database Rollback Procedure.

Database Rollback Procedure

If desired, you can roll back the Acuitas database to a successful backed up version.

CAUTION: Database restore is service affecting. During the process, the Acuitas server is brought down.

To restore the database from a backup:

1 Click the **Settings** icon on the toolbar  and select **Tools** =>

Database Administration.

Acuitas displays the **Database Backup and Restore** window.

2 Use any of the following methods to restore the database to a backup:

Toolbar: Select the backup and click the **Restore Database** icon. 

Pop-up menu: Right-click the backup and click **Restore Database**.

3 At the confirmation prompt, click **OK**.

4 Click **Close** to close the window.

> ***Note:*** *If the restore operation fails, Acuitas displays the Database Restore Operation window. Click Yes to restore the database from the local machine. Click No to restore an existing database.*

---

5 Determine whether the restoration process has finished. To do so, issue the following command from the /opt/Affirmed/NMS/dbrestore directory:

```
[root@qahpems02 dbrestore]# ls -lrt
```

If the command output indicates that the directory contains the sql file, the process *is not* complete.

In the following sample command output, the sql file is shown in red text.

```
total 135096

-rw-r--r-- 1 root root       600 Oct  5 14:19 log4j.xml

drwxr-xr-x 2 root root      4096 Oct 11 10:47 bin

drwxr-xr-x 2 root root      4096 Oct 11 10:47 lib

-rw-r--r-- 1 root root       103 Oct 11 15:41 restore.txt

-rw-r--r-- 1 root root 138315816 Oct 11 16:31
http3_10-11-2012-16-30-03.sql

-rw-r--r-- 1 root root      2038 Oct 11 16:31 restore.log
```

If the command output indicates that the directory no longer contains the sql file and contains the restore.txt file and the restore.log file, the restore is successful.

In the following sample command output, the restore.txt and the restore.log files are shown in red text.

```
total 20

-rw-r--r-- 1 root root  600 Oct  5 14:19 log4j.xml

drwxr-xr-x 2 root root 4096 Oct 11 10:47 bin

drwxr-xr-x 2 root root 4096 Oct 11 10:47 lib

-rw-r--r-- 1 root root  103 Oct 11 16:38 restore.txt

-rw-r--r-- 1 root root 2194 Oct 11 16:38 restore.log
```

6 Once the restoration process has finished, you must manually start the server using starttems.sh with the -u option.

Standalone Disaster Recovery Systems

Overview

The Disaster Recovery feature provides geo-redundant support of the Acuitas system. In the event of a disaster at the physical location of the master system, the slave system is available to become the Acuitas system to provide management support while the master system is out of service.

Disaster Recovery systems are usually *separately* located and require *user intervention* to convert the slave system to master when necessary.

In a Standalone Disaster Recovery system:

- Only a standalone node can serve as the master system. No special configuration of the master system is required.
- Only a standalone node can be deployed as the slave system. That slave system *is not* running as an Acuitas server.

Use the configure_dr script to perform Disaster Recovery required operations. You run the script in the following scenarios:

- **Configuration of Disaster Recovery system** – Run the configure_dr script to set up MySQL slave replication on the Disaster Recovery slave system. Once the script completes, the system is ready to replicate the master database on the master system to the database on the slave system.
- **Configuration when Disaster takes the master system out of service** – When a disaster places the master system out of service, run the configure_dr script to convert the slave system to the standalone master system.
- **Configuration once the Disaster situation has been resolved** – Once the master system is back in service, run the configure_dr script to resume the Disaster Recovery system as it was before the disaster. That is, set the original server to the master role and reset the standalone system to the slave role.
- **Forcibly convert the slave to master** – Run the configure_dr script to forcibly convert the slave system to the standalone master system.
- **Upgrading Disaster Recovery systems** – After upgrading the master and slave machines, run the configure_dr script to set up MySQL slave replication on the Disaster Recovery slave system.
- **Check system status** – Run the **emsstatus** command to check whether the current Acuitas instance is a standalone system or Disaster Recovery slave system.

For configure_dr command syntax, see Acuitas Disaster Recovery Configuration.

Prerequisites

Before you configure the Standalone Disaster Recovery systems, follow this procedure to allow SSH access between the systems during configuration.

To configure SSH access for the affirmed user:

Perform these steps on the machine that will serve as the master server:

1. Log in using the default user account `affirmed` and the account password and issue the following command:

   `#ssh-keygen –t rsa`

2. Press **Enter** for each of the following prompts:

   ```
   Enter file in which to save the key (/home/affirmed/.ssh/id_rsa):
   Enter passphrase (empty for no passphrase):
   Enter same passphrase again:
   ```

3. Issue the following command:

   `ssh-copy-id -i ~/.ssh/id_rsa.pub affirmed@<slave server IP address>`

4. When prompted, enter `affirmed` user password.

Perform these steps on the machine that will serve as the slave server:

1. Log in using the default user account `affirmed` and the account password and issue the following command:

   `#ssh-keygen –t rsa`

2. Press **Enter** for each of the following prompts:

   ```
   Enter file in which to save the key (/home/affirmed/.ssh/id_rsa):
   Enter passphrase (empty for no passphrase):
   Enter same passphrase again:
   ```

3. Issue the following command:

   `ssh-copy-id -i ~/.ssh/id_rsa.pub affirmed@<master server IP address>`

Note: *If the host keys are later changed, you must perform this procedure again to allow SSH access between the systems.*

---

4. When prompted, enter `affirmed` user password.

Fresh Installation and Configuration of Disaster Recovery Slave System

Run the configure_dr script to set up MySQL slave replication on the Disaster Recovery slave system, and maintain the various roles that the slave system must be in during the various maintenance scenerios. Once the script completes, the system is ready to replicate the master database to the database on the slave system.

During this process, the script takes a backup of the master host MySQL databases and imports the database backup into the local MySQL server. These steps can take considerable time, depending on the amount of data in the database.

To configure the slave system:

1. Log in using the default user account `affirmed` and the account password.
2. Once you are logged in, issue the following command to gain super-user privileges:

   `su -`

3. When prompted, enter the super-user password.
4. Copy the ems package *ems-release*.rpm to /data/affirmed.
5. Execute the following command:

   `rpm -ivh /data/affirmed/ems-`*`release`*`.rpm` *(Note: include path of the ems-release*__*.rpm*__**)**

   where *release* is in the form of *x.x.x.y*.

   If the installation fails with an error indicating that the MySQL process *is not* running, follow this procedure:

   a Issue the following command to verify that MySQL is running:

   `systemctl status mysqld`

   ```
   [affirmed@qaems149b bin]$ systemctl status mysqld
   \u25cf mysqld.service - MySQL Server
     Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; vendor preset: disabled)
      Active: active (running) since Tue 2017-10-24 09:15:40 EDT; 2 days ago
        Docs: man:mysqld(8)
              http://dev.mysql.com/doc/refman/en/using-systemd.html
    Main PID: 26663 (mysqld)
      CGroup: /system.slice/mysqld.service
              \u2514\u250026663 /usr/sbin/mysqld --daemonize --pid-file=/var/run/mysqld/mysqld.pid
   ```

   b If the command *does not* indicate active (running), issue the following command to start MySQL:

```
systemctl start mysqld
```

c If MySQL does not start, contact the Affirmed Networks Technical Assistance Center (TAC) for assistance.

6 Make sure that NTP configuration, timezone, and MySQL user password have been correctly set. For instructions, see:

- NTP Server Configuration
- Changing Timezone of the Acuitas Server
- Changing MySQL root/Replication User Password

7 Issue the following command to switch back to the `affirmed` user:

```
exit
```

8 On the slave system, SCP the Acuitas license file (License.dat) to affirmed@<*server_ip*>:/data/affirmed.

9 On the slave system, issue the following command to apply the license:

```
cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/conf/
```

10 On the slave system, use the following syntax:

```
configure_dr convert slave --masterhost <masterhost>
```

where *masterhost* specifies the master hostname or IP address.

When the script completes, you can check the log file for the status.

11 Run the **emsstatus** command to verify that the server is in the expected working state:

```
cd /opt/Affirmed/NMS/bin
./emsstatus
```

```
*********** Acuitas Server Status *************
Node                         : <server name> - <server IP address>
EMS Package                  : AffirmedEms-9.4.0.0-97

EMS Process                  : Stopped
MySQL
    Database               : Running
    Server Id                : 2

Disaster Recovery
    Status                   : Standby
    Master Host              : <master server IP address>
```

```
    Standby IO          : Yes
    Standby SQL         : Yes
***********************************************
```

The output indicates that the node is a DR standby server and lists its server ID and the IP address of the DR master.

12 For an IPv4 server:

a Follow the instructions in Standalone Acuitas Server Startup.

13 If you have configured an IPv6 address for your Acuitas server, you must perform this step to allow the application to communicate over IPv6:

a Log in using the default user account `affirmed` and the account password.

b Enter the following commands:

```
cd /opt/Affirmed/NMS/bin
./set_ipv6_address
  1  eth0  2001:470:8865:601::5e
  2  eth0  2001:470:8865:601:20c:29ff:fef1:7f89
Select the server IP Address from the list above: [1]
```

c Enter the number of the server IP address you want configured.

d At the confirmation prompt, enter **y**.

e You can now start the server with the configured IPv6 address. For instructions, see Standalone Acuitas Server Startup.

Configuration when Master System Out of Service

If a disaster places the master system out of service, run the configure_dr script to convert the slave system to the standalone master system.

To convert the slave system to a master system:

1. On the slave system, use the following syntax:

   `configure_dr convert master`

   When the script completes, you can check the log file for the status.

2. On the new master, run the **emsstatus** command to verify that the server is in the expected working state:

   `cd /opt/Affirmed/NMS/bin`

   `./emsstatus`

```
*********** Acuitas Server Status *************
Node                      : <server name> - <server IP address>
EMS Package               : AffirmedEms-9.4.0.0-97


EMS Process               : Running [PID]
MySQL Database            : Running
    Server Id             : 2
***********************************************
```

Configuration when Master System Back In Service

Once the disaster situation has been resolved, run the configure_dr script on both systems to resume the Disaster Recovery system as it was before the disaster. That is, set the original server to the master role and reset the standalone system to the slave system.

Summary

You execute commands on the master system and the slave system in the order indicated in the following table:

	Master System	Slave System
1	Replicate the database on the standalone system to the database on the master system: **`configure_dr convert slave --masterhost <masterhost>`** where ***`masterhost`*** is the original slave system.	
2		Stop the server and check the status: **`cd /opt/Affirmed/NMS/bin`** **`./emsstop`** **`./emsstatus`**
3	Convert the master system to the master: **`configure_dr convert master`**	
4	Check the status: **`cd /opt/Affirmed/NMS/bin`** **`./emsstatus`**	
5		Set up MySQL slave replication: **`configure_dr convert slave`**
6		Check the status: **`cd /opt/Affirmed/NMS/bin`** **`./emsstatus`**

These steps are detailed in the following procedure.

Procedure

To resume the Disaster Recovery system:

1. Log in using the default user account `affirmed` and the account password.
2. On the master system, use the following syntax to replicate the database on the standalone system to the database on the master system:

cd /opt/Affirmed/NMS/bin

configure_dr convert slave --masterhost *<masterhost>*

where *masterhost* specifies the hostname or IP address of the original slave system.

When the script completes, the system returns the following messages:

```
Taking a backup of masterhost MySQL databases ...

Importing masterhost database and starting replication ...

Exiting DR configuration script
```

3 On the slave system, issue the following commands to stop the server and check the status.

cd /opt/Affirmed/NMS/bin

./emsstop

./emsstatus

```
*********** Acuitas Server Status *************

Node                     : <server name> - <server IP address>

EMS Package              : AffirmedEms-9.4.0.0-97


EMS Process              : Stopped

MySQL Database           : Running

    Server Id            : 2
```

4 On the master system, use the following syntax to convert the system to the master:

cd /opt/Affirmed/NMS/bin

configure_dr convert master

When the script completes, the system returns the following message:

```
Exiting DR configuration script
```

5 On the master system, issue the following commands to check the status.

cd /opt/Affirmed/NMS/bin

./emsstatus

```
*********** Acuitas Server Status *************

Node                    : <server name> - <server IP address>

EMS Package             : AffirmedEms-9.4.0.0-97


EMS Process             : Running [PID]

MySQL Database          : Running
```

```
    Server Id           : 1
***********************************************
```

6 On the slave system, use the following syntax to set up MySQL slave replication:

`cd /opt/Affirmed/NMS/bin`

`configure_dr convert slave`

When the script completes, the system returns the following messages:

```
Taking a backup of masterhost MySQL databases ...

Importing masterhost database and starting replication ...

Exiting DR configuration script
```

7 On the slave system, issue the following commands to check the status.

`cd /opt/Affirmed/NMS/bin`

`./emsstatus`

```
*********** Acuitas Server Status *************
Node                     : <server name> - <server IP address>
EMS Package              : AffirmedEms-9.4.0.0-97

EMS Process              : Stopped
MySQL Database           : Running
    Server Id            : 2

Disaster Recovery
    Status               : Standby
    Master Host          : <master server IP address>
    Standby IO           : Yes
    Standby SQL          : Yes
***********************************************
```

Upgrade of Disaster Recovery Systems

This section describes how to upgrade the Disaster Recovery master and slave machines.

Prerequisite to Backup the Acuitas Server Before Upgrading

If you are upgrading from a release prior to 9.4.0.0 to release 9.4.0.0 or later, that upgrade requires a major upgrade to the OS. Follow the instructions in OS Upgrade Considerations.

Summary

To upgrade Disaster Recovery, you execute commands on the master and the slave in the order indicated in the following table:

	Master	Slave
1	Create a database backup: **`cd /opt/Affirmed/NMS/bin`** **`./emsdb backup --backupfile <file_path>`**	
2		On the server, copy the ems package ems-*new_pkg_name*.rpm to /data/affirmed. Upgrade Acuitas: **`rpm -Uvh /data/affirmed/<new_pkg_name>.rpm`**
3	On the server, copy the ems package ems-*new_pkg_name*.rpm to /data/affirmed. Upgrade Acuitas: **`rpm -Uvh /data/affirmed/<new_pkg_name>.rpm`**	
4		Set up MySQL slave replication: **`configure_dr convert slave`**
5	**`cd /opt/Affirmed/NMS/bin`** **`./emsstatus`**	
6		**`cd /opt/Affirmed/NMS/bin`** **`./emsstatus`**

These steps are detailed in the following procedure.

Procedure

To upgrade the master and slave machines:

1 On the master server, create a database backup:

```
cd /opt/Affirmed/NMS/bin
./emsdb backup --backupfile <file_path>
```

2 On the slave, issue the following command to upgrade Acuitas:

```
rpm –Uvh /data/affirmed/<new_pkg_name>.rpm
```

where *new_pkg_name* is the new Acuitas package.

3 On the master, issue the following command to upgrade and start Acuitas:

```
rpm –Uvh /data/affirmed/<new_pkg_name>.rpm
```

where *new_pkg_name* is the new Acuitas package.

4 On the slave, issue the following command to set up MySQL slave replication:

```
configure_dr convert slave
```

When the script completes, the system returns the following message:

```
Exiting DR configuration script
```

5 On the master system, issue the following commands to check the status.

```
cd /opt/Affirmed/NMS/bin
./emsstatus
*********** Acuitas Server Status *************
Node                     : <server name> - <server IP address>
EMS Package              : AffirmedEms-9.4.0.0-97
EMS Process              : Running [PID]
MySQL Database           : Running
    Server Id            : 1
***********************************************
```

6 On the slave system, issue the following commands to check the status.

```
cd /opt/Affirmed/NMS/bin
./emsstatus
*********** Acuitas Server Status *************
Node                      : <server name> - <server IP address
EMS Package               : AffirmedEms-9.4.0.0-97


EMS Process               : Stopped
```

```
MySQL Database          : Running
    Server Id           : 2

Disaster Recovery
    Status              : Standby
    Master Host         : <master server IP address>
    Standby IO          : Yes
    Standby SQL         : Yes
***********************************************
```

Recovery of a Standalone System

This section provides the recovery procedure for a standalone system. This procedure employs an existing database backup file to perform a point-in-time recovery. Any management information between the time of the backup file and the time of recovery will be lost.

Prerequisite: Scheduling Database Backup

The recovery requires that you have a scheduled database backup file stored on a remote server and the database backup is taken from Acuitas version 5.3.8 or later.

Configure the Acuitas server to backup its database based on a pre-configured schedule. *Do not* schedule backups more than twice a day, as it can impact system performance (backups cause pauses in Acuitas system and affect performance/scalability of the system).

Note: *To schedule a database backup, you must have the Tools > Database Administration > Backup > Schedule Backup security group permission. For instructions, see the Acuitas User's Guide.*

To schedule a database backup:

1. Click the **Settings** icon on the toolbar  and select **Tools =>**

   Database Administration.

   Acuitas displays the **Database Backup and Restore** window.

2. Click the **Scheduled DB-Backup** icon. 
3. Configure the parameters as follows:
   - **Backup Files Prefix Name** – Specify the prefix you want for the backup file, such as DB_backup.
   - **FTP Server** – Specify to have the backup stored on a remote server. *Do not* specify to have the backup stored on the local server, as is will not be available if the entire cluster becomes out of service.
   - **Relative Path in FTP Server** – Specify the path where you want the backup stored.
   - **Full Backup** – Mark this checkbox to trigger a full backup.
   - **Recurrence** – Mark this checkbox to specify a backup recurrence (*do not* schedule more than twice a day, as it can impact system performance):
     – Click the **Recurrence Pattern** icon and configure Recurrence Details.

  – Click the **Recurrence**.

- **Purge Backup** – Mark this checkbox to specify how many backups to retain.

4 Click **Save**.

Recovery Procedure

To recover the standalone system:

1 Perform a fresh installation of the Acuitas software on the standalone system. Follow the instructions in Fresh Installation.

2 Issue the following command to source the database:

```
mysql -e "reset master"
mysqladmin drop AFFIRMED_NMS_DB
mysqladmin create AFFIRMED_NMS_DB
mysql -f AFFIRMED_NMS_DB < <pre-9.4-backupfile>
```

3 Use starttems.sh to start the Acuitas server:

```
cd /opt/Affirmed/NMS/bin
./starttems.sh -u
```

Chapter 3

High Availability Systems

Acuitas offers High Availability (HA) configurations with one Acuitas server and one warm-standby Acuitas server. HA systems are usually *co-located* and allow either *automatic* or *user intervention* switchover from the active to the standby system in case of hardware and/or software failure in the active system.

One server is configured to serve as the primary server; the other is configured to serve as the secondary server. These roles persist across upgrades.

By contrast, the runtime roles of the servers can vary:

- Active server — the server that owns the virtual IP address of the cluster and has the full Acuitas server running.
- Standby server — the server in warm-standby mode that can take over in the event that the active server becomes out of service.

Possible triggers that can cause an automatic failover in the HA system include:

- The active Acuitas server is shut down for some reason, such as because the power supply failed (for example, the rack lost power).
- On the active system, the Acuitas java process hangs, crashes, or otherwise becomes unresponsive.
- The active Acuitas server is taken off the network and cannot be reached by the standby (at which point the standby becomes active).

In addition, you can trigger a manual switchover. For example, to perform maintenance operations on the active Acuitas server (such as, upgrading Acuitas software), manually trigger switchover to standby, then bring down the active Acuitas system.

To configure HA systems, use the configure_ha script. For configure_ha command syntax, see Acuitas High Availability Configuration.

To install/upgrade the Acuitas server and database software, complete the instructions in the section that applies to your system:

- HA Server Fresh Installation and Configuration
- HA System Upgrade
- High Availability + Disaster Recover Systems

Requirement: Use Physical IP Address for HA Administration

Only log into Acuitas using the physical IP address for any HA administration tasks. Never use the virtual IP address for HA administration tasks, as it can cause potential failure in the tasks.

Prerequisites

If you want to remote log into the server, follow this procedure:

To set up remote login:

1. As root, add the IP address of every server you want to access to `/etc/hosts.allow`. At a minimum, add the addresses of the primary and secondary servers to allow access to each other.

   ```
   ALL : <server IP address 1>, <server IP address 2> ...
   ```

   The syntax for IPv6 `/etc/hosts.allow` is as follows:

   ```
   ALL : [<IP Address>]/<netmask>
   ```

   For example:

   ```
   ALL: [2001:470:8865:2000:10:32:1:147]/64
   ```

To gain root access in physical console:

1. From the console, log in using the default user account `affirmed` and the account password.
2. Once you are logged in, issue the following command to gain super-user privileges:

   ```
   su -
   ```

3. When prompted, enter the super-user password.

SSH Access Between Systems

Before you configure the Acuitas HA servers, follow this procedure to allow SSH access between the systems during configuration.

Primary Server

Perform these steps on the primary server.

To configure SSH access for the affirmed user:

1 Log in using the default user account `affirmed` and the account password and issue the following command:

```
#ssh-keygen –t rsa
```

2 Press **Enter** at the following prompt:

```
Enter file in which to save the key (/home/affirmed/.ssh/id_rsa):
```

3 If prompted to overwrite the existing key file, enter **y**:

```
/home/affirmed/.ssh/id_rsa already exists.
Overwrite (y/n)?
```

4 Press **Enter** for each of the following prompts:

```
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
```

5 Issue the following command:

```
ssh-copy-id -i ~/.ssh/id_rsa.pub affirmed@<secondary server IP address>
```

6 When prompted, enter the `affirmed` user password.

Secondary Server

Perform these steps on the secondary server.

To configure SSH access for the affirmed user:

1 Log in using the default user account `affirmed` and the account password and issue the following command:

```
#ssh-keygen –t rsa
```

2 Press **Enter** at the following prompt:

```
Enter file in which to save the key (/home/affirmed/.ssh/id_rsa):
```

3 If prompted to overwrite the existing key file, enter **y**:

```
/home/affirmed/.ssh/id_rsa already exists.
Overwrite (y/n)?
```

4 Press **Enter** for each of the following prompts:

```
Enter passphrase (empty for no passphrase):
```

```
Enter same passphrase again:
```

5 Issue the following command:

`ssh-copy-id -i ~/.ssh/id_rsa.pub affirmed@<primary server IP address>`

6 When prompted, enter the `affirmed` user password.

Note: *If the host keys are later changed, you must perform this procedure again to allow SSH access between the systems.*

HA Server Fresh Installation and Configuration

Affirmed Networks supports Acuitas server installation/upgrade on an Affirmed Networks supported system.

You must download the Acuitas software package from the SFTP server. See Affirmed Networks for instructions on how to access the SFTP server.

Once you download and install the software, you must apply the Acuitas license to activate the software. Obtain the Acuitas license file (License.dat) from Affirmed Networks.

Summary

To configure Acuitas HA servers, you execute commands on the primary server and the secondary server in the order indicated in the following table:

	Primary Server	Secondary Server
1	**a** On the server, copy the ems package `emsha-`***`release`***.rpm to /data/affirmed. **b** Log in using the default user account `affirmed` and the account password. **c** Once you are logged in, issue the following command to gain super-user privileges: su - **d** When prompted, enter the super-user password. **e** Install HA software: **`rpm -ivh /data/affirmed/emsha-`*`release`*`.rpm`** **f** Issue the following command to switch back to the `affirmed` user: **`exit`**	**a** On the server, copy the ems package `emsha-`***`release`***.rpm to /data/affirmed. **b** Log in using the default user account `affirmed` and the account password. **c** Once you are logged in, issue the following command to gain super-user privileges: su - **d** When prompted, enter the super-user password. **e** Install HA software: **`rpm -ivh /data/affirmed/emsha-`*`release`*`.rpm`** **f** Issue the following command to switch back to the `affirmed` user: **`exit`**
2	SCP the Acuitas license file (License.dat) to affirmed@<***server_ip***>:/data/affirmed. Issue the following command to apply the license: **`cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/conf/`**	SCP the Acuitas license file (License.dat) to affirmed@<***server_ip***>:/data/affirmed. Issue the following command to apply the license: **`cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/conf/`**

	Primary Server	Secondary Server
3	Make sure that NTP configuration, timezone, and MySQL user password have been correctly set. For instructions, see: ■ NTP Server Configuration ■ Changing Timezone of the Acuitas Server ■ Changing MySQL root/Replication User Password	Make sure that NTP configuration, timezone, and MySQL user password have been correctly set. For instructions, see: ■ NTP Server Configuration ■ Changing Timezone of the Acuitas Server ■ Changing MySQL root/Replication User Password
4	Start the HA configuration program: **`cd /opt/Affirmed/NMS/bin`** **`./configure_ha [--virtualhostname`** ***`virtual hostname`*** **`] [--virtualip`** ***`virtual IP address`*** **`] [--primaryhostname`** ***`primary hostname`*** **`] [--primaryip`** ***`primary IP address`*** **`] [--secondaryhostname`** ***`secondary hostname`*** **`] [--secondaryip`** ***`secondary IP address`*** **`] [--netmask`** ***`netmask value`*** **`] [--ipinterface`** ***`interface`*** **`] [--backupfile`** ***`file_path`*** **`]`**	
5	Check HA status: **`cd /opt/Affirmed/NMS/bin`** **`./emsstatus`**	Check HA status: **`cd /opt/Affirmed/NMS/bin`** **`./emsstatus`**

The configure_ha script runs in the background. The script prompts for input and then displays the message:

```
The command $COMMAND is running in the background.
Check log file $LOG_FILE for the command result.
```

When you use the --client option, the process runs in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

Procedure

To configure the Acuitas HA servers:

1. On the primary server, copy the ems package emsha-*release* and License.dat to /data/affirmed.
2. On the primary server, log in using the default user account `affirmed` and the account password.
3. Once you are logged in, issue the following command to gain super-user privileges:

   ```
   su -
   ```

   When prompted, enter the super-user password.
4. Install the HA software on the primary server:

   ```
   rpm -ivh /data/affirmed/emsha-release.rpm
   ```

   where *release* is in the form of *x.x.x.y*.

   The system checks available disk space and MySQL version and proceeds with the installation.

   ```
   Installation is done - start the server once the license is installed.
   ```

   If the installation fails with an error indicating that the MySQL process *is not* running, follow this procedure:

   a Issue the following command to verify that MySQL is running:

   ```
   systemctl status mysqld
   ```

   ```
   [root@<server name> log]# systemctl status mysqld
   \u25cf mysqld.service - MySQL Server
     Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled;
   vendor preset: disabled)
      Active: active (running) since Tue 2017-10-24 09:15:40 EDT; 2
   days ago
       Docs: man:mysqld(8)
               http://dev.mysql.com/doc/refman/en/using-systemd.html
    Main PID: 26663 (mysqld)
      CGroup: /system.slice/mysqld.service
               \u2514\u250026663 /usr/sbin/mysqld --daemonize
   --pid-file=/var/run/mysqld/mysqld.pid
   ```

   b If the command *does not* indicate `active (running)`, issue the following command to start MySQL:

   ```
   systemctl start mysqld
   ```

   c If MySQL does not start, contact the Affirmed Networks Technical Assistance Center (TAC) for assistance.
5. Issue the following command to switch back to the `affirmed` user:

```
exit
```

6 On the secondary server, copy the ems package emsha-*release* to /data/affirmed.

7 On the secondary server, log in using the default user account `affirmed` and the account password.

8 Once you are logged in, issue the following command to gain super-user privileges:

```
su -
```

When prompted, enter the super-user password.

9 Install the HA software on the secondary server:

```
rpm -ivh /data/affirmed/emsha-release.rpm
```

where *release* is in the form of *x.x.x.y*.

The system checks available disk space and MySQL version and proceeds with the installation.

```
Installation is done - start the server once the license is installed.
```

10 Issue the following command to switch back to the `affirmed` user:

```
exit
```

11 Make sure that NTP configuration, timezone, and MySQL user password have been correctly set. For instructions, see:

- NTP Server Configuration
- Changing Timezone of the Acuitas Server
- Changing MySQL root/Replication User Password

12 On the primary server, SCP the Acuitas license file (License.dat) to affirmed@<*server_ip*>:/data/affirmed.

13 On the primary server, issue the following command to apply the license:

```
cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/conf/
```

14 On the secondary server, SCP the Acuitas license file (License.dat) to affirmed@<*server_ip*>:/data/affirmed.

15 On the secondary server, issue the following command to apply the license:

```
cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/conf/
```

16 The HA configuration program resides on the server. Change to the following directory on the primary server:

```
cd /opt/Affirmed/NMS/bin
```

17 On the primary server, enter the following command to configure the HA system.

```
./configure_ha [--virtualhostname virtual hostname] [--virtualip
virtual IP address] [--primaryhostname primary hostname] [--primaryip
primary IP address] [--secondaryhostname secondary hostname]
[--secondaryip secondary IP address] [--netmask netmask value]
[--ipinterface interface] [--backupfile file_path]
```

```
Please double check the following configuration options:
{
  virtualhostname => 'virtual_hostname',
  virtualip => ' virtual_IP_address',
  primaryhostname => 'primary_hostname',
  primaryip => 'primary_IP_address',
  secondaryhostname => 'secondary_hostname',
  secondaryip => 'secondary_IP_address',
  primaryslaveid => 'primary_slave_ID',
  secondaryslaveid => 'secondary_slave_ID',
  primaryipinterface => 'primary_IP_interface',
  secondaryipinterface => 'secondary_IP_interface',
  ipinterface => 'IP_interface',
  netmask => 'netmask value'
}
Are you sure you want to continue with this configuration? (y/n):
```

18 After HA configuration set up is complete, the primary server starts as the active server and the secondary server starts as the standby server.

19 Issue the following commands to check the HA deployment status on the primary server:

```
cd /opt/Affirmed/NMS/bin
./emsstatus
```

```
*********** Acuitas Server Status *************
Node                       : <server name> - <server IP address>
EMS Package                : AffirmedEmsHA-9.4.0.0-97

EMS Process                : Running [PID]
MySQL
    Database               : Running
    Server Id              : 1
    Slave IO               : Yes
    Slave SQL              : Yes
```

```
High Availability
    Status                : Active
    Heartbeat             : Running
    Virtual IP           : <virtual IP address> - Owned by this node
***********************************************
```

The output indicates that both the Slave IO and Slave SQL are set to Yes, the node is the active server, the heartbeat service is running, and the Virtual IP is owned by the server.

20 Issue the following commands to check the HA deployment status on the standby node:

```
cd /opt/Affirmed/NMS/bin
./emsstatus
```

```
*********** Acuitas Server Status *************
Node                    : <server name> - <server IP address>
EMS Package             : AffirmedEmsHA-9.4.0.0-97

EMS Process             : Running [PID]
MySQL
    Database            : Running
    Server Id           : 2
    Slave IO            : Yes
    Slave SQL           : Yes

High Availability
    Status              : Standby
    Heartbeat           : Not required by application on standby node
    Virtual IP          : <virtual IP address> - Not owned by this node
***********************************************
```

The output indicates that both the Slave IO and Slave SQL are set to Yes, the node is the standby server, the heartbeat service *is not* required, and the Virtual IP *is not* owned by the server.

HA System Upgrade

This section contains instructions for the following upgrade scenarios:

- **Release Prior to 9.4.0.0 => 9.4.0.0 or Later**

  If you are upgrading from a release prior to 9.4.0.0 to release 9.4.0.0 or later, the upgrade requires a major upgrade to the OS. Follow the instructions in OS Upgrade Considerations.

- **Release 9.4.0.0 => 9.4.x.x or later**

  If you are upgrading from a 9.4.0.0 to 9.4.x.x or later, the upgrade *does not* require a major upgrade to the OS. Use the haupgrade script to upgrade Acuitas on an HA pair.follow the instructions in Standard Upgrade Procedure.

- Rollback Procedure if Upgrade Fails — If the upgrade fails at any point during the upgrade process, the procedure rolls back the software to the last installed version.

- Downgrade Procedure — The procedure reverts the software back to an older version.

Standard Upgrade Procedure

If the upgrade *does not* require a major upgrade to the OS, you can use the standard upgrade procedure. Use the haupgrade script to upgrade Acuitas on an HA pair.

The haupgrade script runs in the background. The script prompts for input and then displays the message:

```
The command $COMMAND is running in the background.
Check log file $LOG_FILE for the command result.
```

When you use the **--**client option, the process runs in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

To upgrade the software with a newer version:

1. On both servers, copy the upgrade package emsha**-***new_pkg_name* to /data/affirmed.

   where *new_pkg_name* is the new Acuitas package.

2. Log into either of the HA servers as user account `affirmed` using the physical IP address (*not* the virtual IP address).

3. On either of the HA servers, issue the following commands:

   ```
   cd /opt/Affirmed/NMS/bin
   ./haupgrade /data/affirmed/emsha-new_pkg_name.rpm
   ```

   ```
   Do you want to upgrade the HA pair on primary <server name> and
   secondary <server name> using /data/affirmed/emsha-new_pkg_name.rpm
   (Y/N)?
   ```

4. Enter **y** to continue with the upgrade.

   ```
   Do you want to take a backup of active node <server name> (Y/N)? This
   is highly recommended if you have not already done so.
   ```

5. Enter **y**. The backup can be used later if you decide to revert the software back to an older version (for information, see Downgrade Procedure).

   ```
   Upgrade will start by upgrading the standby node <server name>.
   Continue (Y/N)?
   ```

6. Enter **y** to continue with the upgrade. The upgrade can take some time to complete. When the upgrade completes, you can check the log file for the status.

Rollback Procedure if Upgrade Fails

If the upgrade fails at any point during the upgrade process, perform the following rollback procedure.

The procedure uses a backup database file from where it was exported to restore to the same system. The backup file was saved during the previous haupgrade and was stored in /opt/Affirmed/NMS/config/db/ with a filename such as preupgradedb_*hostname*.

As an alternative, you can use the emsdb command to obtain a database backup of the primary active node. For details, see Acuitas Database Management.

Note: *If the Acuitas servers have gone through a major OS upgrade as part of the Acuitas upgrade, this rollback procedure does not set the system back to the older OS version. Contact the Affirmed Networks Technical Assistance Center (TAC) for assistance on how to perform a complete rollback following a major OS upgrade.*

To rollback the upgrade:

1. Log in using the default user account `affirmed` and the account password.
2. Once you are logged in, issue the following command to gain super-user privileges:

   `su -`

3. When prompted, enter the super-user password.
4. Issue the following command to obtain the name of the installed Acuitas HA package (for example, AffirmedEmsHA-9.4.0.0-104.x86_64):

   `rpm -qa | grep Ems`

5. On both servers, issue the following command:

   `rpm -ev emsha-`*`failed_pkg_name`*

   where ***`failed_pkg_name`*** is the package name of the failed install.

6. On both servers, issue the following command, specifying the pathname of the package:

   `rpm -ivh emsha-`*`pkg_name`*`.rpm`

   where ***`pkg_name`*** is the pathname of the new Acuitas HA package.

7. Issue the following command to switch back to the `affirmed` user:

   `exit`

8. On the primary server, issue the following commands:

   `cd /opt/Affirmed/NMS/bin`

   `./configure_ha --reload --backupfile <`*`file_path`*`> --restore`

where ***`file_path`*** is the full path to the backup file.

For example, if you use the file from previous upgrade, ***`file_path`*** is /opt/Affirmed/NMS/conf/db/preupgradedb_*hostname*.

```
Please double check the following configuration options:

{

  virtualhostname => 'virtual_hostname',

  virtualip => ' virtual_IP_address',

  primaryhostname => 'primary_hostname',

  primaryip => 'primary_IP_address',

  secondaryhostname => 'secondary_hostname',

  secondaryip => 'secondary_IP_address',

  primaryslaveid => 'primary_slave_ID',

  secondaryslaveid => 'secondary_slave_ID',

  primaryipinterface => 'primary_IP_interface',

  secondaryipinterface => 'secondary_IP_interface',

  ipinterface => 'IP_interface',

  netmask => 'netmask value'

}
Are you sure you want to continue with this configuration? (y/n):
```

9 Enter **y** to continue with this configuration.

```
Loading the existing configuration ..

Please make sure <file_path> is a backup file taken from the local
server. Continue(Y/N)
```

10 Enter **y** to continue.When the script completes, you can check the log file for the status.

Downgrade Procedure

A downgrade reverts the software back to an older version. The procedure requires you to specify a backup file to use for the downgrade. The following rules apply:

- The backup can be one that you created when you last ran the haupgrade script. That backup is stored in /opt/Affirmed/NMS/conf/db/preupgradedb_*hostname*.
- The backup can be one that you created with the emsdb script. For information, see Acuitas Database Management.
- The backup file must be taken from the same release software to which you are downgrading.

Note: *If the backup file* ***is not*** *from the same release, incompatibility issues can result. Make sure that the backup file version and downgrade version match.*

- The backup must be the one on the current standby. If the downgrade cannot locate the backup on the standby, the script quits.

Note: *If the Acuitas servers have gone through a major OS upgrade as part of the Acuitas upgrade, this downgrade procedure does not set the system back to the older OS version. Contact the Affirmed Networks Technical Assistance Center (TAC) for assistance on how to perform a complete downgrade following a major OS upgrade.*

The haupgrade script runs in the background. The script prompts for input and then displays the message:

```
The command $COMMAND is running in the background.
Check log file $LOG_FILE for the command result.
```

When you use the --client option, the process runs in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

To downgrade HA server software:

1. Log in using the default user account `affirmed` and the account password.
2. On either of the HA servers, issue the following commands:

```
cd /opt/Affirmed/NMS/bin
./haupgrade /data/affirmed/emsha-old_pkg_name --backupfile <file_path>
```

where ***`file_path`*** is the full path to the backup file.

For example, if you use the file from previous upgrade, ***`file_path`*** is /opt/Affirmed/NMS/conf/db/preupgradedb_*hostname*.

```
Do you want to upgrade the HA pair on primary <server name> and
secondary <server name> using /data/affirmed/emsha-old_pkg_name (Y/N)?
```

3 Enter **y** to continue with the downgrade.

```
Do you want to use the <file_path> for this upgrade? Note the backup
file has to be from <server name> and from the version you are
upgrading to.
```

4 Enter **y** to continue with the downgrade.

```
... Active node is <server name>

... Standby node is <server name>

Upgrade will start by upgrading the standby node <server name> using
<file_path>. Continue (Y/N)?
```

5 Enter **y** to continue with the downgrade. The downgrade can take some time to complete. The following message appears when HA is downgraded successfully.

```
HA pair has been upgraded successfully.
```

HA Server Switchover

Automatic Switchover

If the active server goes out of service, automatic switchover occurs from the active to the standby system.

Possible triggers that can cause an automatic failover in the HA system include:

- The active Acuitas server is shut down for some reason, such as because the power supply failed (for example, the rack lost power).
- On the active system, the Acuitas java process hangs, crashes, or otherwise becomes unresponsive.
- The active Acuitas server is taken off the network and cannot be reached by the standby (at which point the standby becomes active).
- The active Acuitas system is brought down to perform maintenance operations on the server (such as, upgrading Acuitas software).

Manual Switchover

You can trigger a manual switchover from the active to the standby system. For example, to perform maintenance operations on the active Acuitas server (such as, upgrading Acuitas software), manually trigger switchover to standby, then bring down the active Acuitas system.

The switchover verifies that the standby node is running:

- If the standby server is running, the scripts stops the Acuitas process on active node.
- If the standby server is not running, the scripts quits.

To switchover the active node to the standby node:

1. Log in using the default user account `affirmed` and the account password.
2. On the active server, issue the following commands to switchover the Acuitas server:

```
cd /opt/Affirmed/NMS/bin
./emsha switchover
This is the active node, are you sure you want to switch over? (y/n):
```

3. Enter **y** to switchover to the standby node.

```
Stopping the EMS server..
INFO  [org.jboss.modules] JBoss Modules version 1.3.3.Final
INFO  [org.xnio] XNIO version 3.3.0.Final
```

```
INFO  [org.xnio.nio] XNIO NIO Implementation Version 3.3.0.Final
INFO  [org.jboss.remoting] JBoss Remoting version 4.0.7.Final
INFO  [org.jboss.as.cli.CommandContext] {"outcome" => "success"}
{"outcome" => "success"}
Waiting for JBoss process to exit ...
Waiting for <Secondary IP Address> to become Active ...
Waiting for <Secondary IP Address> to become Active ...
Waiting for <Secondary IP Address> to become Active ...
Waiting for <Secondary IP Address> to become Active ...
Starting the EMS server..
emsnode_name-><Primary server name>
primary node command /opt/Affirmed/NMS/bin/../bin/starttems.sh  -b
0.0.0.0 -n <Primary server name> -j <Primary IP address>
<server name>
nohup: ignoring input
Java HotSpot(TM) 64-Bit Server VM warning: ignoring option
MaxPermSize=512m; support was removed in 8.0

        Affirmed Element Management System
        ==================================

  Checking for License ...
 Checking for License ...         [ Checking for License ... ]
 Checking for License ...         [ Done ]

  Checking for port availability...
  Checking for port availability...       [ Checking for port
availability... ]
 Checking for port availability...    [ Ports are Available ]
 Checking for port availability...    [ Done ]

  Checking for DB connection and Migration...
  Checking for DB connection and Migration...[ Checking for DB
connection... ]
  Checking for DB connection and Migration...[ Checking for DB
migration... ]
 Checking for DB connection and Migration...[ Done ]

 Updating cluster node status...
  Updating cluster node status...         [ Updating cluster node
status... ]
 Updating cluster node status...     [ Done ]

  Checking for Hearbeat Services...
```

```
  Checking for Hearbeat Services...       [ Checking for Hearbeat
Services ... ]
 Checking for Hearbeat Services...    [ Done ]

  Initializing EMS Server ...
```

4 Issue the following commands to check the status of the previous active server:

cd /opt/Affirmed/NMS/bin

./emsstatus

```
*********** Acuitas Server Status *************
Node                    : <server name> - <server IP address>
EMS Package             : AffirmedEmsHA-9.4.0.0-97

EMS Process             : Running [PID]
MySQL
    Database            : Running
    Server Id           : 1
    Slave IO            : Yes
    Slave SQL           : Yes

High Availability
    Status             : Standby
    Heartbeat          : Not required by application on standby node
    Virtual IP         : <virtual IP address> - Not owned by this node
***********************************************
```

5 Issue the following commands to check the status of the current active server:

cd /opt/Affirmed/NMS/bin

./emsstatus

```
*********** Acuitas Server Status *************
Node                    : <server name> - <server IP address>
EMS Package             : AffirmedEmsHA-9.4.0.0-97

EMS Process             : Running [PID]
```

```
MySQL
    Database            : Running
    Server Id           : 2
    Slave IO            : Yes
    Slave SQL           : Yes

High Availability
    Status              : Active
    Heartbeat           : Running
    Virtual IP          : <virtual IP address> - Owned by this node
***********************************************
```

Configuration when Failed Node is Back In Service

Once the failed node is back in service, use the appropriate method to synchronize the databases:

- **Method 1:** If the standby node has been out of service for a short duration (less than several hours), start the Acuitas server and let mysql database replication synchronize the databases.
- **Method 2:** If the standby node has been out of service for a long duration (several hours or more), start the Acuitas server and use the configure_ha script with the `--syncwithactive` option. This method greatly reduces the time to synchronize databases.

Method 1:

1. Log in using the default user account `affirmed` and the account password.
2. Check the status of the system:

   `cd /opt/Affirmed/NMS/bin`

   `./emsstatus`

3. Restart the Acuitas server and allow mysql database replication to occur between the active and standby node:

   `./emsha startAll`

4. At the confirmation prompt, enter **y**.
5. When the system returns the success message:

   `Server started successfully`

   Press **Enter** to return to the operating system prompt.

Method 2:

1. Check the status of the system:

   `cd /opt/Affirmed/NMS/bin`

   `./emsstatus`

2. Issue the following command to synchronize the databases:

   `./configure_ha --synchwithactive`

   The system displays the following messages:

   ```
   Syncing the database with the active node..
   ```

   ```
   Sync with the active node is complete. Checking to see if there is DR
   node to sync ...
   ```

   ```
   slave : <IP Address>
   ```

```
Do you want to start the EMS server on the current node(Y/N)?
```

3 Enter **y** to start the server.

```
Starting the EMS server..

emsnode_name-><Secondary server name>

primary node command /opt/Affirmed/NMS/bin/../bin/starttems.sh  -b
0.0.0.0 -n <Secondary server name> -j <Secondary IP address>

<Secondary server name>

nohup: ignoring input

Java HotSpot(TM) 64-Bit Server VM warning: ignoring option
MaxPermSize=512m; support was removed in 8.0


        Affirmed Element Management System

        ==================================


  Checking for License ...

 Checking for License ...        [ Checking for License ... ]

 Checking for License ...         [ Done ]


  Checking for port availability...

  Checking for port availability...       [ Checking for port
availability... ]

 Checking for port availability...    [ Ports are Available ]

 Checking for port availability...    [ Done ]


  Checking for DB connection and Migration...

  Checking for DB connection and Migration...[ Checking for DB
connection... ]

  Checking for DB connection and Migration...[ Checking for DB
migration... ]

 Checking for DB connection and Migration...[ Done ]


  Updating cluster node status...

  Updating cluster node status...         [ Updating cluster node
status... ]

 Updating cluster node status...     [ Done ]
```

```
  Checking for Hearbeat Services...

  Checking for Hearbeat Services...       [ Checking for Hearbeat
Services ... ]

 Checking for Hearbeat Services...    [ Done ]


  Initializing EMS Server ...
```

4 On the standby node, issue the following commands to check the HA deployment status:

`cd /opt/Affirmed/NMS/bin`

`./emsstatus`

```
*********** Acuitas Server Status *************

Node                    : <server name> - <server IP address>

EMS Package             : AffirmedEmsHA-9.4.0.0-97


EMS Process             : Running [PID]

MySQL

    Database            : Running

    Server Id           : 2

    Slave IO            : Yes

    Slave SQL           : Yes


High Availability

    Status              : Standby

    Heartbeat           : Not required by application on standby node

    Virtual IP          : <virtual IP address> - Not owned by this node

***********************************************
```

High Availability + Disaster Recover Systems

Acuitas supports the High Availability + Disaster Recovery configuration. In a 4-node Acuitas deployment consisting of two HA pairs, one pair acts as the operational HA system while the other pair operates in a standby Disaster Recovery mode.

If the operational HA pair becomes unavailable due to a disaster or other operational failure at its facility, the geographically-separate standby HA pair can be manually activated to take over Acuitas functions.

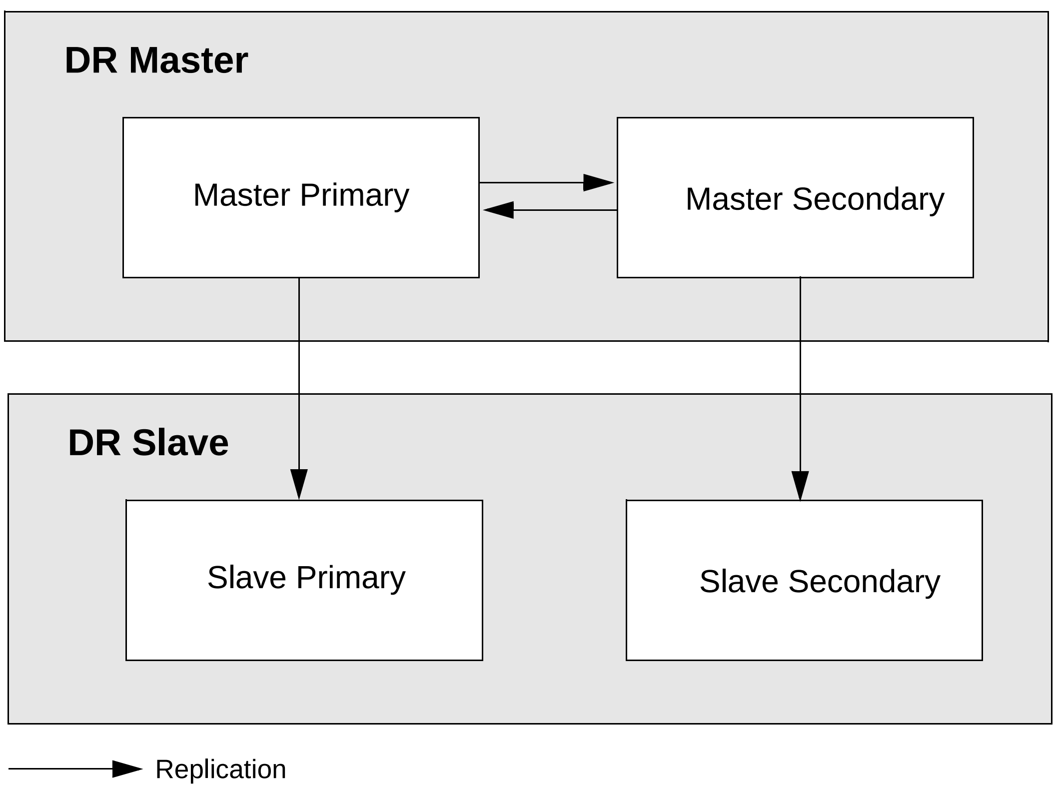
In each HA pair, one server is configured to serve as the primary server; the other is configured to serve as the secondary server. These roles persist across upgrades.

By contrast, the runtime roles of the servers can vary:

- Active server — the server that owns the virtual IP address of the cluster and has the full Acuitas server running. *At the master site only*.
- Standby server — the server in warm-standby mode that automatically takes over in the event that the active server becomes out of service.

In the HA + DR configuration, each node in the slave pair replicates from a node in the master pair. The following figure illustrates one possible replication scenario of the cluster.

Figure 3-1 Sample Replication Scenario in HA + DR Systems



Prerequisites

Before you start configuring HA + DR systems, make sure you have all the required input to the following scripts:

- configure_ha — For command syntax, see Acuitas High Availability Configuration.
- configure_dr — For command syntax, see Acuitas Disaster Recovery Configuration.

Before you start configuring HA + DR systems, follow this procedure to allow SSH access between the systems during configuration.

Perform these steps on the DR Master primary server:

To configure SSH access for the affirmed user:

1. Log in using the default user account `affirmed` and the account password and issue the following command.

   `#ssh-keygen –t rsa`

2. Press **Enter** at the following prompt:

   `Enter file in which to save the key (/home/affirmed/.ssh/id_rsa):`

3. If prompted to overwrite the existing key file, enter **y**:

   `/home/affirmed/.ssh/id_rsa already exists.`

   `Overwrite (y/n)?`

4. Press **Enter** for each of the following prompts:

   `Enter passphrase (empty for no passphrase):`

   `Enter same passphrase again:`

5. Issue the following commands:

   ssh-copy-id -i ~/.ssh/id_rsa.pub affirmed@<*master secondary server IP address*>

   ssh-copy-id -i ~/.ssh/id_rsa.pub affirmed@<*slave primary server IP address*>

   ssh-copy-id -i ~/.ssh/id_rsa.pub affirmed@<*slave secondary server IP address*>

6. When prompted, enter the `affirmed` user password.

7. Add the server IP addresses in the cluster to `/etc/hosts.allow`:

   ALL : <*master primary server IP address*>, <*master secondary server IP address*>, <*slave primary server IP address*>, <*slave secondary server IP address*>

   The syntax for IPv6 `/etc/hosts.allow` is as follows:

   ALL : [<*IP Address*>]/<*netmask*>

For example:

```
ALL: [2001:470:8865:2000:10:32:1:147]/64
```

Note: *By default, the hosts.deny file denies all hosts not specified in hosts.allow. To override deny processing, comment out the ALL: ALL line in hosts:deny.*

Repeat the procedure on each of the other servers. For each server, in step 5, specify the IP addresses of the other three servers.

HA + DR Server Configuration

Summary

To configure Acuitas HA + DR servers, you execute commands on the master HA pair and the slave HA pair, as indicated in the following table. The master HA pair commands can be issued simultaneously with the slave HA pair commands:

	Master HA Pair	Slave HA Pair
1	On both servers, copy the upgrade package emsha-*release*.rpm to /data/affirmed. On either server, perform a fresh install of the HA software: **a** Log in using the default user account affirmed and the account password. **b** Once you are logged in, issue the following command to gain super-user privileges: `su -` **c** When prompted, enter the super-user password. **d** Install the HA software: **`rpm -ivh /data/affirmed/emsha-`*`release`*`.rpm`** **e** Issue the following command to switch back to the `affirmed` user: **`exit`**	On both servers, copy the upgrade package emsha-*release*.rpm to /data/affirmed. On either server, perform a fresh install of the HA software: **a** Log in using the default user account affirmed and the account password. **b** Once you are logged in, issue the following command to gain super-user privileges: `su -` **c** When prompted, enter the super-user password. **d** Install the HA software: **`rpm -ivh /data/affirmed/emsha-`*`release`*`.rpm`** **e** Issue the following command to switch back to the `affirmed` user: **`exit`**

	Master HA Pair	Slave HA Pair
2	On both master nodes, SCP the Acuitas license file (License.dat) to affirmed@<*server_ip*>:/data/affirmed. Issue the following command to apply the license:: **cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/ conf/**	On both slave nodes, SCP the Acuitas license file (License.dat) to affirmed@<*server_ip*>:/data/affirmed. Issue the following command to apply the license:: **cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/ conf/**
3	On both master nodes, make sure that NTP configuration, timezone, and MySQL user password have been correctly set. For instructions, see: ■ NTP Server Configuration ■ Changing Timezone of the Acuitas Server ■ Changing MySQL root/Replication User Password	On both slave nodes, make sure that NTP configuration, timezone, and MySQL user password have been correctly set. For instructions, see: ■ NTP Server Configuration ■ Changing Timezone of the Acuitas Server ■ Changing MySQL root/Replication User Password
4	On the primary master node, start the HA configuration program: **cd /opt/Affirmed/NMS/bin** **./configure_ha [--virtualhostname *virtual hostname*] [--virtualip *virtual IP address*] [--primaryhostname *primary hostname*] [--primaryip *primary IP address*] [--secondaryhostname *secondary hostname*] [--secondaryip *secondary IP address*] [--netmask *netmask value*] [--ipinterface *interface*] [--backupfile *file_path*]** ***Note:*** *Virtual hostname, and virtual IP address must differ from the values for the slave HA pair.*	On the primary slave node, start the HA configuration program: **cd /opt/Affirmed/NMS/bin** **./configure_ha [--virtualhostname *virtual hostname*] [--virtualip *virtual IP address*] [--primaryhostname *primary hostname*] [--primaryip *primary IP address*] [--secondaryhostname *secondary hostname*] [--secondaryip *secondary IP address*] [--netmask *netmask value*] [--ipinterface *interface*] [--backupfile *file_path*]** ***Note:*** *Virtual hostname, and virtual IP address must differ from the values for the master HA pair.*
5	On both master nodes, check HA status: **cd /opt/Affirmed/NMS/bin** **./emsstatus**	On either slave node, set up MySQL slave replication: **configure_dr convert slave --masterhost <*masterhost*>**
6		Check HA status: **cd /opt/Affirmed/NMS/bin** **./emsstatus**

The configure_ha and configure_dr scripts run in the background. The scripts prompt for input and then display the message:

```
The command $COMMAND is running in the background.
Check log file $LOG_FILE for the command result.
```

When you use the --client option, the process runs in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

If desired, run the configure_dr script with the --currenthost option to run the script only on the current host in an HA pair. This option could be used only if the current node is not properly configured as the DR slave of the <*masterhost*>. Use the master primary host as <*masterhost*> for the DR slave primary host and use the master secondary host as <*masterhost*> for the DR slave secondary host.

These steps are detailed in the following procedure.

Procedure

To configure the Acuitas HA + DR servers:

1. On the primary server of the master pair, copy the ems package emsha-*release*.rpm to /data/affirmed.
2. On the master pair, log in using the default user account `affirmed` and the account password.
3. Once you are logged in, issue the following command to gain super-user privileges:

   `su -`
4. When prompted, enter the super-user password.
5. On the master pair, perform a fresh install of the HA software:

   `rpm -ivh emsha-release.rpm` *(Note: include path of the <emsha-release.rpm>)*

   where *release* is in the form of *x.x.x.y*.

   The system checks available disk space and MySQL version and proceeds with the installation.

   ```
   Installation is done - start the server once the license is installed.
   ```

   If the installation fails with an error indicating that the MySQL process *is not* running, follow this procedure:

   a. Issue the following command to verify that MySQL is running:

      `systemctl status mysqld`

      ```
      [affirmed@qaems149b bin]$ systemctl status mysqld
      \u25cf mysqld.service - MySQL Server
        Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; vendor preset: disabled)
         Active: active (running) since Tue 2017-10-24 09:15:40 EDT; 2 days ago
           Docs: man:mysqld(8)
      ```

```
           http://dev.mysql.com/doc/refman/en/using-systemd.html
 Main PID: 26663 (mysqld)
   CGroup: /system.slice/mysqld.service
           \u2514\u250026663 /usr/sbin/mysqld --daemonize
--pid-file=/var/run/mysqld/mysqld.pid
```

b If the command *does not* indicate `active (running)`, issue the following command to start MySQL:

`systemctl start mysqld`

c If MySQL does not start, contact the Affirmed Networks Technical Assistance Center (TAC) for assistance.

6 Issue the following command to switch back to the `affirmed` user:

`exit`

7 Make sure that NTP configuration, timezone, and MySQL user password have been correctly set. For instructions, see:

- NTP Server Configuration
- Changing Timezone of the Acuitas Server
- Changing MySQL root/Replication User Password

8 On the master pair, SCP the Acuitas license file (License.dat) to affirmed@<*server_ip*>:/data/affirmed.

9 On the master pair, issue the following command to apply the license:

`cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/conf/`

10 On the primary server of the master pair, issue the following commands to configure the HA system:

```
cd /opt/Affirmed/NMS/bin
```

```
./configure_ha [--virtualhostname virtual hostname] [--virtualip
virtual IP address] [--primaryhostname primary hostname] [--primaryip
primary IP address] [--secondaryhostname secondary hostname]
[--secondaryip secondary IP address] [--netmask netmask value]
[--ipinterface interface] [--backupfile file_path]
```

Be sure that the virtual hostname, and virtual IP address differ from the values for the slave HA pair.

```
Please double check the following configuration options:
{
  virtualhostname => 'virtual_hostname',
  virtualip => ' virtual_IP_address',
  primaryhostname => 'primary_hostname',
  primaryip => 'primary_IP_address',
```

```
  secondaryhostname => 'secondary_hostname',
  secondaryip => 'secondary_IP_address',
  primaryslaveid => 'primary_slave_ID',
  secondaryslaveid => 'secondary_slave_ID',
  primaryipinterface => 'primary_IP_interface',
  secondaryipinterface => 'secondary_IP_interface',
  ipinterface => 'IP_interface',
  netmask => 'netmask value'
}
Are you sure you want to continue with this configuration? (y/n):
```

11 Enter **y** to continue with this configuration.

The primary server starts as the active server and the secondary server starts as the standby server. When the script completes, you can check the log file for the status.

12 On the master pair, issue the following commands to check the HA deployment status:

cd /opt/Affirmed/NMS/bin

./emsstatus

Primary Server:

```
*********** Acuitas Server Status *************


Node             : <server name> - <server IP address>
EMS Package      : AffirmedEmsHA-9.4.0.0-97


EMS Process      : Running [PID]
MySQL
    Database     : Running
    Server Id    : 1
    Slave IO     : Yes
    Slave SQL    : Yes


High Availability
    Status       : Active
    Heartbeat    : Running
    Virtual IP   : <virtual IP address> - Owned by this node
```

```
***********************************************
```

Secondary Server:

```
*********** Acuitas Server Status *************

Node            : <server name> - <server IP address>
EMS Package     : AffirmedEmsHA-9.4.0.0-97

EMS Process     : Running [PID]
MySQL
    Database    : Running
    Server Id   : 2
    Slave IO    : Yes
    Slave SQL   : Yes

High Availability
    Status      : Standby
    Heartbeat   : Not required by application on standby node
    Virtual IP  : <virtual IP address> - Not owned by this node
```

13 On the slave pair, copy the ems package emsha-*release*.rpm to /data/affirmed.

14 On the slave pair, log in using the default user account `affirmed` and the account password.

15 Once you are logged in, issue the following command to gain super-user privileges:

`su -`

16 When prompted, enter the super-user password.

17 On the slave pair, perform a fresh install of the HA software:

`rpm -ivh emsha-release.rpm` *(Note: include path of the <emsha-release.rpm>)*

where *release* is in the form of *x.x.x.y*.

The system checks available disk space and MySQL version and proceeds with the installation.

```
Installation is done - start the server once the license is installed.
```

18 Issue the following command to switch back to the `affirmed` user:

`exit`

19 Make sure that NTP configuration, timezone, and MySQL user password have been correctly set. For instructions, see:

- NTP Server Configuration
- Changing Timezone of the Acuitas Server
- Changing MySQL root/Replication User Password

20 On the slave pair, SCP the Acuitas license file (License.dat) to affirmed@<*server_ip*>:/data/affirmed.

21 On the slave pair, issue the following command to apply the license:

```
cp /data/affirmed/License.dat /opt/Affirmed/NMS/server/ems/conf/
```

22 On the primary server of the slave pair, issue the following commands to configure the HA system:

```
cd /opt/Affirmed/NMS/bin
```

```
./configure_ha [--virtualhostname virtual hostname] [--virtualip
virtual IP address] [--primaryhostname primary hostname] [--primaryip
primary IP address] [--secondaryhostname secondary hostname]
[--secondaryip secondary IP address] [--netmask netmask value]
[--ipinterface interface] [--backupfile file_path]
```

Be sure that the virtual hostname, and virtual IP address differ from the values for the master HA pair.

```
Please double check the following configuration options:

{

  virtualhostname => 'virtual_hostname',

  virtualip => ' virtual_IP_address',

  primaryhostname => 'primary_hostname',

  primaryip => 'primary_IP_address',

  secondaryhostname => 'secondary_hostname',

  secondaryip => 'secondary_IP_address',

  primaryslaveid => 'primary_slave_ID',

  secondaryslaveid => 'secondary_slave_ID',

  primaryipinterface => 'primary_IP_interface',

  secondaryipinterface => 'secondary_IP_interface',

  ipinterface => 'IP_interface',

  netmask => 'netmask value'

}
Are you sure you want to continue with this configuration? (y/n):
```

When the script completes, you can check the log file for the status. Wait for Master HA Pair to be active/standby before configuring the Slave HA Pair.

23 On either server in the slave pair, issue the following command:

configure_dr convert slave --masterhost <*masterhost*>

where *masterhost* specifies the master primary hostname or IP address.

When the script completes, the system returns the following messages:

```
Replication setup on this HA as slave pair is complete.

Exiting DR configuration script
```

24 Run the **emsstatus** command to verify that the server is in the expected working state:

cd /opt/Affirmed/NMS/bin

./emsstatus

```
*********** Acuitas Server Status *************

Node             : <server name> - <server IP address>
EMS Package      : AffirmedEmsHA-9.4.0.0-97

EMS Process      : Stopped
MySQL
    Database     : Running
    Server Id    : 3

High Availability
    Status       : Standby
    Heartbeat    : Not required by application on standby node

Disaster Recovery
    Status       : Standby
    Master Host  : <master server IP address>
    Standby IO   : Yes
    Standby SQL  : Yes
***********************************************
```

Configuration when Master Pair Out of Service

If a disaster places the master pair out of service, run the configure_dr script to convert the slave pair to the master role to allow Acuitas functions to resume with the converted pair.

The configure_dr script runs in the background. The script prompts for input and then displays the message:

```
The command $COMMAND is running in the background.
Check log file $LOG_FILE for the command result.
```

When you use the --client option, the process runs in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

To convert the slave pair to a master pair:

1. Log in using the default user account `affirmed` and the account password.
2. On the slave pair, use the following syntax:

   `configure_dr convert master`

   The system displays the following messages:

   ```
   ======== Acuitas Disaster Recovery Runtime Information ===========


   MySQL server-id = 3
   MySQL Slave Configuration:
     Master_Host: <host name>
     Slave_IO_Running: Yes
     Slave_SQL_Running: Yes


   JBoss server is not running.


   ==================================================================


   Are you sure you want to disassociate the HA pair (<server name>,
   <server name>) from the master (Y/N)?
   ```
3. Enter **y** to continue with this configuration.

   When the script completes, you can check the log file for the status.
4. On the new master pair, run the **emsstatus** command to verify that the servers are in the expected working state:

   `cd /opt/Affirmed/NMS/bin`

```
./emsstatus
*********** Acuitas Server Status *************
Node                    : <server name> - <server IP address>
EMS Package             : AffirmedEmsHA-9.4.0.0-97

EMS Process             : Running [PID]
MySQL
    Database            : Running
    Server Id           : 3
    Slave IO            : Yes
    Slave SQL           : Yes

High Availability
    Status              : Active
    Heartbeat           : Running
    Virtual IP          : <virtual IP address> - Owned by this node
***********************************************
```

Configuration when Master Pair Back In Service

Once the disaster situation has been resolved, run the configure_dr script on both master and slave pairs to resume the Disaster Recovery system as it was before the disaster. That is, set the original pair to the master role and reset the slave pair to the slave role.

The configure_dr script runs in the background. The script prompts for input and then displays the message:

```
The command $COMMAND is running in the background.
Check log file $LOG_FILE for the command result.
```

When you use the --client option, the process runs in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

If desired, run the configure_dr script with the --currenthost option to run the script only on the current host in an HA pair. This option could be used only if the current node is not properly configured as the DR slave of the <*masterhost*>. Use the master primary host as <*masterhost*> for the DR slave primary host and use the master secondary host as <*masterhost*> for the DR slave secondary host.

Summary

You execute commands on the master pair and the slave pair in the order indicated in the following table:

	Master Pair	Slave Pair
1	On either server, replicate the database on the slave pair to the database on the master pair: **`configure_dr convert slave --masterhost <masterhost>`** where ***`masterhost`*** is the hostname or IP of the original slave node	
2		Stop the original pair and check status: **`cd /opt/Affirmed/NMS/bin`** **`./emsha stopAll standby`** **`./emsha stopAll active`** **`./emsstatus`**
3	On either server or the original pair, convert the master system to the master: **`configure_dr convert master`**	
4		On either server, set up MySQL slave replication: **`configure_dr convert slave`**

	Master Pair	Slave Pair
5		Check the status: **cd /opt/Affirmed/NMS/bin** **./emsstatus**

These steps are detailed in the following procedure.

Procedure

To resume the Disaster Recovery system:

1 Log in using the default user account `affirmed` and the account password.

2 On either server of the master pair, use the following syntax to replicate the database on the slave pair to the database on the master pair:

```
configure_dr convert slave --masterhost <masterhost>
```

where *masterhost* specifies the original active slave hostname or IP address.

When the script completes, the system returns the following messages:

```
Replication setup on this HA as slave pair is complete.


Exiting DR configuration script
```

3 On the original slave pair, issue the following commands to stop the servers and check status:

```
cd /opt/Affirmed/NMS/bin

./emsha stopAll standby

./emsha stopAll active

./emsstatus
```

```
*********** Acuitas Server Status *************

Node                         : <server name> - <server IP address>

EMS Package                  : AffirmedEmsHa-9.4.0.0-97


EMS Process                  : Stopped

MySQL

    Database                : Running

    Server Id               : 3

    Slave IO                : Yes
```

```
    Slave SQL           : Yes
High Availability
    Status              : N/A
    Heartbeat           : Not required by application on standby node
    Virtual IP          : <virtual IP address> - Not owned by this node
***********************************************
```

4 On either server of the original master pair, use the following syntax to convert the system back to the master:

`configure_dr convert master`

```
======== Acuitas Disaster Recovery Runtime Information ===========

MySQL server-id = 1
MySQL Slave Configuration:
  Master_Host: <IP Address>
  Slave_IO_Running: Yes
  Slave_SQL_Running: Yes

JBoss server is not running.

==================================================================
Are you sure you want to disassociate the HA pair (<server name>,
<server name>) from the master (Y/N)?
Enter y to continue with this configuration.
```

5 When the script completes, the system returns the following messages:

```
HA pair (<server name>, <server name>) is operational.

Exiting DR configuration script
```

6 On either server of the slave pair, use the following syntax to set up MySQL slave replication:

`configure_dr convert slave`

When the script completes, the system returns the following messages:

```
Replication setup on this HA as slave pair is complete.
```

```
Exiting DR configuration script
```

7 On the slave system, issue the following commands to check the status.

`cd /opt/Affirmed/NMS/bin`

`./emsstatus`

```
*********** Acuitas Server Status *************
Node                    : <server name> - <server IP address>
EMS Package             : AffirmedEmsHA-9.4.0.0-97

EMS Process             : Stopped
MySQL
    Database          : Running
    Server Id           : 3

High Availability
    Status              : Standby
    Heartbeat           : Not required by application on standby node

Disaster Recovery
    Status              : Standby
    Master Host         : <server name> - <master server IP address>
    Standby IO          : Yes
    Standby SQL         : Yes
***********************************************
```

HA + DR Systems Upgrade

This section contains instructions for the following upgrade scenarios:

- **Release Prior to 9.4.0.0 => 9.4.0.0 or Later**

  If you are upgrading from a release prior to 9.4.0.0 to release 9.4.0.0 or later, the upgrade requires a major upgrade to the OS. Follow the instructions in OS Upgrade Considerations.

- **Release 9.4.0.0 => 9.4.x.x**

  If you are upgrading from a 9.4.0.0 to 9.4.x.x or later, the upgrade *does not* require a major upgrade to the OS. Use the haupgrade script to upgrade Acuitas on an HA pair.follow the instructions in HA + DR Systems Upgrade.

- Rollback if the Upgrade Fails — If the upgrade fails at any point during the upgrade process, the procedure rolls back the software to the last installed version.

- Downgrade Procedure — The procedure reverts the software back to an older version.

Standard Upgrade Procedure

If you are upgrading from 9.4.0.0 to release 9.4.x.x or later, the upgrade *does not* require a major upgrade to the OS. Thus, you can use the standard upgrade procedure. Use the haupgrade script to upgrade the master HA pair and the slave HA pair.

The haupgrade and configure_dr scripts run in the background. The scripts prompt for input and then display the message:

```
The command $COMMAND is running in the background.
Check log file $LOG_FILE for the command result.
```

When you use the --client option, the process runs in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

If desired, run the configure_dr script with the --currenthost option to run the script only on the current host in an HA pair. This option could be used only if the current node is not properly configured as the DR slave of the <*masterhost*>. Use the master primary host as <*masterhost*> for the DR slave primary host and use the master secondary host as <*masterhost*> for the DR slave secondary host.

Summary

To upgrade HA + DR systems, you execute commands on the master HA pair and the slave HA pair in the order indicated in the following table:

	Master HA Pair	Slave HA Pair
1		On both servers, copy the upgrade package emsha-*new_pkg_name*.rpm to /data/affirmed. On either server, install the upgrade package: **cd /opt/Affirmed/NMS/bin** **./haupgrade /data/affirmed/emsha-*new_pkg_name*.rpm**
2	On both servers, copy the upgrade package emsha-*new_pkg_name*.rpm to /data/affirmed. On either server, install the upgrade package: **cd /opt/Affirmed/NMS/bin** **./haupgrade /data/affirmed/emsha-*new_pkg_name***.rpm When prompted to create a database backup of the active server, answer **y**.	
3		On either server, set up MySQL slave replication: **cd /opt/Affirmed/NMS/bin** **./configure_dr convert slave --masterhost <*masterhost*>**
4	**cd /opt/Affirmed/NMS/bin** **./emsstatus**	
5		**cd /opt/Affirmed/NMS/bin** **./emsstatus**

These steps are detailed in the following procedure.

Procedure

To upgrade the master HA pair and slave HA pair:

1. Log in using the default user account `affirmed` and the account password.
2. On both servers in the slave pair, copy the upgrade package emsha-*new_pkg_name*.rpm to /data/affirmed.

   where *new_pkg_name*.rpm is the new Acuitas package.
3. On either server in the slave pair, issue the following commands:

   `cd /opt/Affirmed/NMS/bin`

   `./haupgrade /data/affirmed/emsha-`*`new_pkg_name`*`.rpm`

   `Do you want to upgrade the HA pair on primary` <*server name*> `and secondary` <*server name*> `using /data/affirmed/emsha-`***`new_pkg_name`***.rpm `(Y/N)?`
4. Enter **y** to continue with the upgrade.

   `Do you want to take a backup of active node <server name> (Y/N)? This is highly recommended if you have not already done so.`
5. Enter **n**. You will later take a backup when the master pair is upgraded.

   `Upgrade will start by upgrading the standby node <server name>. Continue (Y/N)?`
6. Enter **y** to continue with the upgrade.

   `Upgrade will prepare for <server name> to take over as active. This will stop the Acuitas server on <server name> and the HA server will not be available until <server name> is up. Press Enter to continue.`
7. Press **Enter** to continue with the upgrade.

   `Upgrade will now upgrade the previous active node <server name>. Press Enter to continue.`
8. Press **Enter** to continue with the upgrade.

   `HA pair has been upgraded successfully.`
9. On both servers in the master pair, copy the upgrade package emsha-*new_pkg_name*.rpm to /data/affirmed.

   where *new_pkg_name*.rpm is the new Acuitas package.
10. On either server in the master pair, issue the following commands:

    `cd /opt/Affirmed/NMS/bin`

    `./haupgrade /data/affirmed/emsha-`*`new_pkg_name`*`.rpm`

    `Do you want to upgrade the HA pair on primary <server name> and secondary <server name> using /data/affirmed/emsha-new_pkg_name.rpm (Y/N)?`
11. Enter **y** to continue with the upgrade.

```
Do you want to take a backup of active node <server name> (Y/N)? This
is highly recommended if you have not already done so.
```

12 Enter **y**. The backup can be used later if you decide to revert the software back to an older version (for information, see Downgrade Procedure).

```
Upgrade will start by upgrading the standby node <server name>.
Continue (Y/N)?
```

13 Enter **y** to continue with the upgrade.

```
Upgrade will prepare for <server name> to take over as active. This
will stop the Acuitas server on <server name> and the HA server will
not be available until <server name> is up. Press Enter to continue.
```

14 Press **Enter** to continue with the upgrade.

```
Upgrade will now upgrade the previous active node <server name>. Press
Enter to continue.
```

15 Press **Enter** to continue with the upgrade.

```
HA pair has been upgraded successfully.
```

16 Once upgrade of the master pair is complete, on either server of the slave pair, re-combine the slave pair with the master pair:

```
cd /opt/Affirmed/NMS/bin
```

```
configure_dr convert slave --masterhost <masterhost>
```

where *masterhost* specifies the master primary hostname or IP address.

Following the Upgrade

Following the upgrade, the original master pair have switched roles:

- The active server is now the standby server
- The standby server is now the active server
- The slave pair points to different master host of original master pair

To check status of the servers:

1 On the master pair, issue the following commands to check the status.

```
cd /opt/Affirmed/NMS/bin
```

```
./emsstatus
```

Primary Server:

```
*********** Acuitas Server Status *************

Node                : <server name> - <server IP address>
```

```
EMS Package     : AffirmedEmsHA-9.4.0.0-97

EMS Process     : Running [PID]
MySQL
    Database    : Running
    Server Id   : 1
    Slave IO    : Yes
    Slave SQL   : Yes

High Availability
    Status      : Standby
    Heartbeat   : Not required by application on standby node
    Virtual IP  : <virtual IP address> - Not owned by this node
```

Secondary Server:

```
*********** Acuitas Server Status *************

Node            : <server name> - <server IP address>
EMS Package     : AffirmedEmsHA-9.4.0.0-97

EMS Process     : Running [PID]

MySQL
    Database    : Running
    Server Id   : 2
    Slave IO    : Yes
    Slave SQL   : Yes

High Availability
    Status      : Active
    Heartbeat   : Running
    Virtual IP  : <virtual IP address> - Owned by this node
***********************************************
```

2 On the slave pair, issue the following commands to check the status.

```
cd /opt/Affirmed/NMS/bin
./emsstatus
```

Primary Server:

```
*********** Acuitas Server Status *************

Node            : <server name> - <server IP address>
EMS Package     : AffirmedEmsHA-9.40.0.0-97


EMS Process     : Stopped


MySQL
    Database    : Running
    Server Id   : 3


High Availability
    Status      : Standby
    Heartbeat   : Not required by application on standby node
Disaster Recovery
    Status      : Standby
    Master Host : <master server IP address>
    Standby IO  : Yes
    Standby SQL : Yes
***********************************************
```

Note: the Master Host is pointing to the current master active node.

Secondary Server:

```
*********** Acuitas Server Status *************

Node            : <server name> - <server IP address>
EMS Package     : AffirmedEmsHA-9.4.0.0-97
```

```
EMS Process     : Stopped


MySQL
    Database    : Running
    Server Id   : 4


High Availability
    Status      : Standby
    Heartbeat   : Not required by application on standby node
Disaster Recovery
    Status      : Standby
    Master Host : <master server IP address>
    Standby IO  : Yes
    Standby SQL : Yes
***********************************************
```

Note: the Master Host is pointing to the current master standby node.

Rollback if the Upgrade Fails

If the upgrade fails during the upgrade process, you can perform a rollback procedure. The steps to follow differ depending on when the upgrade failed and the root cause of the failure.

- **Failure when upgrading the slave pair** — If the upgrade failed when executing haupgrade on the slave HA pair, follow the instructions in Failure During the Slave Pair Upgrade.
- **Failure when upgrading the master pair** — If the upgrade failed when executing haupgrade on the master HA pair, follow the instructions in Failure During the Master Pair Upgrade.
- **Failure when converting the upgraded slave** — If the upgrade failed when converting the upgraded slave site to the slave of the upgraded master site, contact the Affirmed Networks Technical Assistance Center (TAC) for assistance.

Failure During the Slave Pair Upgrade

In this scenario, the haupgrade script attempts to stop replication on both hosts of the HA and run RPM upgrade on those two hosts. First, identify why the script failed.

Check the connection between the HA pair. A good connection is required, as the script uses ssh to execute commands on the remote host. If necessary, fix the connection and run the haupgrade script again, as described in HA + DR Systems Upgrade.

If the RPM upgrade failed, do not proceed with the upgrade, as the new package is the issue. Instead, perform the rollback procedure on the slave pair.

To Rollback the Slave Pair:

1 Log in using the default user account `affirmed` and the account password.

2 Once you are logged in, issue the following command to gain super-user privileges:

```
su -
```

3 When prompted, enter the super-user password.

4 Issue the following command to obtain the name of the installed Acuitas HA package (for example, AffirmedEmsHA-9.4.0.0-44.x86_64):

```
rpm -qa | grep Ems
```

5 On both servers, issue the following command:

```
rpm -ev emsha-failed_pkg_name
```

where ***failed_pkg_name*** is the package name of the failed install.

6 On both servers, issue the following command, specifying the pathname of the old package:

```
rpm -ivh emsha-old_pkg_name
```

where ***old_pkg_name*** is the pathname of the old Acuitas HA package.

7 Issue the following command to switch back to the `affirmed` user:

```
exit
```

8 On the primary server, issue the following commands:

```
cd /opt/Affirmed/NMS/bin

./configure_ha --reload

Please double check the following configuration options:

{

  virtualhostname => 'virtual_hostname',

  virtualip => ' virtual_IP_address',

  primaryhostname => 'primary_hostname',

  primaryip => 'primary_IP_address',
```

```
  secondaryhostname => 'secondary_hostname',
  secondaryip => 'secondary_IP_address',
  primaryslaveid => 'primary_slave_ID',
  secondaryslaveid => 'secondary_slave_ID',
  primaryipinterface => 'primary_IP_interface',
  secondaryipinterface => 'secondary_IP_interface',
  ipinterface => 'IP_interface',
  netmask => 'netmask value'
}
Are you sure you want to continue with this configuration? (y/n):
```

9 Enter **y** to continue with this configuration.

When the script completes, you can check the log file for the status.

10 On either server in the slave pair, issue the following command:

```
configure_dr convert slave --masterhost <masterhost>
```

where *masterhost* specifies the master primary hostname or IP address.

When the script completes, the system returns the following messages:

```
Replication setup on this HA as slave pair is complete.

Exiting DR configuration script
```

Failure During the Master Pair Upgrade

In this scenario, the upgrade is a more complex process. First, identify why the haupgrade script failed.

Check the connection between the HA pair. A good connection is required, as the script uses ssh to execute commands on the remote host. If necessary, fix the connection and run the haupgrade script again, as described in HA + DR Systems Upgrade.

The RPM HA upgrade should not have failed, as the same package upgraded successfully on the slave pair at this point. Perform the rollback procedure on both the master pair and the slave pair (which would have already been upgraded).

To Rollback the Master and Slave Pairs:

1. Perform the rollback procedure on the master pair (first) and the slave pair (second), using the instructions in Rollback Procedure if Upgrade Fails.
2. On either server in the slave pair, issue the following command:

   `configure_dr convert slave --masterhost <masterhost>`

   where *masterhost* specifies the master primary hostname or IP address.

   When the script completes, the system returns the following messages:

```
Replication setup on this HA as slave pair is complete.

Exiting DR configuration script
```

Downgrade Procedure

A downgrade reverts the software back to an older version. The procedure requires you to specify a backup file to use for the downgrade. The following rules apply:

- The backup can be one that you created when you last ran the haupgrade script. That backup is stored in /opt/Affirmed/NMS/conf/db/preupgradedb_*hostname*.
- The backup can be one that you created with the emsdb script. For information, see Acuitas Database Management.
- The backup file must be taken from the same release software to which you are downgrading.

Note: *If the backup file* **is not** *from the same release, incompatibility issues can result. Make sure that the backup file version and downgrade version match.*

- The backup must be the one on the current standby. If the downgrade cannot locate the backup on the standby, the script quits.

Summary

To downgrade HA + DR systems, you execute commands on the master HA pair and the slave HA pair in the order indicated in the following table:

	Master HA Pair	Slave HA Pair
1		On both servers, copy the old package emsha-*old_pkg_name*.rpm to /data/affirmed. On either server, install the old package: **`cd /opt/Affirmed/NMS/bin`** **`./haupgrade /data/affirmed/emsha-`*`old_pkg_name`***.rpm
2	On both servers, copy the old package emsha-*old_pkg_name*.rpm to /data/affirmed. On either server, install the old package: **`cd /opt/Affirmed/NMS/bin`** **`./haupgrade /data/affirmed/emsha-`*`old_pkg_name`***.rpm **`--backupfile`** ***`file_path`*** where ***`file_path`*** is the pathname of the backup file.	

	Master HA Pair	Slave HA Pair
3		On either server, set up MySQL slave replication: **`cd /opt/Affirmed/NMS/bin`** **`./configure_dr convert slave`**
4	**`cd /opt/Affirmed/NMS/bin`** **`./emsstatus`**	
5		**`cd /opt/Affirmed/NMS/bin`** **`./emsstatus`**

These steps are detailed in the following procedure.

Procedure

To downgrade the master HA pair and slave HA pair:

1. Log in using the default user account `affirmed` and the account password.
2. On both slave servers, copy the old package emsha-*old_pkg_name*.rpm to /data/affirmed.

   where *old_pkg_name*.rpm is the old Acuitas package.
3. On either slave server, install the old package:

   ```
   cd /opt/Affirmed/NMS/bin
   ```

   ```
   ./haupgrade /data/affirmed/emsha-old_pkg_name.rpm
   ```

   ```
   Do you want to upgrade the HA pair on primary <server name> and
   secondary <server name> using /data/affirmed/emsha-old_pkg_name.rpm
   (Y/N)?
   ```
4. Enter **y** to continue with the downgrade.
5. On both master servers, copy the old package emsha-*old_pkg_name*.rpm to /data/affirmed.

   where *old_pkg_name*.rpm is the old Acuitas package.
6. On either of the master servers, issue the following commands:

   ```
   cd /opt/Affirmed/NMS/bin
   ```

   ```
   ./haupgrade /data/affirmed/emsha-old_pkg_name --backupfile.rpm
   <file_path>
   ```

   where ***`file_path`*** is the full path to the backup file.

   For example, if you use the file from previous upgrade, ***`file_path`*** is /opt/Affirmed/NMS/conf/db/preupgradedb_*hostname*.

   ```
   Do you want to upgrade the HA pair on primary <server name> and
   secondary <server name> using /data/affirmed/emsha-old_pkg_name (Y/N)?
   ```

7 Enter **y** to continue with the downgrade.

```
Do you want to use the <file_path> for this upgrade? Note the backup
file has to be from <server name> and from the version you are
upgrading to.
```

8 Enter **y** to continue with the downgrade.

```
... Active node is <server name>

... Standby node is <server name>

Upgrade will start by upgrading the standby node <server name> using
<file_path>. Continue (Y/N)?
```

9 Press **Enter** to continue with the downgrade. The upgrade can take some time to complete. The system returns the following message when HA is upgraded successfully.

```
HA pair has been upgraded successfully.
```

10 On either of the slave pair, issue the following command to set up MySQL slave replication:

```
configure_dr convert slave
```

When the script completes, the system returns the following message:

```
Exiting DR configuration script
```

Following the Upgrade

Following the upgrade, the original master pair have switched roles:

- The active server is now the standby server
- The standby server is now the active server
- The slave pair points to different master host of original master pair

To check status of the servers:

1 On the master pair, issue the following commands to check the status.

```
cd /opt/Affirmed/NMS/bin

./emsstatus
```

Primary Server:

```
*********** Acuitas Server Status *************

Node             : <server name> - <server IP address>
EMS Package      : AffirmedEmsHA-9.4.0.0-97

EMS Process      : Running [PID]
MySQL
    Database     : Running
    Server Id    : 1
    Slave IO     : Yes
    Slave SQL    : Yes

High Availability
    Status       : Standby
    Heartbeat    : Not required by application on standby node
    Virtual IP   : <virtual IP address> - Not owned by this node
```

Secondary Server:

```
*********** Acuitas Server Status *************

Node             : <server name> - <server IP address>
EMS Package      : AffirmedEmsHA-9.4.0.0-97

EMS Process      : Running [PID]

MySQL
    Database     : Running
    Server Id    : 2
    Slave IO     : Yes
    Slave SQL    : Yes

High Availability
```

```
    Status      : Active
    Heartbeat   : Running
    Virtual IP  : <virtual IP address> - Owned by this node
***********************************************
```

2 On the slave pair, issue the following commands to check the status.

`cd /opt/Affirmed/NMS/bin`

`./emsstatus`

Primary Server:

```
*********** Acuitas Server Status *************

Node            : <server name> - <server IP address>
EMS Package     : AffirmedEmsHA-9.4.0.0-97
EMS Process     : Stopped

MySQL
    Database    : Running
    Server Id   : 3

High Availability
    Status      : Standby
    Heartbeat   : Not required by application on standby node
Disaster Recovery
    Status      : Standby
    Master Host : <master server IP address>
    Standby IO  : Yes
    Standby SQL : Yes
***********************************************
```

Note: the Master Host is pointing to the current master active node.

Secondary Server:

```
*********** Acuitas Server Status *************

Node            : <server name> - <server IP address>
EMS Package     : AffirmedEmsHA-9.4.0.0-97

EMS Process     : Stopped

MySQL
    Database    : Running
    Server Id   : 4

High Availability
    Status      : Standby
    Heartbeat   : Not required by application on standby node
Disaster Recovery
    Status      : Standby
    Master Host : <master server IP address>
    Standby IO  : Yes
    Standby SQL : Yes
***********************************************
```

Note: the Master Host is pointing to the current master standby node.

Recovery of an HA System

For 2-node HA clusters and 4-node HA + DR clusters, Acuitas increases recoverability by maintaining ongoing replication of the Acuitas database and other data files from the active server to the standby server and DR slave servers. In the case of loss of one or more nodes in the cluster, you should first consider using the remaining nodes in the 2-node or 4-node cluster to recover.

This section provides the recovery procedure for the case where no node in the cluster is available for recovery purposes. This procedure employs an existing database backup file to perform a point-in-time recovery. Any management information between the time of the backup file and the time of recovery will be lost.

Acuitas supports recovery in the following use cases:

- A disaster takes the *entire* cluster out of service, where data cannot be recovered from any node in the cluster.
- The cluster is in service, but you want to revert the configuration and database back to an older database backup.

Prerequisite: Scheduling Database Backup

The recovery requires that you have a scheduled database backup file stored on a remote server and the database backup is taken from Acuitas version 5.3.8 or later.

Configure the Acuitas server to backup its database based on a pre-configured schedule. *Do not* schedule backups more than twice a day, as it can impact system performance (backups cause pauses in Acuitas system and affect performance/scalability of the system).

Note: *To schedule a database backup, you must have the Tools > Database Administration > Backup > Schedule Backup security group permission. For instructions, see the Acuitas User's Guide.*

To schedule a database backup:

1 Click the **Settings** icon on the toolbar  and select **Tools =>**

Database Administration.

Acuitas displays the **Database Backup and Restore** window.

2 Click the **Scheduled DB-Backup** icon. 

3 Configure the parameters as follows:

- **Backup Files Prefix Name** – Specify the prefix you want for the backup file, such as DB_backup.
- **FTP Server** – Specify to have the backup stored on a remote server. *Do not* specify to have the backup stored on the local server, as is will not be available if the entire cluster becomes out of service.
- **Relative Path in FTP Server** – Specify the path where you want the backup stored.
- **Full Backup** – Mark this checkbox to trigger a full backup.
- **Recurrence** – Mark this checkbox to specify a backup recurrence (*do not* schedule more than twice a day, as it can impact system performance):
  - Click the **Recurrence Pattern** icon and configure Recurrence Details.
  - Click the **Recurrence**.
- **Purge Backup** – Mark this checkbox to specify how many backups to retain.

4 Click **Save**.

Recovery Procedure

To recover the cluster:

1 Perform a fresh installation of the HA software on every node of the cluster. Follow the instructions in HA Server Fresh Installation and Configuration.

2 Use configure_ha --backupfile <*backupfile*> --restore to restore the database. If the previous HA configuration is available on the system and you want to use it, use the --reload option. If the configuration is not available or if you want to use a different configuration, provide all HA configuration at the command line.

```
configure_ha [--virtualhostname virtual hostname] [--virtualip virtual
IP address] [--primaryhostname primary hostname] [--primaryip primary
IP address] [--secondaryhostname secondary hostname] [--secondaryip
secondary IP address] [--netmask netmask value] [--ipinterface
interface] [--force][--reload] [--backupfile file_path]
[--synchwithactive] --restore
```

Argument	Description
Host Configuration Options: When host configuration settings are omitted, the script looks for any existing HA configuration and uses it to reconfigure the HA pair.	
`--virtualhostname <virtual hostname>`	Specifies the hostname of the HA pair. ***Note:*** *The configure_ha script accepts a hostname containing only characters, numbers, hyphen, and dot(.). It cannot start with a number, hyphen, or dot. It cannot end with a hyphen or dot.*

Argument	Description
--virtualip ***<virtual IP address>***	Specifies the logical IP address of the HA pair
--primaryhostname ***<primary hostname>***	Specifies the hostname of the primary server of the HA pair. ***Note:*** *The configure_ha script accepts a hostname containing only characters, numbers, hyphen, and dot(.). It cannot start with a number, hyphen, or dot. It cannot end with a hyphen or dot.*
--primaryip ***<primary IP address>***	Specifies the IP address of the primary server of the HA pair
--secondaryhostname ***<primary hostname>***	Specifies the hostname of the secondary server of the HA pair ***Note:*** *The configure_ha script accepts a hostname containing only characters, numbers, hyphen, and dot(.). It cannot start with a number, hyphen, or dot. It cannot end with a hyphen or dot.*
--secondaryip ***<secondary IP address>***	Specifies the IP address of the secondary server of the HA pair
--ipinterface ***<interface>***	Use only if auto detection of the IP interface fails. Specifies the interface on both the primary or secondary server, *if different than the default eth0.*
--netmask ***<netmask value>***	The netmask value to use for virtual IP address on both the primary and secondary node. Required for IPv6 and optional for IPv4. When not specified, the default is 24 for IPv4.
Optional Options:	
-help	Prints a brief help message and exits.
-man	Prints the manual page and exits.
--force	Specifies to override the existing configure_ha.conf and enter new settings as command line input.
--reload	Specifies to use configuration settings from the existing configure_ha.conf instead of from command line input.
--backupfile ***<file_path>***	Specifies the path to the MySQL database file to restore the database. You can use this option with the --reload option or when providing all the Host Configuration Options.

Argument	Description
`--syncwithactive`	Specifies to synchronize the database from the active node when bringing up the standby node. Once the database is synchronized, the system prompts to start the Acuitas server on the standby node. You can use this option when bringing up the standby node that has been down for some amount of time (such as, several hours). The option greatly reduces the time to synchronize databases (compared to the mysql synchronization). For instructions, see Configuration when Failed Node is Back In Service.
`--restore`	Restores the database from the specified path to the MySQL database file.

```
Please double check the following configuration options:
{
  virtualhostname => 'virtual_hostname',
  virtualip => ' virtual_IP_address',
  primaryhostname => 'primary_hostname',
  primaryip => 'primary_IP_address',
  secondaryhostname => 'secondary_hostname',
  secondaryip => 'secondary_IP_address',
  primaryslaveid => 'primary_slave_ID',
  secondaryslaveid => 'secondary_slave_ID',
  primaryipinterface => 'primary_IP_interface',
  secondaryipinterface => 'secondary_IP_interface',
  ipinterface => 'IP_interface',
  netmask => 'netmask value'
}
Are you sure you want to continue with this configuration? (y/n):
```

3 Enter **y** to continue with this configuration.

```
Loading the existing configuration ..
```

```
Please make sure /opt/Backup_prefix_ems-version_date-time.sql is a
scheduled backup file generated from Acuitas 5.3.8.0 and later.
Continue(Y/N)?
```

4 Enter **y** to continue. The configuration can take some time to complete. The system returns the following message when HA is configured successfully.

```
----- HA configuration set up is complete---

Do you want to start the EMS server on primary and secondary nodes?
(y/n)
```

5 Enter **y** to start the servers. The primary server starts as the active server and the secondary server starts as the standby server.

Chapter 4

Other Administrative Information

This chapter provides the following administrative information:

- Acuitas Server Administration Scripts — Describes the various scripts you use to administer the Acuitas server:
  - Configure the Server IPv6 IP Address
  - Standalone Acuitas Server Startup
  - Standalone Acuitas Server Shutdown
  - Acuitas Server Status
  - Acuitas Database Management
  - Acuitas Disaster Recovery Configuration
  - Acuitas High Availability Configuration
  - Script to Report TWAG Interval Peak Credited Sessions
  - Script to Adjust MySQL BinLog Purging Threshold
  - Script to Configure Thresholds for HA/DR Replication Lagging Alarms
  - Backup Performance Management Statistics Files to Destination Server
- NTP Server Configuration
- Configuring Acuitas to Communicate Using Public Key Infrastructure
- Changing Hostname/IP Address of the Acuitas Server
- Changing Timezone of the Acuitas Server
- Changing MySQL root/Replication User Password
- Enabling IPv6
- Increasing Acuitas Server Resources on a Virtual Machine
- Optional: Channel Bonding on Acuitas Physical Interfaces

Acuitas Server Administration Scripts

To run Acuitas server administration scripts, log in using the default user account `affirmed` and the account password.

Configure the Server IPv6 IP Address

If desired to have the server configured to be started in IPv6 mode, use set_ipv6_address to configure the Acuitas server with an IPv6 IP address. The script sets the IPv6 IP address only once.

Otherwise, you can start the server in the default IPv4 mode.

To configure the Acuitas Server with an IPv6 IP address:

1 Enter the following commands:

```
cd /opt/Affirmed/NMS/bin

./set_ipv6_address

  1  eth0  2001:470:8865:601::5e

  2  eth0  2001:470:8865:601:20c:29ff:fef1:7f89

Select the server IP Address from the list above: [1]
```

2 Enter the number of the server IP address you want configured.

3 At the confirmation prompt, enter **y**.

Standalone Acuitas Server Startup

Use starttems.sh to start the Acuitas server:

```
cd /opt/Affirmed/NMS/bin

./starttems.sh
```

Standalone Acuitas Server Shutdown

Use emsstop to shut down the Acuitas server:

```
cd /opt/Affirmed/NMS/bin

./emsstop
```

Acuitas Server Status

Use emsstatus to determine the current status of the Acuitas server:

`cd /opt/Affirmed/NMS/bin`

`./emsstatus`

Acuitas displays the server status. The display depending on the server's role, as shown in the following examples.

Standalone Server

```
*********** Acuitas Server Status *************

Node                    : <server name> - <server IP address>

EMS Package             : AffirmedEms-9.4.0.0-97



EMS Process             : Running [PID]

MySQL

    Database          : Running

    Server Id           : 1

***********************************************
```

Standalone Disaster Recovery Slave Server

```
*********** Acuitas Server Status *************

Node                    : <server name> - <server IP address>

EMS Package             : AffirmedEms-9.4.0.0-97



EMS Process             : Stopped

MySQL

    Database            : Running

    Server Id           : 2



Disaster Recovery
```

```
    Status              : Standby
    Master Host         : <master server IP address>
    Standby IO          : Yes
    Standby SQL         : Yes
***********************************************
```

HA Active Server

```
*********** Acuitas Server Status *************
Node                     : <server name> - <server IP address>
EMS Package              : AffirmedEmsHA-9.4.0.0-97

EMS Process              : Running [PID]
MySQL
    Database             : Running
    Server Id            : 1
    Slave IO             : Yes
    Slave SQL            : Yes

High Availability
    Status               : Active
    Heartbeat            : Running
    Virtual IP           : <virtual IP address> - Owned by this node
***********************************************
```

HA Standby Server

```
*********** Acuitas Server Status *************

Node                     : <server name> - <server IP address>

EMS Package              : AffirmedEmsHA-9.4.0.0-97


EMS Process              : Running [PID]

MySQL

    Database             : Running

    Server Id            : 2

    Slave IO             : Yes

    Slave SQL            : Yes


High Availability

    Status               : Standby

    Heartbeat            : Not required by application on standby node

    Virtual IP           : <virtual IP address> - Not owned by this node

***********************************************
```

HA Disaster Recovery Slave Servers

Primary Server

```
*********** Acuitas Server Status *************

Node                        : <server name> - <server IP address>

EMS Package                 : AffirmedEmsHA-9.4.0.0-97



EMS Process                 : Stopped

MySQL

    Database                : Running

    Server Id               : 3



High Availability

    Status                  : Standby

    Heartbeat               : Not required by application on standby node



Disaster Recovery

    Status                  : Standby

    Master Host             : <master server IP address>

    Standby IO              : Yes

    Standby SQL             : Yes

***********************************************
```

Secondary Server

```
*********** Acuitas Server Status *************


Node                        : <server name> - <server IP address>
EMS Package                 : AffirmedEmsHA-9.4.0.0-97


EMS Process                 : Stopped
MySQL
    Database                : Running
    Server Id               : 4
High Availability
    Status                  : Standby
    Heartbeat               : Not required by application on standby node


Disaster Recovery
    Status                  : Standby
    Master Host             : <master server IP address>
    Standby IO              : Yes
    Standby SQL             : Yes
***********************************************
```

Acuitas Database Management

Use emsdb to perform common database tasks.

Command Syntax

emsdb <*args*> [--host *host*] [--port *port*] [--user *user*] [--password *password*]

Argument	Description
args	
backup --backupfile <*file_path*>	Backs up the MySQL database.
stopreplication	Stops replication of the MySQL database from the configured master.
-help	Prints a brief help message and exits.
-man	Prints the manual page and exits.
Optional Local Database Connection Options: Use to connect to the database at host:port, using credentials user:password.	
--host <*host*>	Specifies the hostname
--port <*port*>	Specifies the port
--user <*user*>	Specifies the user
--password <*password*>	Specifies the user password

Acuitas Disaster Recovery Configuration

Use configure_dr to set up Disaster Recovery operations on either standalone DR systems or HA + DR systems.

The configure_dr script runs in the background. The script prompts for input and then displays the message:

```
The command $COMMAND is running in the background.
Check log file $LOG_FILE for the command result.
```

When you use the --client option, the process runs in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

Command Syntax

```
configure_dr <args> [--host host] [--port port] [--user user]
[--password password] [--masterhost masterhost] [--repluser repluser]
[--replpassword replpassword] [--slaveid slaveid] [--force] [--client]
[--currenthost]
```

Argument	Description
Required Options:	
args	
convert master	Converts the server to be a standalone master.
convert slave	Converts the server to be a slave of masterhost.
--masterhost <*masterhost*>	Specifies the master hostname. Required only when specifying a new masterhost. Not required if the masterhost was previously configured and no change is needed.
Optional Local Database Connection Options: Use to connect to the database at host:port, using credentials user:password.	
--host <*host*>	Specifies the hostname
--port <*port*>	Specifies the port
--user <*user*>	Specifies the user
--password <*password*>	Specifies the user password
Optional Master Database Connection Options: Use to connect to the Master database at masterhost, using credentials repluser:replpassword.	
--repluser <*repluser*>	Specifies the replicate user
--replpassword <*replpassword*>	Specifies the replicate user password

Argument	Description
`--slaveid <slaveid>`	Specifies the slave when converting server to slave. Use to override default server-id (2) when converting server to slave. Default is 2 for standalone slave only.
Optional Options:	
`--forceconvert`	Specifies to skip checking of master pair status and forcefully convert the slave into master.
`-help`	Prints a brief help message and exits.
`-man`	Prints the manual page and exits.
`--client`	Runs the script in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.
`--currenthost`	Runs the script only on the current host in an HA pair. This option could be used only if the current node is not properly configured as the DR slave of the <*masterhost*>. Use the master primary host as <*masterhost*> for the DR slave primary host and use the master secondary host as <*masterhost*> for the DR slave secondary host.

Acuitas High Availability Configuration

Use configure_ha to set up High Availability operations on either HA systems or HA + DR systems. Run the script *only* on the primary node(s):

- Primary node of an HA system
- Primary master node and primary slave node of an HA + DR system

The configure_ha script runs in the background. The script prompts for input and then displays the message:

```
The command $COMMAND is running in the background.
Check log file $LOG_FILE for the command result.
```

The configuration entered at the command line is saved in the configure_ha file:

- Use the **`--reload`** argument to reload the existing configuration and re-set up the HA pair. The command usually does not require any specific options.

  It can be used in combination with the **`–backupfile`** argument:

  ```
  ./configure_ha --reload [--backupfile <backupdb>]
  ```

- Use the **`--force`** argument to override the configuration and enter new settings at the command line:

  ```
  ./configure_ha --force [--virtualhostname virtual hostname]
  [--virtualip virtual IP address] [--primaryhostname primary
  hostname] [--primaryip primary IP address] [--secondaryhostname
  secondary hostname] [--secondaryip secondary IP address] [--netmask
  netmask value] [--ipinterface interface] [--backupfile file_path]
  --restore
  ```

- Use the **`--syncwithactive`** argument to resynchronize a standby server's database with the active server. The command does not change existing configuration. The command is intended for a standby server that has been out of service for an extended period of time. Once the standby server is back in service, use this command to bring the server up to speed.

- Use the **`--restore`** argument to revert the configuration back to an older database backup.

Use the **`--client`** option to run the script in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

For sample command output of each of these arguments, see Sample Command Output.

Command Syntax

configure_ha [--virtualhostname *virtual hostname*] [--virtualip *virtual IP address*] [--primaryhostname *primary hostname*] [--primaryip *primary IP address*] [--secondaryhostname *secondary hostname*] [--secondaryip *secondary IP address*] [--netmask *netmask value*] [--ipinterface *interface*] [--force][--reload] [--backupfile *file_path*] [--synchwithactive] --restore [--client]

Argument	Description
Host Configuration Options: When host configuration settings are omitted, the script looks for any existing HA configuration and uses it to reconfigure the HA pair.	
--virtualhostname <*virtual hostname*>	Specifies the hostname of the HA pair. ***Note:*** *The configure_ha script accepts a hostname containing only characters, numbers, hyphen, and dot(.). It cannot start with a number, hyphen, or dot. It cannot end with a hyphen or dot.*
--virtualip <*virtual IP address*>	Specifies the logical IP address of the HA pair
--primaryhostname <*primary hostname*>	Specifies the hostname of the primary server of the HA pair. ***Note:*** *The configure_ha script accepts a hostname containing only characters, numbers, hyphen, and dot(.). It cannot start with a number, hyphen, or dot. It cannot end with a hyphen or dot.*
--primaryip <*primary IP address*>	Specifies the IP address of the primary server of the HA pair
--secondaryhostname <*primary hostname*>	Specifies the hostname of the secondary server of the HA pair. ***Note:*** *The configure_ha script accepts a hostname containing only characters, numbers, hyphen, and dot(.). It cannot start with a number, hyphen, or dot. It cannot end with a hyphen or dot.*
--secondaryip <*secondary IP address*>	Specifies the IP address of the secondary server of the HA pair
--ipinterface <*interface*>	Use only if auto detection of the IP interface fails. Specifies the interface on both the primary or secondary server, *if different than the default eth0.*
--netmask <*netmask value*>	The netmask value to use for virtual IP address on both the primary and secondary node. Required for IPv6 and optional for IPv4. When not specified, the default is 24 for IPv4.
Optional Options:	
-help	Prints a brief help message and exits.

Argument	Description
-man	Prints the manual page and exits.
--force	Specifies to override the existing configure_ha.conf and enter new settings as command line input.
--reload	Specifies to use configuration settings from the existing configure_ha.conf instead of from command line input.
--backupfile <*file_path*>	Specifies the path to the MySQL database file to restore the database. You can use this option with the --reload option or with all the Host Configuration Options.
--syncwithactive	Specifies to synchronize the database from the active node when bringing up the standby node. Once the database is synchronized, the system prompts to start the Acuitas server on the standby node. You can use this option when bringing up the standby node that has been down for some amount of time (such as, several hours). The option greatly reduces the time to synchronize databases (compared to the MySQL synchronization). For instructions, see Configuration when Failed Node is Back In Service.
--restore	Restores the database from the specified path to the MySQL database file.
--client	Runs the script in the foreground. Interrupting the script with <CTRL>-C terminates the operation, and is not recommended. The process *will not* continue to run in the background.

Sample Command Output

configure_ha with --reload argument

```
./configure_ha --reload

Please double check the following configuration options:
{
  virtualhostname => 'HA9177',
  virtualip => '10.32.9.177',
  primaryhostname => 'qahpg8ems03',
  primaryip => '10.32.7.177',
  secondaryhostname => 'qahpg8ems04',
  secondaryip => '10.32.8.177',
  primaryslaveid => '1',
  secondaryslaveid => '2',
  primaryipinterface => 'eth1',
  secondaryipinterface => 'eth1',
  ipinterface => 'eth1',
  netmask => '24'
}

Are you sure you want to continue with this configuration? (Y/N): Y
Loading the existing configuration ..
RUN in background

Please double check the following configuration options:
{
  virtualhostname => 'HA9177',
  virtualip => '10.32.9.177',
  primaryhostname => 'qahpg8ems03',
  primaryip => '10.32.7.177',
  secondaryhostname => 'qahpg8ems04',
  secondaryip => '10.32.8.177',
  primaryslaveid => '1',
  secondaryslaveid => '2',
  primaryipinterface => 'eth1',
  secondaryipinterface => 'eth1',
  ipinterface => 'eth1',
  netmask => '24'
}
```

```
Loading the existing configuration ..
Setting up the host files on the primary and secondary nodes...
Setting up the sshd_config files on the primary and secondary nodes...
Add 10.32.7.177,10.32.8.177 to the local node 10.32.7.177 ...
Add 10.32.7.177,10.32.8.177 to the node 10.32.8.177 ...
Configuring iptables on the primary and secondary nodes...
iptables: Saving firewall rules to /etc/sysconfig/iptables: [  OK  ]
iptables: Saving firewall rules to /etc/sysconfig/iptables: [  OK  ]
Clearing out any prior pacemaker configuration ...
Stopping Cluster (pacemaker)...
Stopping Cluster (corosync)...
Warning: Unable to load CIB to get guest and remote nodes from it,
those nodes will not be deconfigured.
qahpg8ems03: Stopping Cluster (pacemaker)...
qahpg8ems04: Stopping Cluster (pacemaker)...
qahpg8ems03: Successfully destroyed cluster
qahpg8ems04: Successfully destroyed cluster
Creating the pacemaker cluster ...
Creating cluster of qahpg8ems03 and qahpg8ems04 ...
qahpg8ems03: Already authorized
qahpg8ems04: Already authorized
Destroying cluster on nodes: qahpg8ems03, qahpg8ems04...
qahpg8ems04: Stopping Cluster (pacemaker)...
qahpg8ems03: Stopping Cluster (pacemaker)...
qahpg8ems04: Successfully destroyed cluster
qahpg8ems03: Successfully destroyed cluster

Sending 'pacemaker_remote authkey' to 'qahpg8ems03', 'qahpg8ems04'
qahpg8ems03: successful distribution of the file 'pacemaker_remote
authkey'
qahpg8ems04: successful distribution of the file 'pacemaker_remote
authkey'
Sending cluster config files to the nodes...
qahpg8ems03: Succeeded
qahpg8ems04: Succeeded

Starting cluster on nodes: qahpg8ems03, qahpg8ems04...
qahpg8ems04: Starting Cluster...
qahpg8ems03: Starting Cluster...
qahpg8ems03: Cluster Enabled
qahpg8ems04: Cluster Enabled

Synchronizing pcsd certificates on nodes qahpg8ems03, qahpg8ems04...
```

```
qahpg8ems03: Success
qahpg8ems04: Success
Restarting pcsd on the nodes in order to reload the certificates...
qahpg8ems03: Success
qahpg8ems04: Success
waiting for qahpg8ems03 to complete pcs startup ...
Creating virtualip resource: 10.32.9.177 ...
pcs active:
pcs active:
Waiting for resource to start ...
pcs active: qahpg8ems03
Setting up mysql replication on the primary and secondary nodes ...
Can't open /tmp/tmp_my.cnf: No such file or directory.
Restarting Mysql on primary and secondary nodes...


Taking a backup of database on primary node...

Importing primary database into 10.32.8.177 and starting replication
...
Replication complete...

----- HA configuration set up is complete---
Starting the EMS server..
emsnode_name->qahpg8ems03, muticast_address->224.0.0.72
primary node command /opt/Affirmed/NMS/bin/../bin/starttems.sh  -b
0.0.0.0 -u 224.0.0.72 -n qahpg8ems03 -j 10.32.7.177
qahpg8ems03
Java HotSpot(TM) 64-Bit Server VM warning: ignoring option
MaxPermSize=512m; support was removed in 8.0

        Affirmed Element Management System
        ==================================

  Checking for License ...
 Checking for License ...         [ Checking for License ... ]
 Checking for License ...          [ Done ]

  Checking for port availability...
  Checking for port availability...       [ Checking for port
availability... ]
 Checking for port availability...    [ Ports are Available ]
 Checking for port availability...    [ Done ]
```

```
  Checking for DB connection and Migration...
  Checking for DB connection and Migration...[ Checking for DB
connection... ]
  Checking for DB connection and Migration...[ Checking for DB
migration... ]
 Checking for DB connection and Migration...[ Done ]

  Updating cluster node status...
  Updating cluster node status...         [ Updating cluster node
status... ]
 Updating cluster node status...     [ Done ]

  Checking for Hearbeat Services...
  Checking for Hearbeat Services...       [ Checking for Hearbeat
Services ... ]
 Checking for Hearbeat Services...    [ Done ]

  Initializing EMS Server ...
Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Active

Starting services on secondary node..
Starting the EMS server..
emsnode_name->qahpg8ems04, muticast_address->224.0.0.72
secondary node command /opt/Affirmed/NMS/bin/../bin/starttems.sh  -b
0.0.0.0 -u 224.0.0.72 -n qahpg8ems04 -j 10.32.8.177
Server successfully started on Standby node ..
```

configure_ha with --force argument

```
./configure_ha --virtualhostname HA9177 --virtualip 10.32.9.177
--primaryhostname qahpg8ems03 --primaryip 10.32.7.177
--secondaryhostname qahpg8ems04 --secondaryip 10.32.8.177
--ipinterface eth1 –force
{
  virtualhostname => 'HA9177',
  virtualip => '10.32.9.177',
  primaryhostname => 'qahpg8ems03',
  primaryip => '10.32.7.177',
  secondaryhostname => 'qahpg8ems04',
  secondaryip => '10.32.8.177',
  primaryslaveid => '1',
  secondaryslaveid => '2',
  primaryipinterface => 'eth0',
  secondaryipinterface => 'eth0',
  ipinterface => 'eth1',
  netmask => '24'
}Do you want to override the above existing configuration? (Y/N): Y

Please double check the following configuration options:
{
  virtualhostname => 'HA9177',
  virtualip => '10.32.9.177',
  primaryhostname => 'qahpg8ems03',
  primaryip => '10.32.7.177',
  secondaryhostname => 'qahpg8ems04',
  secondaryip => '10.32.8.177',
  primaryslaveid => '1',
  secondaryslaveid => '2',
  primaryipinterface => 'eth1',
  secondaryipinterface => 'eth1',
  ipinterface => 'eth1',
  netmask => '24'
}
Are you sure you want to continue with this configuration? (Y/N): Y

Setting up the host files on the primary and secondary nodes...
Setting up the sshd_config files on the primary and secondary nodes...
Add 10.32.7.177,10.32.8.177 to the local node 10.32.7.177 ...
Add 10.32.7.177,10.32.8.177 to the node 10.32.8.177 ...
Configuring iptables on the primary and secondary nodes...
```

```
iptables: Saving firewall rules to /etc/sysconfig/iptables: [  OK  ]
iptables: Saving firewall rules to /etc/sysconfig/iptables: [  OK  ]
Clearing out any prior pacemaker configuration ...
Stopping Cluster (pacemaker)...
Stopping Cluster (corosync)...
Warning: Unable to load CIB to get guest and remote nodes from it,
those nodes will not be deconfigured.
qahpg8ems03: Stopping Cluster (pacemaker)...
qahpg8ems04: Stopping Cluster (pacemaker)...
qahpg8ems03: Successfully destroyed cluster
qahpg8ems04: Successfully destroyed cluster
Creating the pacemaker cluster ...
Creating cluster of qahpg8ems03 and qahpg8ems04 ...
qahpg8ems03: Already authorized
qahpg8ems04: Already authorized
Destroying cluster on nodes: qahpg8ems03, qahpg8ems04...
qahpg8ems04: Stopping Cluster (pacemaker)...
qahpg8ems03: Stopping Cluster (pacemaker)...
qahpg8ems04: Successfully destroyed cluster
qahpg8ems03: Successfully destroyed cluster

Sending 'pacemaker_remote authkey' to 'qahpg8ems03', 'qahpg8ems04'
qahpg8ems04: successful distribution of the file 'pacemaker_remote
authkey'
qahpg8ems03: successful distribution of the file 'pacemaker_remote
authkey'
Sending cluster config files to the nodes...
qahpg8ems03: Succeeded
qahpg8ems04: Succeeded

Starting cluster on nodes: qahpg8ems03, qahpg8ems04...
qahpg8ems03: Starting Cluster...
qahpg8ems04: Starting Cluster...
qahpg8ems03: Cluster Enabled
qahpg8ems04: Cluster Enabled

Synchronizing pcsd certificates on nodes qahpg8ems03, qahpg8ems04...
qahpg8ems03: Success
qahpg8ems04: Success
Restarting pcsd on the nodes in order to reload the certificates...
qahpg8ems03: Success
qahpg8ems04: Success
waiting for qahpg8ems03 to complete pcs startup ...
```

```
Creating virtualip resource: 10.32.9.177 ...
pcs active:
pcs active:
Waiting for resource to start ...
pcs active: qahpg8ems03
Setting up mysql replication on the primary and secondary nodes ...
Can't open /tmp/tmp_my.cnf: No such file or directory.
Restarting Mysql on primary and secondary nodes...


Taking a backup of database on primary node...

Importing primary database into 10.32.8.177 and starting replication
...
Replication complete...

----- HA configuration set up is complete---
Starting the EMS server..
emsnode_name->qahpg8ems03, muticast_address->224.0.0.72
primary node command /opt/Affirmed/NMS/bin/../bin/starttems.sh  -b
0.0.0.0 -u 224.0.0.72 -n qahpg8ems03 -j 10.32.7.177
qahpg8ems03
Java HotSpot(TM) 64-Bit Server VM warning: ignoring option
MaxPermSize=512m; support was removed in 8.0

        Affirmed Element Management System
        ==================================

  Checking for License ...
 Checking for License ...        [ Checking for License ... ]
 Checking for License ...         [ Done ]

  Checking for port availability...
  Checking for port availability...       [ Checking for port
availability... ]
 Checking for port availability...    [ Ports are Available ]
 Checking for port availability...    [ Done ]

  Checking for DB connection and Migration...
  Checking for DB connection and Migration...[ Checking for DB
connection... ]
  Checking for DB connection and Migration...[ Checking for DB
migration... ]
 Checking for DB connection and Migration...[ Done ]
```

```
  Updating cluster node status...
  Updating cluster node status...        [ Updating cluster node
status... ]
 Updating cluster node status...     [ Done ]

  Checking for Hearbeat Services...
  Checking for Hearbeat Services...      [ Checking for Hearbeat
Services ... ]
 Checking for Hearbeat Services...    [ Done ]

  Initializing EMS Server ...
Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Active

Starting services on secondary node..
Starting the EMS server..
emsnode_name->qahpg8ems04, muticast_address->224.0.0.72
secondary node command /opt/Affirmed/NMS/bin/../bin/starttems.sh  -b
0.0.0.0 -u 224.0.0.72 -n qahpg8ems04 -j 10.32.8.177
Server successfully started on Standby node ..
```

configure_ha with --syncwithactive argument

```
./configure_ha --syncwithactive
Syncing the database with the active node..
Sync with the active node is complete. Checking to see if there is DR
node to sync ...
slave : 10.32.3.33
Do you want to start the EMS server on the current node(Y/N)?Y
Starting the EMS server..
emsnode_name->qahpg8ems02.affirmed.local, muticast_address->224.0.0.88
primary node command /opt/Affirmed/NMS/bin/../bin/starttems.sh -b
0.0.0.0 -u 224.0.0.88 -n qahpg8ems02.affirmed.local -j 10.32.4.33
qahpg8ems02.affirmed.local
nohup: ignoring input
[root@qahpg8ems02 bin]# Java HotSpot(TM) 64-Bit Server VM warning:
ignoring option MaxPermSize=512m; support was removed in 8.0

        Affirmed Element Management System
        ==================================

  Checking for License ...
 Checking for License ...        [ Checking for License ... ]
 Checking for License ...         [ Done ]

  Checking for port availability...
  Checking for port availability...       [ Checking for port
availability... ]
 Checking for port availability...    [ Ports are Available ]
 Checking for port availability...    [ Done ]

  Checking for DB connection and Migration...
  Checking for DB connection and Migration...[ Checking for DB
connection... ]
  Checking for DB connection and Migration...[ Checking for DB
migration... ]
 Checking for DB connection and Migration...[ Done ]

  Updating cluster node status...
  Updating cluster node status...         [ Updating cluster node
status... ]
 Updating cluster node status...     [ Done ]

  Checking for Hearbeat Services...
```

```
  Checking for Hearbeat Services...       [ Checking for Hearbeat
Services ... ]
 Checking for Hearbeat Services...    [ Done ]

  Initializing EMS Server ...
[root@qahpg8ems02 bin]#
[root@qahpg8ems02 bin]# ./emsstatus

*********** Acuitas Server Status *************

Node            : qahpg8ems02.affirmed.local - 10.32.4.33
EMS Package     : AffirmedEmsHA-9.4.0.0-97

EMS Process     : Running [4780]
MySQL
    Database    : Running
    Server Id   : 2
    Slave IO    : Yes
    Slave SQL   : Yes

High Availability
    Status      : Standby
    Heartbeat   : Not required by application on standby node
    Virtual IP  : 10.32.4.33 - Not owned by this node

***********************************************
```

configure_ha with --reload --restore arguments

```
./configure_ha --reload --restore --backupfile
/opt/Affirmed/NMS/server/databaseBackup/system/Database_Backup_ems-7.0
.0.0-28_10-17-2016-16-45-24.sql

Please double check the following configuration options:
{
  virtualhostname => 'HA1233',
  virtualip => '10.32.12.33',
  primaryhostname => 'qahpg8ems01.affirmed.local',
  primaryip => '10.32.3.33',
  secondaryhostname => 'qahpg8ems02.affirmed.local',
  secondaryip => '10.32.4.33',
  primaryslaveid => '1',
  secondaryslaveid => '2',
  primaryipinterface => 'eth0',
  secondaryipinterface => 'eth0',
  ipinterface => 'eth0',
  netmask => '24'
}

Are you sure you want to continue with this configuration? (Y/N):
Loading the existing configuration ..
Please make sure
/opt/Affirmed/NMS/server/databaseBackup/system/Database_Backup_ems-7.0
.0.0-28_10-17-2016-16-45-24.sql is a scheduled backup file generated
from Acuitas 5.3.8.0 and later. Continue(Y/N)?Setting up the host
files on the primary and secondary nodes...
Configuring iptables on the primary and secondary nodes...
Clearing out any prior pacemaker configuration ...
Stopping Cluster (pacemaker)...
Stopping Cluster (cman)...
qahpg8ems01.affirmed.local: Stopping Cluster (pacemaker)...
qahpg8ems02.affirmed.local: Stopping Cluster (pacemaker)...
qahpg8ems01.affirmed.local: Successfully destroyed cluster
qahpg8ems02.affirmed.local: Successfully destroyed cluster
Creating the pacemaker cluster ...
Creating cluster of qahpg8ems01.affirmed.local and
qahpg8ems02.affirmed.local ...
qahpg8ems01.affirmed.local: Authorized
qahpg8ems02.affirmed.local: Authorized
Destroying cluster on nodes: qahpg8ems01.affirmed.local,
qahpg8ems02.affirmed.local...
```

```
qahpg8ems01.affirmed.local: Stopping Cluster (pacemaker)...
qahpg8ems02.affirmed.local: Stopping Cluster (pacemaker)...
qahpg8ems02.affirmed.local: Successfully destroyed cluster
qahpg8ems01.affirmed.local: Successfully destroyed cluster

Sending cluster config files to the nodes...
qahpg8ems01.affirmed.local: Updated cluster.conf...
qahpg8ems02.affirmed.local: Updated cluster.conf...

Starting cluster on nodes: qahpg8ems01.affirmed.local,
qahpg8ems02.affirmed.local...
qahpg8ems02.affirmed.local: Starting Cluster...
qahpg8ems01.affirmed.local: Starting Cluster...
qahpg8ems01.affirmed.local: Cluster Enabled
qahpg8ems02.affirmed.local: Cluster Enabled

Synchronizing pcsd certificates on nodes qahpg8ems01.affirmed.local,
qahpg8ems02.affirmed.local...
qahpg8ems01.affirmed.local: Success
qahpg8ems02.affirmed.local: Success

Restarting pcsd on the nodes in order to reload the certificates...
qahpg8ems01.affirmed.local: Success
qahpg8ems02.affirmed.local: Success
waiting for qahpg8ems01.affirmed.local to complete pcs startup ...
waiting for qahpg8ems01.affirmed.local to complete pcs startup ...
waiting for qahpg8ems01.affirmed.local to complete pcs startup ...
Creating virtualip resource: 2001:470:8865:2000:10:32:12:33 ...
pcs active:
pcs active:
Waiting for resource to start ...
pcs active: qahpg8ems01.affirmed.local
Setting up mysql replication on the primary and secondary nodes ...
Restarting Mysql on primary and secondary nodes...
Shutting down MySQL..... SUCCESS!
Starting MySQL......... SUCCESS!
Shutting down MySQL..... SUCCESS!
Starting MySQL......... SUCCESS!
Importing the data to primary node using the backup file specified...
Importing
/opt/Affirmed/NMS/server/databaseBackup/system/Database_Backup_ems-7.0
.0.0-28_10-17-2016-16-45-24.sql into localhost database ...
Running database upgrade..
Taking a backup of database on primary node...
```

```
Replication complete...

----- HA configuration set up is complete---

Do you want to start the EMS server on primary and secondary nodes?
(Y/N): Starting the EMS server..
emsnode_name->qahpg8ems01.affirmed.local, muticast_address->224.0.0.88
primary node command /opt/Affirmed/NMS/bin/../bin/starttems.sh -b
0.0.0.0 -u 224.0.0.88 -n qahpg8ems01.affirmed.local -j 10.32.3.33
qahpg8ems01.affirmed.local
Java HotSpot(TM) 64-Bit Server VM warning: ignoring option
MaxPermSize=512m; support was removed in 8.0

        Affirmed Element Management System
        ==================================

  Checking for License ...
 Checking for License ...        [ Checking for License ... ]
 Checking for License ...         [ Done ]

  Checking for port availability...
  Checking for port availability...       [ Checking for port
availability... ]
 Checking for port availability...    [ Ports are Available ]
 Checking for port availability...    [ Done ]

  Checking for DB connection and Migration...
  Checking for DB connection and Migration...[ Checking for DB
connection... ]
  Checking for DB connection and Migration...[ Checking for DB
migration... ]
 Checking for DB connection and Migration...[ Done ]

  Updating cluster node status...
  Updating cluster node status...         [ Updating cluster node
status... ]
 Updating cluster node status...     [ Done ]

  Checking for Hearbeat Services...
  Checking for Hearbeat Services...       [ Checking for Hearbeat
Services ... ]
 Checking for Hearbeat Services...    [ Done ]

  Initializing EMS Server ...
Waiting for primary to be up fully: Standby
```

```
Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Standby

Waiting for primary to be up fully: Active

Starting services on secondary node..
Starting the EMS server..
emsnode_name->qahpg8ems02.affirmed.local, muticast_address->224.0.0.88
secondary node command /opt/Affirmed/NMS/bin/../bin/starttems.sh -b
0.0.0.0 -u 224.0.0.88 -n qahpg8ems02.affirmed.local -j 10.32.4.33
Server successfully started on Standby node ..

  Initializing EMS Server ...            [ Server started successfully
]
 Initializing EMS Server ...       [ Done ]

Server started successfully
```

Script to Report TWAG Interval Peak Credited Sessions

Use gen_TWAG_usage_report.sh to generate a report of TWAG Interval Peak number of credited sessions to a CSV file:

```
cd /opt/Affirmed/NMS/bin
```

```
./gen_TWAG_usage_report.sh <options>
```

Argument	Description
`-s <start date>`	Report's start date Default: First day of last month Format: yyyy-mm-dd
`-e <end date>`	Report's end date Default: last day of last month Format: yyyy-mm-dd
`-d <directory path>`	Directory path to save the report Default: /opt/Affirmed/NMS/server/ems/report
`-h`	Prints a brief help message and exits.

For example, the following command generates a report from 2015-01-01 to 2015-12-31 and saves the CSV file in the /home/ems directory:

```
gen_TWAG_usage_report -d /home/ems -s 2015-01-01 -e 2015-12-31
```

You can set up a cron job to run the script periodically. For example, for a monthly cron job, add the following line to the cronjob table:

```
30 1 1 * * root sh /opt/Affirmed/NMS/bin/gen_TWAG_usage_report.sh >
/dev/null 2>&1
```

Sample Command Output

```
#resourcename,date,zone,wag,PeakCreditedSessions
PMSimulator0,2016-01-20,default,wagSim,3000
PMSimulator0,2016-01-21,default,wagSim,3000
PMSimulator0,2016-01-22,default,wagSim,3000
PMSimulator0,2016-01-23,default,wagSim,3000
PMSimulator0,2016-01-24,default,wagSim,3000
PMSimulator0,2016-01-25,default,gwSim,13500
PMSimulator0,2016-01-25,default,wagSim,3000
```

Script to Adjust MySQL BinLog Purging Threshold

By default, the MySQL BinLog purging threshold is set to 85, indicating that the MySQL BinLogs are retained until the disk usage reaches 85%. The purging threshold is configurable.

Use the getsysconfig command to determine the purging threshold of the sysconfig with attribute name `purgebinlogthreshold`:

```
cd /opt/Affirmed/NMS/bin

./emsdb getsysconfig --purgebinlogthreshold

purgebinlogthreshold = 85
```

Use the setsysconfig command to set the purging threshold to a different value:

```
./emsdb setsysconfig --purgebinlogthreshold=<threshold value>
```

For example:

```
./emsdb setsysconfig --purgebinlogthreshold=70
```

Script to Configure Thresholds for HA/DR Replication Lagging Alarms

Replication monitoring uses thresholds for sending HA/DR replication lagging alarms. The default alarm threshold values are:

Threshold Name	Default Value (secs)
LAG_BEHIND_WARNING_THRESHOLD: HA Replication Warning Alarm Threshold	5
LAG_BEHIND_CRITICAL_THRESHOLD: HA Replication Critical Alarm Threshold	300
SLAVE_LAG_BEHIND_WARNING_THRESHOLD: DR Replication Warning Alarm Threshold	30
SLAVE_LAG_BEHIND_CRITICAL_THRESHOLD: DR Replication Critical Alarm Threshold	300

These threshold values are configurable. Use the emsdb script to list/set the replication lagging threshold values.

To list current threshold settings:

1 Enter the following commands (the script must be run from the /bin directory):

```
cd /opt/Affirmed/NMS/bin

./emsdb showThresholds

   Threshold Name                                        Value
---------------------------------------------------------------------
    LAG_BEHIND_WARNING_THRESHOLD                           5
    SLAVE_LAG_BEHIND_WARNING_THRESHOLD                     30
    LAG_BEHIND_CRITICAL_THRESHOLD                          300
    SLAVE_LAG_BEHIND_CRITICAL_THRESHOLD                    300
```

To set a threshold setting:

1 Enter the following commands (the script must be run from the /bin directory):

```
cd /opt/Affirmed/NMS/bin

./emsdb setThreshold --name <threshold name> --value <threshold value>
```

For example:

```
./emsdb setThreshold --name LAG_BEHIND_WARNING_THRESHOLD --value 60
```

2 Reissue the shoe command to verify the change was made:

```
cd /opt/Affirmed/NMS/bin

./emsdb showThresholds

   Threshold Name                                        Value
---------------------------------------------------------------------
    LAG_BEHIND_WARNING_THRESHOLD                           60
    SLAVE_LAG_BEHIND_WARNING_THRESHOLD                     30
    LAG_BEHIND_CRITICAL_THRESHOLD                          300
    SLAVE_LAG_BEHIND_CRITICAL_THRESHOLD                    300
```

In addition, the Replication alarms indicate the replication direction:

- primary => secondary examples:

  ```
  Replication server (10.32.7.177 => 10.32.8.177) is 241 seconds
  behind the master

  Replication server (10.32.7.177 => 10.32.8.177) is down
  ```

- secondary => primary examples:

  ```
  Replication server (10.32.8.177 => 10.32.7.177) is 825 seconds
  behind the master
  ```

  ```
  Replication server (10.32.8.177 => 10.32.7.177) is down
  ```

Script to Add/Remove Virtual IP Addresses to/from Acuitas HA Cluster

Use configure_vip to add/remove additional virtual IP addresses to/from the Acuitas HA cluster. To add/remove additional virtual IP addresses, run the script from the primary server. When you configure additional virtual IP addresses, Acuitas validates that they are co-located on the same active node to allow proper functions of the HA, as only the active node runs the Acuitas application.

Each additional VIP configuration is saved to the specified file in the following directory on both servers:

`/opt/Affirmed/NMS/conf/cluster/vips`

The script is supported on IPv4 only.

Command Syntax

`configure_vip` ***`<args>`*** **`[--vipname`** ***`file name`*** **`] [--vipaddr`** ***`virtual IP address`*** **`] [--nic`** ***`NIC number`*** **`] [--netmask`** ***`netmask value`*** **`]`**

Argument	Description
args	
create	Adds a virtual IP address to the HA cluster. During validation, the script ensures the VIP configuration file name and virtual IP address have not previously been configured. Following successful configuration in pacemaker, the new VIP configuration file is created on both servers to retain configuration of the VIP resource.
delete	Deletes a virtual IP address from the HA cluster. Following successful deletion of the VIP resource in pacemaker, the corresponding VIP configuration file is removed from both servers. ***Note:*** *You cannot use this script to remove the virtual IP address created as part of the base HA configuration.*
info	Lists pacemaker details of the VIP resources configured on the HA cluster, including the base Virtual IP address created as part of the HA configuration.
--vipname ***<file name>***	Specifies the VIP configuration file name. The name must be unique among all the configured virtual IP addresses.
--vipaddr ***<virtual IP address>***	Specifies the virtual IP address to be created.

Argument	Description
--nic <*NIC number*>	Specifies the network interface to which to connect the newly-created virtual IP address.
--netmask <*netmask value*>	Specifies the netmask value to use for virtual IP address. When not specified, the default is 24.

Sample Command Output

configure_vip with --create argument

```
./configure_vip create --vipname mytest2 --vipaddr 10.33.7.135 --nic
eth1

Please double check the following configuration options:

{

  vipname => 'mytest2',

  vipaddr => '10.33.7.135',

  nic => 'eth1',

  netmask => '24'

}

Are you sure you want to continue with this configuration? (Y/N): y

pcs active: emsload3

pcs active: emsload2

Added VIP resource mytest2 in pacemaker.
```

configure_vip with --delete argument

```
./configure_vip delete --vipname mytest2

Please double check the following configuration options:

{

  vipname => 'mytest2',

  vipaddr => '10.33.7.135',

  nic => 'eth1',

  netmask => '24'
```

```
}

Are you sure you want to delete this configuration? (Y/N): y

Attempting to stop: mytest2... Stopped

Deleted VIP resource mytest2 in pacemaker.
```

configure_vip with --info argument

```
./configure_vip info


=========== HA VIP configuration ===========



 Resource: virtualip (class=ocf provider=heartbeat type=IPaddr2)

  Attributes: cidr_netmask=24 ip=10.32.7.189 nic=eth0

  Operations: monitor interval=300 (virtualip-monitor-interval-300)

              start interval=0s timeout=20s
(virtualip-start-interval-0s)

             stop interval=0s timeout=20s (virtualip-stop-interval-0s)



========== Additional VIP mytest2 ==========



 Resource: mytest2 (class=ocf provider=heartbeat type=IPaddr2)

  Attributes: cidr_netmask=24 ip=10.33.7.135 nic=eth1

  Operations: monitor interval=300 (mytest2-monitor-interval-300)

             start interval=0s timeout=20s (mytest2-start-interval-0s)

              stop interval=0s timeout=20s (mytest2-stop-interval-0s)



===========================================
```

Backup Performance Management Statistics Files to Destination Server

It is highly recommended that you backup your performance management statistics files to either a local or remote server. These files can be useful for the Affirmed TAC to troubleshoot and identify the root cause issues occurring on the network elements.

For instructions on how to use the configureRemoteServerForCsvPush.sh script for the backup, refer to the section *Copying Performance Management Statistics Files to Destination Server* in the *Acuitas User's Guide*.

NTP Server Configuration

A Network Timing Protocol server is required to synchronize timing between Acuitas and the AN devices.

Bare Metal Systems

Use the configure_ntp.sh script to configure a system NTP server and enable its administrative state.

To configure a system NTP server and enable its administrative state:

1 Enter the following commands (the script must be run from the /bin directory):

```
cd /opt/Affirmed/NMS/bin
```

```
./configure_ntp.sh
```

2 When prompted, enter the IP address of the NTP time server.

The script updates the NTP configuration file and brings up the configured NTP server.

3 Enter the following command to verify that the NTP server is successfully started:

```
ps -ef | grep -i ntpd
```

```
ntp      13760     1  0 10:59 ?        00:00:00 ntpd -u ntp:ntp -p
/var/run/ntpd.pid -g
```

4 Reboot the Acuitas server.

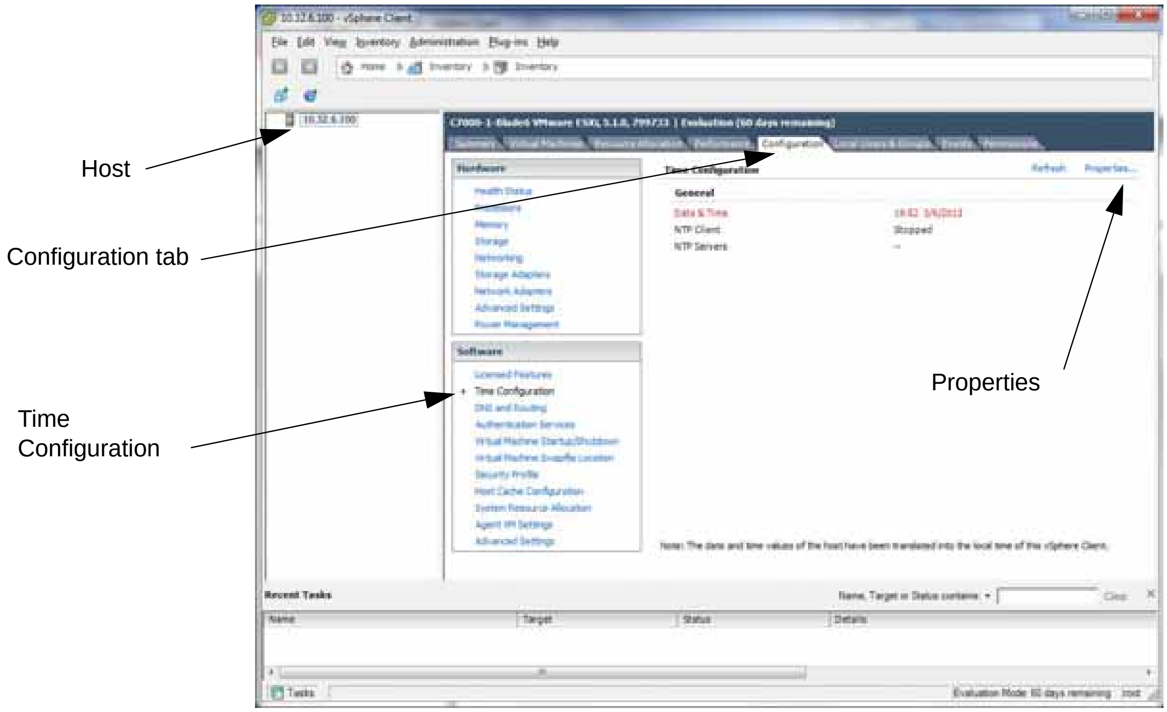
```
# reboot
```

Virtual Machine

To enable NTP:

1 Log into your vSphere client.

Figure 4-1 Time Configuration



2 Perform the following steps at the initial screen:

a Click the host.

b Click the **Configuration** tab.

c In the **Software** list, click **Time Configuration**.

d Click **Properties**.

3 In Time Configuration:

a Set the Time and Date for the host.

b Mark the **NTP Client Enabled** checkbox and click **Options**.

4 In the NTP daemon:

a Click **NTP Settings**.

b Click **Add**.

c Enter an NTP server IP address.

d Mark the **Restart NTP service to apply changes** checkbox.

e Click **OK**.

5 In the NTP daemon:

   a Click **General**.

   b Click **Start**.

   c Click **OK**.

   d In Time Configuration, click **Options**.

   e Mark the **Start and stop with host** checkbox and click **Start**.

   f Click **OK**.

6 Verify that all settings are listed correctly in the **Time Configuration** screen.

Configuring Acuitas to Communicate Using Public Key Infrastructure

When a MCC has been configured to use pki (public key infrastructure) for ssh access, you must configure Acuitas to communicate with the MCC. The configuration includes the following procedures:

- Generate the SSH keys
- Install the generated keys (public key on MCC clusters and private key on Acuitas server)

To generate the ssh keys:

1 Log in using the default user account `affirmed` and the account password.

2 Issue the following Linux command:

```
#ssh-keygen –t rsa
```

(Alternative: Use **`ssh-keygen -t dsa`** to generate ssh dsa keys)

3 Complete the following prompts:

```
Enter file in which to save the key (/home/affirmed/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
```

When a passphrase it specified, enter that value in the password field when adding the MCC to Acuitas (in the **Add Network Element** window).

The key generation procedure produces the following private/public key pairs:

- private key: id_dsa (or id_dsa for dsa keys)
- public key: id_dsa.pub (or id_dsa.pub for dsa keys)

To install the generated keys:

1. Install the public key on the MCC cluster (replace id_rsa.pub with id_dsa.pub if using dsa keys):

   a. Place the public key id_rsa.pub (or id_dsa.pub) in the /public/images/ directory of the MCC cluster.

   b. Log into the CLI of the MCC and issue the following commands in sequence:

   ```
   pki install type user source /public/images/id_rsa.pub
   ```

   ```
   pki ssh authorizedkey add id_rsa.pub user emsadmin
   ```

2. Install the private key on the Acuitas server as root (replace id_rsa with id_dsa ifusing dsa keys):

   a. Place the generated private key id_rsa under /opt/Affirmed/NMS/conf/.ssh directory.

   b. Issue the following command:

   ```
   chmod 600 /opt/Affirmed/NMS/conf/.ssh/id_rsa
   ```

   c. If the private key has a passphrase, enter that value in the password field when adding the MCC to Acuitas (in the **Add Network Element** window).

The installed private key are preserved following Acuitas upgrades.

See the *Affirmed Networks System Administration Guide* for details on how to administer pki authentication for MCC users. Some useful MCC CLI commands are:

- **`show pki ssh summary`** to display pki configurations.
- **`show user-access users`** to display user access controls.
- **`user-access ssh-password-authentication user emsadmin`** to administer pki access control for user `emsadmin`.

Changing Hostname/IP Address of the Acuitas Server

This section describes how to change the hostname and/or IP address on the Acuitas server.

Procedure for a Standalone System

To change the IP address of the Acuitas server:

1. Stop Acuitas:

   ```
   cd /opt/Affirmed/NMS/bin
   ./emsstop
   ```

2. Issue the following commands to clear out the EMSNODE and other related tables:

   ```
   cd /opt/Affirmed/NMS/db/bin
   ./dbupgrade --execute reset_emsnode
   ```

3. Stop MySQL:

   ```
   systemctl stop mysqld
   ```

4. As `root`, change the Acuitas server IP address in the following file:

   /etc/sysconfig/network-scripts/ifcfg-eth0

5. Update the entries in /etc/hosts with the new IP address.

6. As `root`, reboot the server or enter the following command:

   ```
   systemctl restart network
   ```

7. Start MySQL:

   ```
   systemctl start mysqld
   ```

8. Restart the server:

   ```
   cd /opt/Affirmed/NMS/bin
   ./starttems.sh
   ```

To change the Hostname of the Acuitas server:

1. Stop Acuitas:

   ```
   cd /opt/Affirmed/NMS/bin
   ./emsstop
   ```

2. Issue the following commands to clear out the EMSNODE and other related tables:

   ```
   cd /opt/Affirmed/NMS/db/bin
   ./dbupgrade --execute reset_emsnode
   ```

3. Stop MySQL:

   ```
   systemctl stop mysqld
   ```

4. Update the entries in /etc/hosts with the new hostname.

5. As `root`, change hostname using the following command:

   ```
   hostnamectl set-hostname <new_hostname>
   ```

6 Verify the new hostname using the following command:

```
hostnamectl status
```

7 Start MySQL:

```
systemctl start mysqld
```

8 Restart the server:

```
cd /opt/Affirmed/NMS/bin
./starttems.sh
```

Procedure for an HA System

Preparation

1 Verify that the primary node of the HA is the active node. If it is not, issue the following command to perform a manual switchover (to set the primary node as the active node):

```
cd /opt/Affirmed/NMS/bin
./emsha switchover
```

2 At the confirmation prompt, enter **y**.

3 Issue the following commands to verify that the HA status is operational and up:

```
cd /opt/Affirmed/NMS/bin
./emsstatus
```

4 Issue the following commands on the primary active node to backup the database:

```
cd /opt/Affirmed/NMS/bin
./emsdb backup --backupfile /opt/backup/backup.db
Taking a backup of localhost MySQL databases ...
Backup file is /opt/backup/backup.db
```

5 Move the backup of /opt/backup/backup.db to a remote server:

```
scp /opt/backup/backup.db <remote server path>
```

Procedure

1 Shutdown Acuitas on both hosts, starting with the standby node:

a Stop Acuitas on the standby node:

```
cd /opt/Affirmed/NMS/bin
./emsha stopAll
```

b At the confirmation prompt, enter **y**.

c Stop Acuitas on the active node:

```
cd /opt/Affirmed/NMS/bin
```

```
./emsha stopAll
```

d At the confirmation prompt, enter **y**.

2 Issue the following commands to clear out the EMSNODE and other related tables:

```
cd /opt/Affirmed/NMS/db/bin
./dbupgrade --execute reset_emsnode
```

3 Shutdown MySQL on both hosts, starting with the standby node:

a Stop MySQL on the standby node:

```
systemctl stop mysqld
```

b Stop MySQL on the active node:

```
systemctl stop mysqld
```

4 As `root`, issue the following commands to clear out the cluster configuration:

```
pcs cluster stop --force
pcs cluster destroy --all
```

5 As `root`, change IP address on both hosts (*skip this step if you do not need to change the IP addresses*):

a Remove the old entries created by HA script from /etc/hosts.

b Change the Acuitas server IP address in the following file:

/etc/sysconfig/network-scripts/ifcfg-<eth0 | 1 | 2 | 3>

c Restart the server:

```
systemctl restart network
```

6 Reset the keyless ssh login between the hosts to allow SSH access between the systems during configuration:

a Log in using the default user account `affirmed` and the account password and issue the following command:

```
#ssh-keygen –t rsa
```

b Press **Enter** at the following prompt:

```
Enter file in which to save the key (/home/affirmed/.ssh/id_rsa):
```

c If prompted to overwrite the existing key file, enter **y**:

```
/home/affirmed/.ssh/id_rsa already exists.
Overwrite (y/n)?
```

d Press **Enter** for each of the following prompts:

```
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
```

e Issue the following command:

```
ssh-copy-id -i ~/.ssh/id_rsa.pub affirmed@<IP address of another server in HA>
```

  f When prompted, enter the `affirmed` password.

7 As `root`, change hostname on both hosts (*skip this step if you do not need to change the hostnames*):

  a Remove the old entries created by configure_ha script from /etc/hosts.

  b Change hostname using the following command on both hosts:

```
hostnamectl set-hostname <new_hostname>
```

  c Verify the new hostname using the following command on both hosts:

```
hostnamectl status
```

  d Restart the server:

```
systemctl restart network
```

8 Start MySQL on both hosts, starting with the active node:

  a Start MySQL on primary/active:

Active Node:

```
systemctl start mysqld
```

If you have changed the hostname:

```
mysql -uroot -p –e “update mysql.user set Host=’<new hostname>’
where Host=’<old hostname>’”
```

```
mysql -uroot -p –e “flush privileges”
```

When prompted, enter the root password.

If you have changed the IP address:

```
mysql -uroot -p –e “update mysql.user set Host=’<new IP address>’
where Host=’<old IP address>’”
```

```
mysql -uroot -p –e “flush privileges”
```

When prompted, enter the root password.

  b Start MySQL on secondary/standby:

Standby Node:

```
systemctl start mysqld
```

If you have changed the hostname:

```
mysql -uroot -p –e “update mysql.user set Host=’<new hostname>’
where Host=’<old hostname>’”
```

```
mysql -uroot -p –e “flush privileges”
```

When prompted, enter the root password.

If you have changed the IP address:

```
mysql -uroot -p –e “update mysql.user set Host=’<new IP address>’
where Host=’<old IP address>’”
```

```
mysql -uroot -p –e “flush privileges”
```

When prompted, enter the root password.

9 Re-configure HA (to use the backed up database, use the --backupfile option):

a Clean up previous HA configuration on both hosts:

```
rm /opt/Affirmed/NMS/conf/cluster/configure_ha.conf
```

b Run configure_ha from the primary node, using the new hostname and/or the new IP address:

```
cd /opt/Affirmed/NMS/bin
```

```
./configure_ha [--virtualhostname virtual hostname] [--virtualip
virtual IP address] [--primaryhostname primary hostname]
[--primaryip primary IP address] [--secondaryhostname secondary
hostname] [--secondaryip secondary IP address] [--netmask netmask
value] [--ipinterface interface] [--backupfile file_path] --restore
```

Changing Timezone of the Acuitas Server

Prerequisite

If the Acuitas server is running on a virtual machine, you must have NTP enabled on the host. For instructions, see Virtual Machine.

This procedure requires you to `su` to `root`.

Procedure for a Standalone System

To change the timezone of the Acuitas Server:

1 Issue the following command to configure the timezone.

`tzselect`

```
Please identify a location so that time zone rules can be set
correctly.
Please select a continent or ocean.
 1) Africa
 2) Americas
 3) Antarctica
 4) Arctic Ocean
 5) Asia
 6) Atlantic Ocean
 7) Australia
 8) Europe
 9) Indian Ocean
10) Pacific Ocean
11) none - I want to specify the time zone using the Posix TZ format.
```

2 Enter a location number.

```
Please select a country.
 1) Anguilla              19) Dominican Republic     37) Peru
 2) Antigua & Barbuda     20) Ecuador                38) Puerto Rico
 3) Argentina             21) El Salvador            39) St Barthelemy
 4) Aruba               22) French Guiana         40) St Kitts & Nevis
 5) Bahamas               23) Greenland              41) St Lucia
 6) Barbados              24) Grenada                42) St Maarten
(Dutch)
 7) Belize                25) Guadeloupe             43) St Martin
(French)
 8) Bolivia               26) Guatemala              44) St Pierre &
Miquelon
 9) Brazil                27) Guyana                 45) St Vincent
```

```
10) Canada              28) Haiti              46) Suriname
11) Caribbean NL        29) Honduras           47) Trinidad & Tobago
12) Cayman Islands      30) Jamaica            48) Turks & Caicos Is
13) Chile               31) Martinique         49) United States
14) Colombia            32) Mexico             50) Uruguay
15) Costa Rica          33) Montserrat         51) Venezuela
16) Cuba                34) Nicaragua          52) Virgin Islands
(UK)
17) Curaçao             35) Panama             53) Virgin Islands
(US)
18) Dominica            36) Paraguay
```

3 Enter a country number.

```
Please select one of the following time zone regions.
 1) Eastern (most areas)              16) Central - ND (Morton rural)
 2) Eastern - MI (most areas)         17) Central - ND (Mercer)
 3) Eastern - KY (Louisville area)    18) Mountain (most areas)
 4) Eastern - KY (Wayne)              19) Mountain - ID (south); OR
(east)
 5) Eastern - IN (most areas)        20) MST - Arizona (except Navajo)
 6) Eastern - IN (Da, Du, K, Mn)      21) Pacific
 7) Eastern - IN (Pulaski)            22) Alaska (most areas)
 8) Eastern - IN (Crawford)           23) Alaska - Juneau area
 9) Eastern - IN (Pike)               24) Alaska - Sitka area
10) Eastern - IN (Switzerland)        25) Alaska - Annette Island
11) Central (most areas)              26) Alaska - Yakutat
12) Central - IN (Perry)              27) Alaska (west)
13) Central - IN (Starke)             28) Aleutian Islands
14) Central - MI (Wisconsin border)   29) Hawaii
15) Central - ND (Oliver)
```

4 Enter a time zone region number.

```
The following information has been given:

        United States
        Eastern (most areas)

Therefore TZ='America/New_York' will be used.
Local time is now:      Tue Dec 12 17:24:55 EST 2017.
Universal Time is now:  Tue Dec 12 22:24:55 UTC 2017.
Is the above information OK?
```

5 At the confirmation prompt, enter **1** for Yes.

```
You can make this change permanent for yourself by appending the line
        TZ='America/New_York'; export TZ
```

```
to the file '.profile' in your home directory; then log out and log in
again.

Here is that TZ value again, this time on standard output so that you
can use the /usr/bin/tzselect command in shell scripts:
America/New_York
```

6 Issue the following command followed by the output from step 5:

Note: *This example sets the time zone to eastern time. Change your command accordingly.*

```
timedatectl set-timezone America/New_York
```

7 Reboot the Acuitas Server:

```
# reboot
```

When the server comes back up, it is using the new timezone.

Procedure for a Cluster

This section describes how to change timezone in an HA + DR cluster.

To change the timezone of the cluster:

1 On the master pair, issue the following commands to shutdown the standby server and the active server, in that order:

Standby server:

```
cd /opt/Affirmed/NMS/bin
./emsha stopAll
```

Active server:

```
cd /opt/Affirmed/NMS/bin
./emsha stopAll
```

2 At each confirmation prompt, enter **y**.

3 No action is required on the slave pair, as long as no Acuitas process is running. Issue the following command on all servers in the cluster to ensure that no Acuitas process is running.

```
cd /opt/Affirmed/NMS/bin
./emsstatus
```

4 Issue the following command to configure the timezone.

```
tzselect
```

```
Please identify a location so that time zone rules can be set
correctly.
Please select a continent or ocean.
 1) Africa
 2) Americas
 3) Antarctica
 4) Arctic Ocean
 5) Asia
 6) Atlantic Ocean
 7) Australia
 8) Europe
 9) Indian Ocean
10) Pacific Ocean
11) none - I want to specify the time zone using the Posix TZ format.
```

5 Enter a location number.

```
Please select a country.
 1) Anguilla            19) Dominican Republic    37) Peru
 2) Antigua & Barbuda   20) Ecuador               38) Puerto Rico
 3) Argentina           21) El Salvador           39) St Barthelemy
```

```
 4) Aruba                 22) French Guiana          40) St Kitts & Nevis
 5) Bahamas               23) Greenland              41) St Lucia
 6) Barbados              24) Grenada                42) St Maarten
(Dutch)
 7) Belize                25) Guadeloupe             43) St Martin
(French)
 8) Bolivia               26) Guatemala              44) St Pierre &
Miquelon
 9) Brazil                27) Guyana                 45) St Vincent
10) Canada                28) Haiti                  46) Suriname
11) Caribbean NL         29) Honduras               47) Trinidad & Tobago
12) Cayman Islands       30) Jamaica                48) Turks & Caicos Is
13) Chile                 31) Martinique             49) United States
14) Colombia              32) Mexico                 50) Uruguay
15) Costa Rica            33) Montserrat             51) Venezuela
16) Cuba                  34) Nicaragua              52) Virgin Islands
(UK)
17) Curaçao               35) Panama                 53) Virgin Islands
(US)
18) Dominica              36) Paraguay
```

6 Enter a country number.

```
Please select one of the following time zone regions.
 1) Eastern (most areas)            16) Central - ND (Morton rural)
 2) Eastern - MI (most areas)       17) Central - ND (Mercer)
 3) Eastern - KY (Louisville area)  18) Mountain (most areas)
 4) Eastern - KY (Wayne)            19) Mountain - ID (south); OR
(east)
 5) Eastern - IN (most areas)      20) MST - Arizona (except Navajo)
 6) Eastern - IN (Da, Du, K, Mn)    21) Pacific
 7) Eastern - IN (Pulaski)          22) Alaska (most areas)
 8) Eastern - IN (Crawford)         23) Alaska - Juneau area
 9) Eastern - IN (Pike)             24) Alaska - Sitka area
10) Eastern - IN (Switzerland)      25) Alaska - Annette Island
11) Central (most areas)            26) Alaska - Yakutat
12) Central - IN (Perry)            27) Alaska (west)
13) Central - IN (Starke)           28) Aleutian Islands
14) Central - MI (Wisconsin border) 29) Hawaii
15) Central - ND (Oliver)
```

7 Enter a time zone region number.

```
The following information has been given:

        United States
        Eastern (most areas)
```

```
Therefore TZ='America/New_York' will be used.
Local time is now:      Tue Dec 12 17:24:55 EST 2017.
Universal Time is now:  Tue Dec 12 22:24:55 UTC 2017.
Is the above information OK?
```

8 At the confirmation prompt, enter **1** for Yes.

```
You can make this change permanent for yourself by appending the line
        TZ='America/New_York'; export TZ
to the file '.profile' in your home directory; then log out and log in
again.

Here is that TZ value again, this time on standard output so that you
can use the /usr/bin/tzselect command in shell scripts:
America/New_York
```

9 Issue the following command followed by the output from step 8:

Note: *This example sets the time zone to eastern time. Change your command accordingly.*

```
timedatectl set-timezone America/New_York
```

10 Reboot the Acuitas Server:

```
# reboot
```

When the server comes back up, it is using the new timezone.

11 Issue the following command on all servers in the cluster to ensure that everything is operational.

```
cd /opt/Affirmed/NMS/bin
./emsstatus
```

Setting Up MySQL root/Replication User Password

This section describes how to set up the MySQL root user and/or MySQL replication user password(s) on the Acuitas server.

Note: *Anytime you perform a fresh installation of the Acuitas software and anytime you change the MySQL root or replication user password, you need to change the replication user password again. See Changing MySQL root/Replication User Password.*

Procedure for a Standalone System

This section describes how to set up the MySQL root user and/or MySQL replication user password(s) for a standalone system. Setting the MySQL replication user password applies only on the slave server of DR standalone system.

To set up the MySQL root user and/or MySQL replication user password(s):

1 Enter the following command:

```
./configureserver.sh

Configure Server Details
        ------------------------

        1.DatabaseDetails

        2.PortDetails

        3.Authentication

        4.ProtocolDetails

        5.PurgeConfigurationDetails

        6.Exit

        Please select your option from (1 | 2 | 3 | 4 | 5 | 6) [1] :
```

2 Enter **1**.

3 At each of the following prompts, press **Enter** to keep the default value.

At the root user Password and Confirm Password prompts, enter the new root user password.

At the Replication Password and Confirm Replication Password prompts, enter the new replication password (*Only for a DR standalone system*).

```
Database Details
----------------
Database (MySQL) [MySQL]:

    Server Name/IP [localhost]:

    Repln Name/IP []:

    Port [3306]:

    Database Name [AFFIRMED_NMS_DB]:

    User Name [root]:

    Password [********]:

    Confirm Password:

    Replication Password [********]:

    Confirm Replication Password:

Would you like to save the changes (Y | N) [N]:
```

4 Enter **Y** to save changes.

```
MySQL user's password changed

MySQL replication user's password changed

Configure Server Details
        ------------------------

        1.DatabaseDetails

        2.PortDetails

        3.Authentication

        4.ProtocolDetails
```

```
        5.PurgeConfigurationDetails

        6.Exit


        Please select your option from (1 | 2 | 3 | 4 | 5 | 6) [1] :
```

5 To exit, enter **6**.

6 As `affirmed` user, perform these steps in order:

   a Enter the following command:

   ```
   mysql_config_editor set --login-path=client --user=root
   --host=localhost --password
   ```

   b When prompted for password, enter the MySQL root password you configured with **`configureserver.sh`**.

   c Enter the following command:

   ```
   mysql_config_editor set --login-path=replication --user=acuitas
   --password
   ```

   d When prompted for password, enter the MySQL replication password you configured with **`configureserver.sh`**.

7 As `root` user, perform these steps:

   a Enter the following command:

   ```
   mysql_config_editor set --login-path=client --user=root
   --host=localhost --password
   ```

   b When prompted for password, enter the MySQL root password you configured with **`configureserver.sh`**.

Procedure for a Cluster

This section describes how to set up the MySQL root user and/or MySQL replication user password(s) in an HA and HA + DR cluster.

To set up the MySQL root user and/or MySQL replication user password(s):

1 Enter the following command:

```
./configureserver.sh
```

```
Configure Server Details
        ------------------------


        1.DatabaseDetails


        2.PortDetails
```

```
3.Authentication

4.ProtocolDetails

5.PurgeConfigurationDetails

6.Exit


Please select your option from (1 | 2 | 3 | 4 | 5 | 6) [1] :
```

2. Enter **1**.
3. At each of the following prompts, press **Enter** to keep the default value.

   At the root user Password and Confirm Password prompts, enter the new root user password.

   At the Replication Password and Confirm Replication Password prompts, enter the new replication password.

```
Database Details
----------------
Database (MySQL) [MySQL]:

    Server Name/IP [localhost]:

    Repln Name/IP []:

    Port [3306]:

    Database Name [AFFIRMED_NMS_DB]:

    User Name [root]:

    Password [********]:

    Confirm Password:

    Replication Password [********]:

    Confirm Replication Password:

Would you like to save the changes (Y | N) [N]:
```

4. Enter **Y** to save changes.

```
MySQL user's password changed

MySQL replication user's password changed

Configure Server Details
        ------------------------

        1.DatabaseDetails

        2.PortDetails

        3.Authentication

        4.ProtocolDetails

        5.PurgeConfigurationDetails

        6.Exit


        Please select your option from (1 | 2 | 3 | 4 | 5 | 6) [1] :
```

5 To exit, enter **6**.

6 As `affirmed` user, perform these steps in order:

   a Enter the following command:

   `mysql_config_editor set --login-path=client --user=root --host=localhost --password`

   b When prompted for password, enter the MySQL root password you configured with **`configureserver.sh`**.

   c Enter the following command:

   `mysql_config_editor set --login-path=replication --user=acuitas --password`

   d When prompted for password, enter the MySQL replication password you configured with **`configureserver.sh`**.

7 As `root` user, perform these steps:

   a Enter the following command:

   `mysql_config_editor set --login-path=client --user=root --host=localhost --password`

   b When prompted for password, enter the MySQL root password you configured with **`configureserver.sh`**.

Changing MySQL root/Replication User Password

This section describes how to change the MySQL root user and/or MySQL replication user password(s) on the Acuitas server.

Anytime you perform a fresh installation of the Acuitas software and anytime you change the MySQL root or replication user password, you need to change the replication user password again.

Procedure for a Standalone System

This section describes how to change the MySQL root user and/or MySQL replication user password(s) for a standalone system. Changing the MySQL replication user password applies only on the slave server of DR standalone system.

To change the MySQL root user and/or MySQL replication user password(s):

1. Log in using the default user account `affirmed` and the account password.
2. Enter the following commands to stop the Acuitas server:

   ```
   cd /opt/Affirmed/NMS/bin
   ./emsstop
   ```
3. Perform the setup instructions in Procedure for a Standalone System.
4. If you have changed the replication password on the slave server of DR standalone system, enter the following commands:

   ```
   cd /opt/Affirmed/NMS/bin
   ./configure_dr --changereplpwd
   ```
5. Enter the following commands to start the Acuitas server *(Not for a slave server of DR standalone system)*:

   ```
   cd /opt/Affirmed/NMS/bin
   ./starttems.sh
   ```

Procedure for a Cluster

This section describes how to change the MySQL root user and/or MySQL replication user password(s) in an HA and HA + DR cluster.

To change the MySQL root user and/or MySQL replication user password(s):

1. Issue the following steps, first on the standby server, then on the active server:
   a. Log in using the default user account `affirmed` and the account password.
   b. Enter the following commands to stop the Acuitas server:

      ```
      cd /opt/Affirmed/NMS/bin
      ./emsha stopAll
      ```

2. Perform the setup instructions in Procedure for a Cluster to change the password using configureserver.sh. Perform the steps on both servers as `affirmed` user (in any order). Be sure to use the same password on all servers.
3. If you have changed the replication password, enter the following commands on the primary server only:

   ```
   cd /opt/Affirmed/NMS/bin
   ./configure_ha --changereplpwd
   ```

4. Enter the following commands on primary first and then secondary server to start the Acuitas server:

   ```
   cd /opt/Affirmed/NMS/bin
   ./emsha startAll
   ```

5. If the system is HA + DR:
   a. On the slave pair, log in using the default user account `affirmed` and the account password.
   b. Complete step 2 through step 3 on both servers.

Enabling IPv6

The an-ems-3.0.*x*-rhel-7.*y*-mysql-8.0.*z* OS image disables IPv6 by default. Use this procedure to enable it:

To enable IPv6:

1 As `root`, edit /etc/sysctl.conf and comment out the following line:

```
#net.ipv6.conf.all.disable_ipv6 = 1
```

2 As `root`, create a new file named /etc/sysctl.d/enable_ipv6.conf and add the following lines:

```
net.ipv6.conf.all.disable_ipv6 = 0
net.ipv6.conf.default.disable_ipv6 = 0
net.ipv6.conf.lo.disable_ipv6 = 0
```

3 As `root`, execute the following commands to activate the configuration and have it persist upon reboot:

```
sysctl -p /etc/sysctl.d/ipv6.conf
dracut -f
```

Increasing Acuitas Server Resources on a Virtual Machine

This section describes how to increase the number of CPUs used by the Acuitas VM.

To increase the number of CPUs used by the Acuitas VM:

1. Stop the Acuitas server:

   ```
   cd /opt/Affirmed/NMS/bin
   ```

   ```
   ./emsstop
   ```

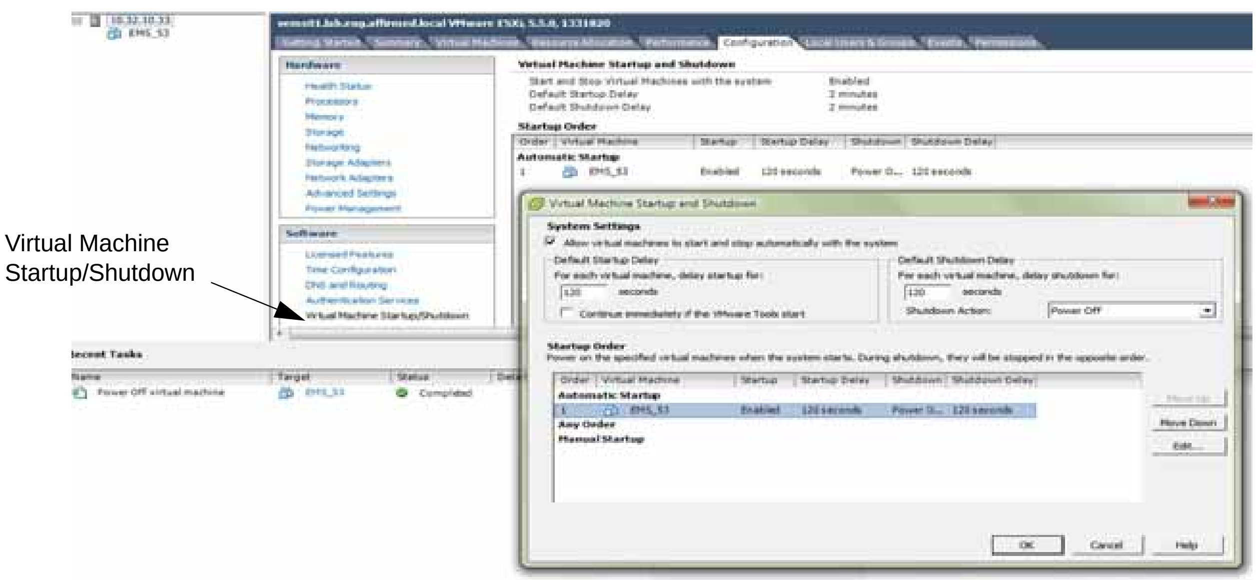
2. Log into your vSphere client.
3. Under Host, right-click the Acuitas guest VM and select **Power** > **Power Off**.
4. If the Acuitas guest VM is not setup to automatic start up when the host is started, perform these steps:

   a. Click the **Configuration** tab.

   b. In the **Sofware** list, click **Virtual Machine Startup/Shutdown**.

   c. Click **Properties**.

Figure 4-2 Virtual Machine Startup/Shutdown



   d. Mark the **Allow virtual machines to start and stop automatically with the machine** checkbox.

   e. In **Startup Order**, click the Acuitas guest VM and click **Move Up** to move it to Automatic Startup.

   f. Click **OK**.

5 To change the CPU setting for the Acuitas guest VM, perform these steps:

a Right-click Acuitas guest VM and select **Edit Settings**.

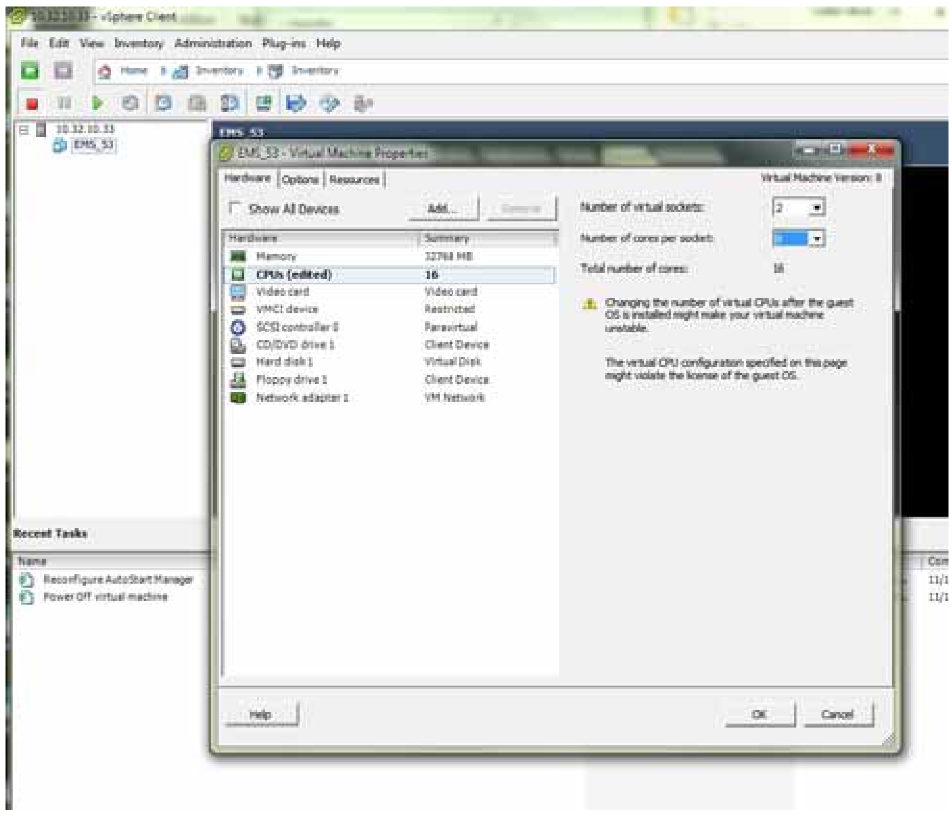
b Click the **Hardware** tab.

c Click **CPUs**.

d Keep the number of virtual sockets at 2. Increase the number of virtual cores to 8, so the total number of cores is 16.

e Click **OK**.

Figure 4-3 CPUs Settings



6 Under Host, right-click the Acuitas guest VM and select **Power** > **Power On**.

7 Log onto the Acuitas server as `root`.

8 Start the Acuitas server:

Standalone Server:

```
cd /opt/Affirmed/NMS/bin
./starttems.sh
```

2-node HA Cluster:

Issue the following commands from one of the servers in the cluster:

```
cd /opt/Affirmed/NMS/bin
./emsha startAll  primary
./emsha startAll secondary
```

4-node HA Cluster:

Issue the following commands from one of the servers in the master pair:

```
cd /opt/Affirmed/NMS/bin
./emsha startAll  primary
./emsha startAll secondary
```

9. At each confirmation prompt, enter **y**.
10. When each system returns the success message:

    ```
    Server started successfully
    ```

    Press **Enter** to return to the operating system prompt.

Optional: Channel Bonding on Acuitas Physical Interfaces

Channel bonding enables multiple network interfaces to act as a single interface, thereby increasing bandwidth and providing redundancy. You can configure channel bonding to operate with Acuitas servers (standalone and High Availability) to remove a single point of failure for the Acuitas management network.

To set up Channel Bonding:

1. To create a channel bonding interface, create a file in the /etc/sysconfig/network-scripts/directory named ifcfg-bond*N*, where *N* is the interface number.
2. Edit the ifcfg-bond*N* file and add the following lines:

```
DEVICE=bondN
IPADDR=<IP address of the device>
PREFIX=<Netmask>
GATEWAY=<Gateway IP address>
ONBOOT=yes
BOOTPROTO=none
USERCTL=no
BONDING_OPTS="mode=active-backup primary=<eth#> miimon=100"
```

   Note: The `primary=<eth#>` setting is optional. Use it to specify the port associated with the mac-address used to produce the EMS license, but only in the case where this specific port should always be the active port.

3. Save the file.
4. After the channel bonding interface is created, configure the network interfaces to be bound together by adding the MASTER and SLAVE directives to their corresponding configuration files. To do so, add the following lines to each network interface configuration file you are bonding.

```
ONBOOT=yes
MASTER=bondN
SLAVE=yes
USERCTL=no
mode=active-backup
```

   For example, to bond eth0 and eth1, add these lines to /etc/sysconfig/network-scripts/ifcfg-eth0 and /etc/sysconfig/network-scripts/ifcfg-eth1

5. As root, create a new file in the /etc/modprobe.d/ directory named <*bonding*>.conf with any filename with the required .conf extension. Add the following line to the file:

```
alias bondN bonding
```

   Where *N* is the interface number, such as 0.

6. Execute the following commands to restart the network:

```
service network restart
```

OS Upgrade Considerations

This appendix contains instructions that apply to an OS upgrade. See the *Software Release Notes for the Acuitas Service Management System* to determine if an upgrade to the OS is required.

Prerequisite to Backup the Acuitas Server if Needed for Disaster Recovery

Before performing the upgrade, backup the Acuitas Server in case you need to later restore the OS to the pre-upgrade level. (For instructions on restoring the OS, see Restoring the OS to Pre-upgrade Level).

Complete the instructions in the section that applies to your system:

- Bare Metal System
- Virtual Machine

Bare Metal System

To backup the Acuitas Server:

1. Log into the Acuitas server:
2. Issue the following commands to backup the database:

```
cd /opt/Affirmed/NMS/bin/
./emsdb backup –backupfile <path_to_backup_file>
```

3. Issue the following commands to archive the Config/Data/Export directories:

```
cd /opt/Affirmed/NMS
tar -czvf <file_name>.tar.gz conf/cluster/configure_ha.conf
server/ems/conf server/ems/data/config/ server/ems/data/gr/
server/ems/data/hornetq/ server/ems/data/nc server/ems/data/nebackups/
server/ems/data/report/ server/ems/data/tx-object-store/
server/ems/archive/ server/ems/export/
```

4. Copy the database backup and archived files to a remote location.

Virtual Machine

The procedure applies to an ESXi system.

To backup the Acuitas Server:

1. To start the vSphere client, select **Start** > **Programs** > **VMware** > **VMware vSphere Client** and log in.
2. Select the VM from the list and power off the VM:
   a. Click the **Summary** tab.
   b. Click **Power Off** in the **Commands** list.
3. Select **File** > **Export** > **Export OVF Template**.
4. Specify the directory on the local machine (or external storage device) to store the OVF file. This file can be very large depending on the VM specifications, so make sure the specified location has enough storage space. If you do not have a location with enough storage space, use the backup instructions in Bare Metal System.
5. Click **OK**.
6. After the export is complete, power the VM back on:
   a. Click the **Summary** tab.
   b. Click **Power On** in the **Commands** list.

Upgrading the OS

To upgrade the underlying Operating System, perform the following tasks:

- Back up the existing system
- Create a new installation with the desired OS Image
- Restore the backup into the new environment

These tasks are described in detail in the following sections.

Backup the Existing Installation

To backup the existing installation:

1 Trigger a database backup:

a Click the **Settings** icon on the toolbar and select **Tools** => **Database Administration**.

Acuitas displays the **Database Backup and Restore** window.

b Click the **User Triggered DB-Backup** icon.

c Configure the parameters as follows:

– **Backup Files Prefix Name** – Specify the prefix you want for the backup file, such as DB_backup.

– **FTP Server** – Specify whether you want the backup stored on the local or remote server. It is recommended that you store the backup on the remote server, as a local backup will not be available if the system becomes out of service.

– **Relative Path in FTP Server** – If you select to have the backup stored on remote server, specify the path where you want it stored. If you select to have the backup stored on the local server, it is stored at: */opt/Affirmed/NMS/server/databaseBackup*.

– **Full Backup** – Mark this checkbox to trigger a full backup.

d Click **Save**.

e At the confirmation prompt, click **OK**.

f Click **Close**.

2 Move the backup to a remote file server for restoring on the VM.

3 Save the license file from /opt/Affirmed/NMS/server/ems/conf/License.dat to a remote file server (Do so once per standalone server or HA cluster).

4 *VMs Only:* Record the MAC address of the primary interface of each server in the deployment to be used on the new installation (Alternative: obtain a new license file).

5 *Optional*: Copy the following files to a remote file server:

- /opt/Affirmed/NMS/server/ems/resources/.keystore
- /etc/hosts
- /etc/hosts.allow or /etc/hosts.deny

Note: You can also back up historical information, such as logs and statistics on an as-needed basis. Contact the TAC for further information.

Create the New Installation

VM Recommendation

Power down the existing system, leaving it intact until the upgrade is verified. Create new VMs for the new installation. Do not delete the existing VMs until you are sure the upgrade has been verified.

VM Only

After creating the VMs using the instructions below, you need to modify the MAC address to match that of the original installation. Failure to do so will invalidate your license, preventing startup of the Acuitas application.

Edit the VM settings (VMWare) or adjust the flavor (OpenStack) to increase disk size from the default 200GB. For VMWare, this value must be in multiples of 50GB, and must be between 250 and 2000 GB. In either case, the recommended minimum is 550GB.

Prepare for the New Installation

To prepare for installation of the new system, follow the the appropriate instructions in Preparing the System for Installation/Upgrade:

- Bare Metal System
- ESXi System
- OpenStack System

Install the New Application

To install the new application:

1. From the console, log into the new installation.
2. Once you are logged in, issue the following command to gain super-user privileges:

   ```
   su -
   ```

3. Configure the system to allow network access by adding its IP address to /etc/hosts.allow.
4. Run /root/ems_setup.sh and configure the following settings:
   - Hostname
   - IP Address/Netmask/Gateway
   - DNS
5. *VMWare Only:* Exit the ems_setup.sh application. When prompted, enter the new disk size that you specified when you created the VM.
6. Verify network connectivity. Bare Metal systems may require manually enabling the appropriate Ethernet interface.
7. Copy the new Acuitas rpm to /data/affirmed on the new installation, and as root, issue the following command to install it:

   ```
   rpm -ivh /data/affirmed/ems-x.x.x.x.rpm
   ```

8. *Optional*: Once the installation is compete, restore /etc/hosts, /etc/hosts.allow, and/or /etc/hosts.deny.
9. Change to user `affirmed` and configure the following:
   - Changing Timezone of the Acuitas Server
   - NTP Server Configuration
10. Restore the License.dat file into /opt/Affirmed/NMS/server/ems/conf.
11. *Optional*: Restore the .keystore file to /opt/Affirmed/NMS/server/ems/resources.

Establish MySQL Passwords

Acuitas Release 9.4 and later requires you to establish the MySQL root password and replication password if using HA or DR. Although it was not required, if you made changes to either of these passwords in your previous installation, you must establish the *same* password(s) so that the upcoming database restore can function correctly. If you *did not* change the default root or replication MySQL passwords, you must execute the following procedure to set the passwords to the previous defaults so that the upcoming database restore can function correctly (Contact the TAC for the previous default passwords if needed). If you were previously using the default passwords, it is recommended that you change them once the restore operation is complete.

Execute the following procedure on each server in the Acuitas deployment. If using HA and/or DR, all passwords *must match* among all servers.

To establish MySQL passwords:

1. Login to the Acuitas server as the `affirmed` user.
2. Execute the configureserver.sh script.
3. Set the root MySQL password as desired, and confirm.
4. *HA and DR only*: Set the replication password as desired, and confirm.
5. Save the settings and exit the script.
6. Issue the following command to store the root MySQL user credentials:

   ```
   mysql_config_editor set --login-path=client --user=root
   --host=localhost --password
   ```

7. When prompted for password, enter the MySQL root password.
8. *HA or DR Only*: Execute the following command to store the MySQL replication user credentials:

   ```
   mysql_config_editor set --login-path=replication --user=acuitas
   --password
   ```

9. When prompted for password, enter the MySQL replication password.
10. Issue the following command to gain super-user privileges:

    ```
    su -
    ```

11. Issue the following command:

    ```
    mysql_config_editor set --login-path=client --user=root
    --host=localhost --password
    ```

12. When prompted for password, enter the MySQL root password.

Restore the Database and Start the Application

If you are using DR (either standalone or HA), the slave site is installed as per the fresh installation instructions — only the Master site has the database restored. Once the restore is complete, convert the slave site to replicate from the master site.

Standalone (Master Site Only if DR)

To restore the database:

1. Copy the database backup to the server.
2. Issue the following commands to prepare MySQL for the database import:
   ```
   mysql -e "reset master":
   mysqladmin drop AFFIRMED_NMS_DB
   mysqladmin create AFFIRMED_NMS_DB
   ```
3. Issue the following command to import the database backup (**Note:** Expected errors occur during this step):
   ```
   mysql -f AFFIRMED_NMS_DB < <path to pre-9.4 backup file>
   ```
4. Issue the following command to start the standalone Acuitas application with the upgrade flag (to correct errors encountered in the previous step):
   ```
   /opt/Affirmed/NMS/bin/starttems.sh -u
   ```

HA (Master Site Only if DR)

To restore the database:

1. Copy the database backup to the *Primary* server.
2. Execute the configure_ha script with required options to restore the database:
   - If you have restored configure_ha.conf in /opt/Affirmed/NMS/conf/cluster/, issue this command:
     ```
     /opt/Affirmed/NMS/bin/configure_ha --reload --restore --backupfile
     <file_path>
     ```
   - If this is first setup of HA, issue this command with the full set of <*HA parameters*>:
     ```
     /opt/Affirmed/NMS/bin/configure_ha <HA parameters> --restore
     --backupfile <file_path>
     ```

Restoring the OS to Pre-upgrade Level

If you encounter post upgrade issues, you can restore the OS to the pre-upgrade level.

Complete the instructions in the section that applies to your system:

- Bare Metal Systems
- Virtual Machine

Bare Metal Systems

To restore the Acuitas Server:

1. Perform a fresh installation:
   - On a Standalone system, follow the instructions in Fresh Installation.
   - On an HA system or HA + DR system, follow the instructions in Recovery of an HA System.
2. Copy the database backup files (that you saved in Prerequisite to Backup the Acuitas Server if Needed for Disaster Recovery) to the Acuitas server.
3. Restore the database using the appropriate procedure:
   - Recovery of a Standalone System
   - Recovery of an HA System

Virtual Machine

The procedure applies to an ESXi system.

To restore the Acuitas Server:

1. Perform the steps in Install the OVA Template Package using the file you exported and saved in Prerequisite to Backup the Acuitas Server if Needed for Disaster Recovery.
2. Power the VM back on:
   a. Click the **Summary** tab.
   b. Click **Power On** in the **Commands** list.